Bitdoin Academy — Monthly Academic Score Tracking, AI Learning Analysis & Career Guidance

Product Requirements Document (PRD) · Version 1.0 Draft for Owner Review

23 October 2026

Prepared for: Bitdoin platform owner and delivery team
Scope: Monthly Academic Score Tracking, AI Learning Analysis & Career Guidance for Bitdoin Academy (Lao PDR)
Status: Draft for owner review — open decisions in Section 40
Companion files: Bitdoin_MVP_Backlog.xlsx (requirements matrix, WBS) · PDF rendering

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1. Document Control and Version History

Field	Value
Document	Product Requirements Document — Academic Tracking, AI Learning Analysis & Career Guidance (“the Module”)
Product	Bitdoin Academy (https://bitdoin.store/academy), part of the Bitdoin platform
Version	1.0 — Draft for owner review
Date	23 October 2026
Status	Draft. Requires owner decisions in Section 40 before development estimates are committed
Prepared for	Bitdoin platform owner and delivery team
Prepared by	Product team (PM, EdTech strategy, BA, Lao education, UX, architecture, AI/OCR, data, privacy, QA, commercial) — compiled with Claude
Inputs	Supplied master brief; supplied concept note “Monthly Score Tracking & Career Guidance: Idea Evaluation” (20 Oct 2026); public inspection of bitdoin.store on 23 Oct 2026; external research listed in Section 41
Companion files	*Bitdoin_MVP_Backlog.xlsx* (requirements matrix and work breakdown structure); PDF rendering of this document

1.1 Version history

Version	Date	Author	Change
0.1	20 Oct 2026	Owner	Concept note and idea evaluation
0.9	23 Oct 2026	Product team	Platform inspection, Lao context research, full PRD draft
1.0	23 Oct 2026	Product team	Draft issued for owner review

1.2 Approvals required

Role	Name	Decision	Date
Platform owner	*To be completed*	Approve scope, MVP and open decisions (Section 40)	
Engineering lead	*To be completed*	Confirm architecture option after discovery (PM-001)	
Lao legal counsel	*To be completed*	Opinion on minors’ consent and data transfer (LEG-001)	
Education reviewer	*To be completed*	Validate grading model and Lao terminology	

1.3 How to read this document — evidence labels

Every material statement in this PRD is labelled so readers can see how much weight it can bear.

Label	Meaning	Example
[CONFIRMED-SITE]	Verified by inspecting the public Bitdoin website on 23 Oct 2026	Academy Premium is LAK 39,000/month
[CONFIRMED-BRIEF]	Stated as a requirement in the master brief or concept note	Manual entry must remain an equal option to OCR
[EXTERNAL]	External fact from a cited source (Section 41), with the date it was checked	NUOL entrance exams are split into Group A (STEM) and Group B (non-STEM)
[PROPOSED]	A product or design recommendation made in this PRD	Embed academic Q&A inside the existing AI Coach
[ASSUMPTION]	Believed reasonable but not verified; needs confirmation	Most upper-secondary schools report monthly scores out of 10
Unverified — requires confirmation	Facts that could not be verified and must be checked before use	Any internal detail of Bitdoin’s backend

Requirement identifiers (e.g. **SCORE-006**) are stable and are used across the requirements matrix (Section 15), the screen specifications (Section 18), APIs (Section 27), tests (Section 35) and the backlog (Section 38 and the spreadsheet). Open decisions are referenced as **OD-xx** (Section 40). Sources are referenced as **[S-xx]** (Section 41).

2. Executive Summary

What we are building. A new area inside Bitdoin Academy — working name **“My Scores” (ຄະແນນຂອງຂ້ອຍ)** — where Lao secondary students record the monthly scores they already receive in their school score books, see clear trends, understand relatively stronger and weaker subjects, receive practical next steps linked to real Academy and Bookstore content, and explore university majors and careers using their interests as well as their results. Parents and guardians can be invited by the student to follow progress constructively.

Why it matters. The concept note identifies the core weakness of a pure exam-preparation product: students need exam preparation for only a few months of the year. Monthly score tracking gives a reason to return every month from Grade 10 (or earlier), builds a longitudinal record that no single diagnostic test can match, and creates the moments when families value guidance most — monthly results, end of term, and Grade 11–12 decisions about majors **[CONFIRMED-BRIEF]**. The NUOL entrance examination remains high stakes: 7,867 candidates sat the 2026–2027 examinations for 123 bachelor’s programmes **[EXTERNAL, S-08]**.

What we found on the existing platform. Bitdoin already presents “One account, two platforms” with an EN/ລາວ toggle; Academy already includes an **AI Coach**, **weekly challenges**, **habit tracker**, **XP & streaks**, and a **Career Explorer** with 14 career paths; Academy Premium costs **LAK 39,000/month or LAK 390,000/year** with **manual activation** and auto-renew off; Bookstore checkout supports **QR payment and bank transfer [CONFIRMED-SITE, S-01–S-06]**. The Module must therefore *extend* the AI Coach, Career Explorer, XP and Premium rather than build parallel versions. No public information exists about Bitdoin’s backend, database or authentication implementation; these are listed as discovery items (Section 5.4).

How we will build it responsibly.

- Manual entry is fast and equal to photo scanning; OCR output is never saved until the student reviews and confirms it (OCR-006).
- Every number shown is reproducible from documented deterministic formulas (Section 20); AI explains numbers but never invents them (AI-003, AI-009).
- Scores are stored in their original grading scale, and different schools or scales are never mixed without an explicit, labelled conversion (SCORE-008).
- The Module never ranks students against each other in the MVP (ANALYSIS-008) and never presents a questionnaire as a validated psychometric test (CAREER-001).
- Career suggestions show several options with transparent reasons, never a single answer or suitability percentage, and never use gender, ethnicity, province or income as inputs (CAREER-002, CAREER-004).
- Every university or admission fact carries a source, a verification date and a review-by date (UNI-002).
- Minors’ data are minimised (age band, not date of birth), guardian access requires two-sided confirmation, and consent rules are finalised by Lao legal review under Law on Electronic Data Protection No. 25/NA (2017) and any successor law **[EXTERNAL, S-15–S-17]**.

Recommended delivery sequence. Phase 0 research and validation (score-book samples, grading survey, legal review, OCR benchmark) → Phase 1 Core MVP (account integration, profile, manual monthly entry, configurable subjects/scales, history, trend dashboard, privacy controls, admin configuration) → Phase 2 OCR and analysis → Phase 3 parent, AI Coach academic mode and premium entitlements → Phase 4 career guidance with a verified pathways database → Phase 5 mock exams and ecosystem integration. Indicative Phase 1 effort is **≈ 105–180 ideal person-days** for the MVP work packages in Section 38 (excluding Phase 0, before a 20–30% contingency), which a team of 4–6 people could deliver in roughly 3–4 months after discovery, subject to the codebase review.

Monetisation. Basic tracking stays free (the hook and data engine). Premium value is the insight layer — detailed analysis, AI Coach academic mode, study plans, parent reports — delivered through the **existing Academy Premium** plan, plus a **one-time Career Report** and a **Family plan** as candidates for price testing (Section 30). Candidate prices are hypotheses anchored on the current LAK 39,000/month Premium, not market facts.

Decisions needed before development (full list in Section 40): grant codebase access and confirm the identity model (OD-01); confirm the MVP grade scope (OD-02); approve the AI and data-residency policy (OD-03); legal position on minors’ consent (OD-07); and how the Module fits inside the existing Premium plan (OD-11).

3. Product Vision

> **Help Lao students understand their academic performance, identify strengths and weaknesses, improve through personalised learning, and explore suitable educational and career opportunities based on evidence, interests and aspirations. [CONFIRMED-BRIEF]**

The Module answers five questions for every student:

#	Student question	Module capability	Key requirements
1	How am I performing academically?	Latest period summary, subject comparison	DASH-001, DASH-004
2	Am I improving or declining over time?	Trend charts with confidence-aware trend labels	DASH-002, ANALYSIS-003
3	Which subjects and specific topics should I focus on?	Strength/weakness classification + topic diagnostics	ANALYSIS-002, QUIZ-002
4	What practical steps can help me improve?	Rules-based recommendations, study planner, AI Coach	LEARN-001, PLAN-003, AI-001
5	Which majors or careers might suit me?	Interest questionnaire, explainable options, verified pathways	CAREER-002, UNI-003

Vision guardrail. Scores are one source of evidence about learning at one school at one time. The Module must never present scores as a measure of intelligence, talent, potential or future success (Section 12, principle P-03).

Three-year picture [PROPOSED]. A Lao student opens Bitdoin each month after receiving results, records scores in under 90 seconds (or by photo), sees what changed, takes a 10-minute topic quiz when a subject dips, follows a short weekly plan using Academy lessons, shares a monthly summary with a parent, and in Grade 11–12 uses a verified pathways database to compare majors at Lao universities — with mock exams in Grade 12 feeding the same learning profile.

4. Business Context and Opportunity

4.1 Context

- **Bitdoin today [CONFIRMED-SITE].** Bitdoin presents two areas under one account: a physical-book marketplace for Lao PDR with price comparison, multi-bookstore cart, local delivery and LAK/USD display (S-05, S-06), and Bitdoin Academy, a subscription learning and career-coaching platform with an AI study coach, learning paths and a career explorer (S-02).
- **Upper-secondary demand [EXTERNAL].** Grade 12 graduates, other than holders of provincial government scholarships, must pass the NUOL entrance examination to be admitted (S-09). NUOL ran its 2026–2027 entrance examinations on 8 August 2026 with 7,867 candidates for 123 bachelor's programmes (S-08).
- **Free general content exists [EXTERNAL].** The Ministry of Education and Sports' Khang Panya Lao platform, adapted from UNICEF's Learning Passport, provides digital teaching and learning content and works offline (S-25). Competing on generic free lessons is weak; the Module's differentiation is personal longitudinal data and guidance **[CONFIRMED-BRIEF]**.
- **Connectivity [EXTERNAL].** DataReportal reports 5.03 million internet users in Laos in October 2025 and 6.73 million cellular connections at end-2025, with 96.7% of connections classed as broadband (S-20). Mobile-first, low-data design is therefore appropriate.
- **System change [EXTERNAL].** Laos is transitioning from a 5+4+3 to a 6+3+3 general education structure (S-13); the Prime Minister confirmed introduction starting with Grade 6 in the 2028–2029 academic year (S-12). Upper secondary remains three years (Grades 10–12), but lower-secondary grade numbering will change during the Module's lifetime — the data model must handle it (Section 23.2).

4.2 Opportunity

Opportunity	Rationale	Evidence level
Monthly habit	Students already receive monthly scores; recording them is a low-effort, recurring reason to return	[CONFIRMED-BRIEF]; monthly practice per school is [ASSUMPTION] until Phase 0
Compounding data advantage	A year of monthly data plus topic quizzes personalises exam prep better than any one-off test	[CONFIRMED-BRIEF]
Parent willingness to pay	Parents of secondary students typically pay; a constructive monthly report justifies Premium	[CONFIRMED-BRIEF]; willingness to pay is [ASSUMPTION] to be tested (RES-003, RES-004)
Career guidance gap	Many Lao families have limited guidance on which major leads where	[CONFIRMED-BRIEF]; to be validated in interviews
Ecosystem pull-through	Recommendations can route to existing Academy lessons/challenges and Education-category books	[CONFIRMED-SITE] that these exist; conversion is [ASSUMPTION]

4.3 Business objectives

ID	Objective	Measure (Section 8)
BO-1	Increase recurring monthly engagement with Bitdoin Academy	Monthly score-entry completion rate; M3 retention
BO-2	Create a free-to-paid path for the Academy Premium plan	Free-to-paid conversion among active trackers
BO-3	Establish Bitdoin as a trusted, responsible source of academic and pathway guidance in Lao	Student-reported usefulness; complaint and correction rates
BO-4	Build a data foundation for future exam preparation	Students with ≥ 6 months of history entering Grade 12

5. Existing Bitdoin Platform Assessment

5.1 Method

The public website was inspected on **23 October 2026** by fetching https://bitdoin.store, /bookstore, /bookstore/faq, /academy, /academy/subscription and /academy/careers. The sign-in page (/auth) **could not be inspected** because the site's robots policy disallows automated access; no authentication details are therefore recorded. No source code, API documentation or database access was available.

5.2 Verified findings

Area	What was verified (23 Oct 2026)	Implication for the Module	Source
Brand & shell	Title "Bitdoin — Bookstore & Academy for Lao PDR"; tagline "One account, two platforms"; theme colour #1e3a5f; installable-web-app meta tags; "Switch platform" link	Module lives inside the Academy shell and reuses the theme and switcher	S-01, S-02
Language	EN/ລາວ toggle on all inspected pages; Bookstore FAQ states storefront supports Lao and English	Reuse the same preference (AUTH-003)	S-01, S-06
Account	One account across Bookstore and Academy; FAQ: Academy "is a separate subscription platform on the same account"	No new registration (AUTH-001)	S-01, S-06
Sign-in	/auth used for "Sign in", "Start free", "Create free account"	Return-URL redirect to /auth for unauthenticated users	S-02
Academy features	Learning paths (Reading summaries, Money lessons, AI skills, English, Productivity, Career advice, Scholarships, Business ideas); knowledge checks; AI Coach; weekly challenges; habit tracker; XP & streaks; Play & Learn; Career Explorer	Integrate, do not duplicate: AI Coach (AI-001), challenges/XP (PLAN-005), Career Explorer (CAREER-008)	S-02
AI Coach	Routes such as /academy/coach?career=software-engineer exist (deep-linking with context)	The Module can deep-link into AI Coach with academic context	S-04
Career Explorer	14 career paths with demand labels, majors, skills, certificate links, study-to-career steps, Laos/Asia/Global indicative salary ranges, employer and job-board links; several marked "Premium career"; page states job-market signals are "directional sample data in this MVP"	Personalised guidance must link to these records; provenance for salary/market data must be added before use in paid reports (CAREER-005)	S-04
Membership	Free: LAK 0; Premium Monthly LAK 39,000/month; Premium Yearly LAK 390,000/year ("2 months free"); "Manual activation supports Lao payment workflows"; auto-renew Off; payment history view	Module premium features should map onto this plan via entitlements (BILLING-002); renewal reminders needed (NOTIF-005)	S-03
Bookstore	Physical-book marketplace; search by title, author, publisher, category, language, ISBN; categories include **Education**; book pages at /bookstore/books/{UUID}; LAK and USD display	Recommendations reference real book IDs (LEARN-006)	S-05, S-06
Payments	Checkout supports QR payment and bank transfer; cash on delivery defined but not enabled	Reuse existing payment workflow for one-time purchases (BILLING-004)	S-06
Support	WhatsApp, Messenger, phone, email	Support playbook for the Module uses the same channels	S-05, S-06
Infrastructure hints	Email-obfuscation path under /cdn-cgi/ indicates a Cloudflare edge in front of the site	Edge caching and rate limiting may be available — **Unverified — requires confirmation** of plan/configuration	S-05

5.3 Capabilities to reuse vs build

Capability	Exists?	Decision
Account and sign-in	Yes [CONFIRMED-SITE]	Reuse (AUTH-001). Internal implementation **Unverified — requires confirmation**
Language preference	Yes	Reuse (AUTH-003)

Capability	Exists?	Decision
AI Coach	Yes	Extend with an “academic mode” scoped to the student’s data (AI-001) rather than a second chatbot
Weekly challenges, XP & streaks	Yes	Integrate study tasks with XP only if the owner approves, with non-punitive rules (PLAN-005, OD-09)
Career Explorer	Yes	Extend with personalisation and verified Lao programme data (CAREER-008)
Premium subscription	Yes (manual activation)	Reuse through an entitlement service (BILLING-001/002)
Payments (QR, bank transfer)	Yes (Bookstore checkout)	Reuse for one-time purchases; automated QR confirmation is a later enhancement (Section 30.7)
Score tracking, analysis, OCR, guardian links, pathways DB	No	Build in the Module

5.4 Unknowns that must be confirmed with the owner

Each item maps to an open decision (Section 40) and to discovery task PM-001.

#	Unknown	Why it matters	Ref
U-1	Technology stack, hosting, repository structure	Determines integration option (Section 24.2)	OD-01
U-2	Authentication and session mechanism; user ID format; whether roles exist	Identity adapter design; guardian/admin roles	OD-01
U-3	Database engine and whether module tables may share it	Schema placement; row-level security availability	OD-01
U-4	Where the EN/ลาว preference is stored	AUTH-003	OD-04
U-5	Academy content model (lessons, challenges, knowledge checks), IDs and completion events	Recommendations and completion tracking (LEARN-004, LEARN-007)	OD-05
U-6	Bookstore catalogue API, stock/offer model	Book recommendations (LEARN-006)	OD-06
U-7	Subscription records and manual-activation process	Entitlements (BILLING-002, BILLING-007)	OD-11
U-8	AI Coach architecture: model provider, prompt framework, logging, safety	Academic mode (AI-001..009)	OD-08
U-9	Career Explorer data model and the sources behind salary/demand figures	CAREER-005, CAREER-008	OD-10
U-10	Existing privacy policy, terms, minimum age, cookie consent	PRIV-001, PRIV-002	OD-07
U-11	Analytics, logging and monitoring tools in use	Section 32, 33	OD-01
U-12	Whether a native Android app exists or is planned	Camera, push, offline strategy	OD-02

6. Problem Statement

For students. Lao secondary students typically receive scores per subject at regular intervals, often recorded by hand in a school score book **[ASSUMPTION — format and frequency to be verified in Phase 0]**. The book shows numbers but not patterns: students cannot easily see whether they are improving, which subjects are consistently lower, or what to do about it. A single low mark can feel like failure while a slow decline goes unnoticed.

For parents. Parents often see results occasionally and without context, which can lead to anxiety or punitive reactions rather than specific support.

For career decisions. In Grades 11–12, students and families choose between entrance-exam groups, majors and institutions with limited structured, dated, local information. Generic global career lists do not reflect Lao programmes, entrance requirements or labour-market realities **[CONFIRMED-BRIEF]**.

For Bitdoin. Academy has strong learning tools but no recurring, personal academic record that tells it what each student needs, so recommendations and Premium value are generic.

Problem statement. *Lao secondary students and their families lack a simple, trustworthy way to turn the monthly scores they already receive into an accurate picture of progress, specific next steps and well-sourced guidance on study and career pathways — and Bitdoin Academy lacks the longitudinal data needed to personalise learning.*

7. Product Objectives

ID	Objective	Measurable target (initial, to be re-baselined after pilot)	Linked
PO-1	Make monthly recording effortless	Median time to submit a period ≤ 90 s for returning users; onboarding ≤ 120 s	SCORE-001, PROFILE-001
PO-2	Make the record trustworthy	100% of displayed scores show source and verification; 0 OCR values saved without confirmation	SCORE-012, OCR-006
PO-3	Make analysis explainable	100% of labels show the rule and confidence; golden-test accuracy 100%	DASH-005, ANALYSIS-001
PO-4	Turn insight into action	≥ 30% of students with a “needs attention” subject start a recommended action within 14 days	LEARN-001
PO-5	Involve families constructively	≥ 40% of linked guardians open ≥ 2 of 3 monthly reports	PARENT-006
PO-6	Provide responsible pathway guidance	100% of time-sensitive facts sourced and dated; fairness evaluation passes	UNI-002, CAREER-004
PO-7	Protect minors	Zero cross-student data exposure incidents; legal review closed before launch	AUTH-002, LEG-001
PO-8	Create sustainable revenue	Targets set after pilot using the method in Section 32.5	BILLING-*

8. Success Metrics

The full analytics framework, formulas and event taxonomy are in Section 32. The headline metrics are:

Tier	Metric	Definition (summary)	Initial target approach
North Star	**Monthly Recording Students (MRS)**	Students who submitted ≥ 1 reporting period in the calendar month	Baseline in pilot; target growth set per Section 32.5
Activation	First-entry rate	% of new profiles that submit a first period within 7 days	Pilot hypothesis ≥ 60%
Engagement	Period completion rate	Submitted periods ÷ expected periods for active enrolments	Pilot hypothesis ≥ 70%
Retention	M3 tracker retention	% of a monthly cohort recording in month 3	Pilot hypothesis ≥ 40%
Learning	Topic re-test change	Mean change in topic mastery between diagnostic and re-test (descriptive; not causal)	Report only; no causal claim
Value	Usefulness rating	% rating monthly summary “useful” or “very useful”	≥ 70%
Commercial	Free-to-paid conversion	Paying users ÷ active trackers (rolling 90 days)	Set after price tests
Trust	Correction rate after OCR	% OCR-imported fields edited by the student	Monitor; alert > 15%
Safety	Cross-student exposure incidents	Count of confirmed incidents	0

Learning-outcome metrics are **descriptive** only. The Module will not claim it caused score improvements without a controlled evaluation design (Section 32.4).

9. Stakeholder Analysis

Stakeholder	Interest	Influence	Key concerns	Engagement
Bitdoin owner	Growth, revenue, brand trust	Decides	Cost, time to market, legal risk	Decision workshop (PM-002); monthly steering
Students (Grades 10–12; later 6–9)	Understand progress, improve, choose a path	High (adoption)	Effort, privacy from peers/parents, being judged	Interviews, pilot, in-app feedback
Parents / guardians	Child's success; value for money	High (payment)	Accuracy, understanding, cost	Interviews, report usability tests
Teachers / tutors	Student support; potential content partners	Medium	Workload; misrepresentation of school grading	Grading survey; future teacher tools (Phase 5)
Schools / provincial education offices	Accurate representation; student welfare	Medium–high (credibility)	Data use, comparisons between schools	No school data sharing in MVP; future partnership discovery
Universities (NUOL, UHS, SU, SKU, CU and others)	Accurate programme information	Medium	Misstated requirements or cut-offs	Source-only publication; correction channel
Ministry of Education and Sports	Policy alignment; curriculum	High (context)	Alignment with curriculum and structural reform	Monitor policy; use official sources
Ministry of Technology and Communications	Data protection authority under Law 25/NA (S-16)	High (regulatory)	Compliance	Via legal counsel
AI/OCR vendors	Commercial	Medium	—	Vendor assessment (Section 29.5)
Content partners (tutors, authors)	Revenue share, reach	Medium	Licensing, attribution	Content licensing register
Support team	Resolve issues	Medium	Workload, sensitive cases	Playbook, break-glass process

10. Target Users and Personas

10.1 Segments and priority

Segment	Phase	Priority
Upper-secondary students, Grades 10–12	1	Primary
Parents / guardians of those students	3	Primary payer
Lower-secondary students (Grades 6–9; renumbered as 6+3+3 is introduced)	Evaluate after Phase 1, gated by OD-07	Secondary
Platform administrators (sub-roles)	1	Required
Teachers / tutors	5 (discovery earlier)	Future
Schools, university advisors, education partners	Post-Phase 5	Exploratory

10.2 Persona 1 — Noy, Grade 11 student, Vientiane Capital (primary)

Attribute	Description
Background	16, public upper-secondary school; science-leaning; uses Facebook and YouTube daily; shares an Android phone plan with prepaid data
Needs	Quick way to see whether she is improving in Mathematics and Physics; ideas for what to study; information on Engineering vs IT at NUOL
Goals	Pass the Group A entrance exam; keep Mathematics ≥ 8
Motivations	Curiosity, family expectations, wanting to feel in control
Challenges	Limited data budget; worries about parents seeing every low score; little time
Digital literacy	High for social apps; moderate for forms and settings
Device & connectivity	Mid-range Android (3–4 GB RAM), 4G in the city, Wi-Fi at home
Typical journey	Receives October results → opens Bitdoin → enters 8 scores in 75 s → sees Physics dipped for the second month → takes a Physics diagnostic → completes 2 Academy lessons → asks AI Coach “which topic should I practise first?”
Permissions	Full control of own data; decides what her guardian sees
Free vs Premium	Free: profile, entry, history, dashboard, basic labels. Premium (via parent): detailed analysis, AI Coach academic mode, study plan, career report

10.3 Persona 2 — Kham, Grade 10 student, rural district, Savannakhet Province

Attribute	Description
Background	15, district school; helps family on weekends; first in family likely to attend university
Needs	Very simple Lao interface; works on weak signal; no cost
Goals	Stay in school; improve Lao Language and English
Motivations	Teacher encouragement; wanting a stable job
Challenges	Intermittent 3G/4G; older phone with limited storage; low confidence with English UI
Digital literacy	Basic–moderate
Device & connectivity	Low-end Android (2 GB RAM), prepaid data, frequent offline periods
Typical journey	Enters scores while offline → draft saves → syncs at school Wi-Fi → sees “Not enough data yet” message with encouragement → follows free recommendation to an English lesson
Permissions	Full control of own data

Attribute	Description
Free vs Premium	Entirely free path must be valuable; OCR allowance not needed

10.4 Persona 3 — Mrs Vongsa, parent of two (Grades 9 and 12), Luang Prabang

Attribute	Description
Background	44, small business owner; pays for tutoring; uses BCEL One and WhatsApp
Needs	Clear monthly picture without arguing with children; guidance on majors for her Grade 12 son
Goals	Both children continue to university
Motivations	Investment in children's future
Challenges	Limited time; not familiar with grading nuances; may over-react to a single low mark
Digital literacy	Moderate
Device & connectivity	Android, reliable 4G
Typical journey	Receives invite code from son → signs in with her Bitdoin account → son confirms link → reads monthly report → follows "how to help" suggestions → buys a career report
Permissions	Sees only scopes each child enables; cannot see AI chats; cannot edit scores
Free vs Premium	Free: link and basic trend view. Premium/Family: monthly reports, email delivery, career report

10.5 Persona 4 — Teacher Phonesavanh, upper-secondary Mathematics teacher and private tutor (future)

Attribute	Description
Background	35, teaches Grades 10–12, runs weekend tutoring
Needs	See which topics her tutees struggle with; assign practice
Goals	Better results for students; possible income from content
Challenges	Workload; must not see students outside her group
Digital literacy	Moderate–high
Typical journey (Phase 5)	Student shares a "tutor view" for Mathematics only → teacher sees topic results → recommends an exercise set
Permissions	Only student-granted, subject-scoped, time-limited access; no school-wide visibility
Free vs Premium	Future teacher plan or content revenue share (OD-17)

10.6 Persona 5 — Admin team (Super Administrator, Education Content Manager, Academic Reviewer, Career Content Reviewer, Support Agent, Financial Administrator)

Attribute	Description
Background	Small Bitdoin team; some members part-time
Needs	Configure subjects, scales, schools and calendars; review OCR quality and AI outputs; maintain sourced career data; resolve tickets; reconcile manual payments
Challenges	Limited staff; Lao/English bilingual content; need clear approval workflows
Digital literacy	High
Device	Desktop browser primarily
Permissions	Per role, least privilege, MFA (Section 28)

10.7 Optional future stakeholders — evaluation

Stakeholder	Possible value	Risks	Recommendation
Schools	Verified score import; aggregated insights	Data-sharing agreements, comparisons between schools, consent	Explore after Phase 3 via a pilot MoU; verified imports set `verification=school_verified` only through signed agreements
University advisors	Correct programme data; outreach	Commercial pressure on students	Data partnership only; no student data sharing
Education NGOs / partners	Content, reach to rural students	Mission alignment	Content partnerships; aggregated, anonymised impact reports only

11. Scope and Non-Scope

11.1 In scope (across all phases)

Modules A–N as specified in Section 14: academic profile; monthly score entry; optional photo/OCR; dashboard; strengths/weaknesses analysis; diagnostic quizzes; recommendations; AI Coach academic mode; study planner; guardian access and reports; major and career guidance; universities and pathways database; mock-exam integration hooks; notifications; administration; entitlements; privacy and consent.

11.2 MVP scope (Phase 1)

See Section 36. In short: account integration, profile and enrolments, manual monthly entry with drafts and corrections, configurable subjects/scales/calendars, history, descriptive dashboard, privacy controls (notice, consent record, export, delete), essential admin configuration, in-app reminders.

11.3 Explicitly out of scope

Item	Reason
Ranking students against peers, schools or provinces	Fairness and cross-school incomparability (ANALYSIS-008)
Predicting entrance-exam outcomes or admission probability	Insufficient validated data; high harm if wrong
Psychometric personality or aptitude testing	Not validated; interest questionnaire only (CAREER-001)
Diagnosing learning difficulties or mental-health conditions	Clinical domain; out of remit (ANALYSIS-004)
Selling or sharing student data with third parties	Trust and legal risk; never
Teacher-wide or school-wide dashboards in Phases 1–4	Consent and governance not yet designed
Native Android app in MVP	Responsive web/PWA first; revisit after OD-02
“Leaked” or official exam questions	Integrity; original or licensed content only (QUIZ-005)
Automatic payment collection via a new gateway in MVP	Existing manual-activation workflow is reused first

12. Product Principles

ID	Principle	What it means in practice
P-01	**Student first, student in control**	The student owns the record, chooses what guardians see, can export and delete
P-02	**Evidence before advice**	Every insight states the data it rests on and its confidence; “not enough data yet” is a valid answer
P-03	**Scores are not the person**	No language about intelligence, talent or potential; growth-oriented copy
P-04	**Manual is first-class**	Typing is as fast and visible as scanning; AI is a convenience, not a gate
P-05	**Never silently trust AI**	OCR needs confirmation; generated text is checked against the snapshot
P-06	**Original scale, explicit conversion**	Store raw values and scales; convert only when defensible and labelled
P-07	**Options, not verdicts**	Several pathways, transparent reasons, explore any career
P-08	**Sourced and dated**	University and career facts show source and verification date; expired facts are flagged
P-09	**Minimum data, maximum care for minors**	Age band not birth date; consent recorded; least-privilege access
P-10	**One Bitdoin**	Reuse account, language, AI Coach, Career Explorer, Premium and design system
P-11	**Built for real networks**	Small payloads, drafts that survive disconnection, compressed uploads
P-12	**Kind engagement**	No shaming, no punitive streaks, quiet hours, frequency caps

13. Feature and Epic Overview

Module	Epic (ID prefix)	Summary	Phase	Free / Premium (proposed)
A	Academic Profile (PROFILE)	Profile, enrolments by year, school/grade changes	1	Free
B	Monthly Score Entry (SCORE)	Fast manual entry, drafts, validation, corrections, history, provenance	1	Free
C	Photo & OCR (OCR)	Optional scan with quality checks, extraction, mandatory review	2	Free allowance + Premium allowance
D	Dashboard (DASH)	Summary, trends, subject charts, comparisons	1 (basic) / 2 (labels)	Free (basic)
E	Strengths & Weaknesses (ANALYSIS, QUIZ)	Rule-based classification, trend tests, diagnostics	2	Basic labels free; detailed analysis Premium
F	Recommendations (LEARN)	Rules-based next steps linked to Academy/Bookstore	2	Limited free; full Premium
G	AI Academic Assistant (AI)	Academic mode of the existing AI Coach	3	Limited free quota; Premium
H	Study Planner (PLAN)	Goals, weekly tasks, flexible rescheduling	3	Premium (basic goal free)
I	Parent / Guardian (PARENT)	Invite, scopes, family dashboard, monthly reports	3	Link free; reports Premium/Family
J	Major & Career Guidance (CAREER)	Questionnaire, explainable options, report	4	Exploration free; detailed report paid
K	Pathways Database (UNI)	Universities, programmes, requirements, scholarships with provenance	4	Free to browse
L	Mock Exams (EXAM)	Blueprint-aligned mocks as separate evidence stream	5	Premium
M	Notifications (NOTIF)	In-app first, push/email opt-in, caps, quiet hours	1→3	Free
N	Administration (ADMIN)	Configuration, content workflows, quality monitoring, audit	1→4	Internal
—	Identity & Access (AUTH)	Account reuse, authorisation, language	1	—
—	Entitlements (BILLING)	Plan mapping, quotas, one-time and family purchases	3	—
—	Privacy & Safety (PRIV)	Notice, consent, export, deletion, AI data controls	1→2	—

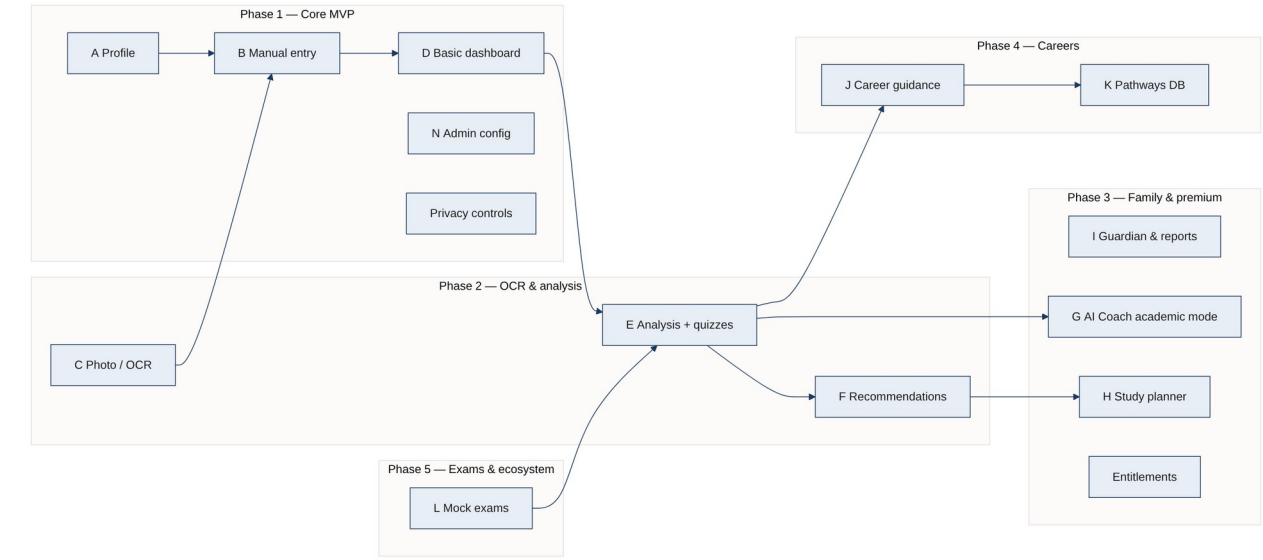


Figure 1: 13. Feature and Epic Overview

14. Detailed Functional Requirements

This section specifies each module's behaviour, data, rules and backend responsibilities. User stories with acceptance criteria are in Section 15; calculations in Sections 19–20; OCR/AI in Section 21; career methodology in Section 22.

14.1 Module A — Student Onboarding and Academic Profile

Purpose. Create a minimal, accurate academic context (grade, year, school, subjects) linked to the existing Bitdoin account, and keep it correct across promotions, repetitions and transfers.

14.1.1 Data fields

Field	Type / format	Required?	Validation	Rationale / data-minimisation note
Display name	Text, 1–40 chars, Lao/Latin	Required	No URLs, no phone numbers (regex), profanity filter	Defaults to Bitdoin account first name if available; can differ
Age band	Enum: under 13 / 13–15 / 16–17 / 18+	Required	One selection	Used only for consent rules (PRIV-002). **Date of birth is not collected**
Grade level	Enum configured per education structure (e.g. Upper Secondary 1–3 = Grades 10–12)	Required	Must exist in the active structure	Stored with `education_structure_id` to survive 6+3+3 renumbering
Academic year	Enum from configured years (e.g. 2026–2027)	Required	Current or previous year only at onboarding	Allows catching up on the year in progress
Province	Enum of 18 Lao provinces incl. Vientiane Capital	Required	Must be in list	Needed for school search and regional calendars; never used in career ranking (CAREER-004)
District	Enum filtered by province	Optional	Must belong to province	Improves school search only
School	Search-select from catalogue OR free text	Required (either)	Free text 3–80 chars	Free text creates admin review task (PROFILE-002)
Subjects	Multi-select pre-filled from grade template	Required (≥ 1)	≤ 25	Must mirror the student's score book
Preferred language	Inherits Bitdoin EN/ลาว	Inherited	—	No duplicate setting (AUTH-003)
Academic goals	Up to 3 chips (e.g. "Pass entrance exam", "Improve English") + optional 200-char text	Optional	Length	Not visible to guardians unless shared (PARENT-004)
Subjects of interest	Multi-select from catalogue	Optional	≤ 5	Feeds recommendations and career questionnaire
Future interests	Free text or career chips from Career Explorer	Optional	≤ 5 chips	Feeds CAREER only with student consent
Not collected	Full legal name, date of birth, student ID number, home address, phone, ethnicity, religion, family income, photos of the student	—	—	Unnecessary for purpose (P-09)

14.1.2 Business rules

1. **BR-A1 — One profile per account.** A Bitdoin user has at most one student profile. A guardian account may also hold a student profile (e.g. an adult learner), but guardian and student roles are separate records.
2. **BR-A2 — Enrolment per academic year.** The profile holds one *enrolment* per academic year (or part-year for transfers). Scores attach to an enrolment, never directly to the profile, so grade, school and grading scale are preserved historically.
3. **BR-A3 — Promotion workflow.** When the configured academic year rolls over (ADMIN-004), the next session shows the "New school year" sheet (PROFILE-004). Options: *Promoted* (grade + 1, same school

by default), *Repeating* (same grade, new enrolment flagged `repeat=true`), *Changed school* (pick school and grade), *Left school / finished* (profile becomes read-only for scores; history and career features remain).

4. **BR-A4 — Closed enrolments are read-only.** Scores in a closed enrolment can be corrected only through an explicit "Correct a past year" action that requires a reason and creates revisions (SCORE-007).
5. **BR-A5 — Mid-year transfer.** Ends the current enrolment at the last full period, creates a new one starting at the effective period, inherits subjects where codes match, and requires the new school's grading scale (PROFILE-005).
6. **BR-A6 — Subject changes.** Adding or dropping a subject sets start/end periods on the subject enrolment; history is never deleted (SCORE-013).
7. **BR-A7 — Education structure versioning.** Each grade value references an `education_structure` (e.g. `LA-5-4-3` or `LA-6-3-3`). Upper secondary labels remain Grades 10–12 in both; lower-secondary mappings are configured by admins when the reform reaches those grades (Section 23.2).
8. **BR-A8 — Unlisted schools.** Free-text schools are visible only to the student and admins; admins either map them to a catalogue school or create a new one (ADMIN-003).

14.1.3 Backend requirements

- **BE-A1.** `POST /v1/students/me/profile` is idempotent per user; a second call returns the existing profile (200) rather than creating another.
- **BE-A2.** Promotion/transfer operations execute in one transaction: close enrolment → create enrolment → copy subject enrolments → write audit event.
- **BE-A3.** School search endpoint supports Lao Unicode and romanised queries with diacritic- and tone-mark-insensitive matching (Section 23.5) and returns ≤ 20 results in ≤ 300 ms p95.

14.2 Module B — Monthly Academic Score Entry

Purpose. The foundation: accurate, fast entry of each reporting period's scores, with drafts, corrections and provenance.

14.2.1 Concepts

Concept	Definition
Reporting period	A configured unit for which a school reports scores: usually a month; also `term_official`, `semester_official`, `annual_official`, `exam` types (ADMIN-004)
Score entry	One submission for one enrolment and one reporting period; contains one score line per subject enrolment
Score line	Value or status for one subject in one period, with raw value, scale and provenance
Grading scale	Configured min, max, step, display format and optional pass mark; versioned (ADMIN-002)
Source	`manual`, `ocr_confirmed`, `school_import` (future)
Verification	`self_reported` (default), `guardian_acknowledged` (future, guardian taps "I've seen the score book"), `school_verified` (only via signed school agreements)

14.2.2 Manual-entry flow

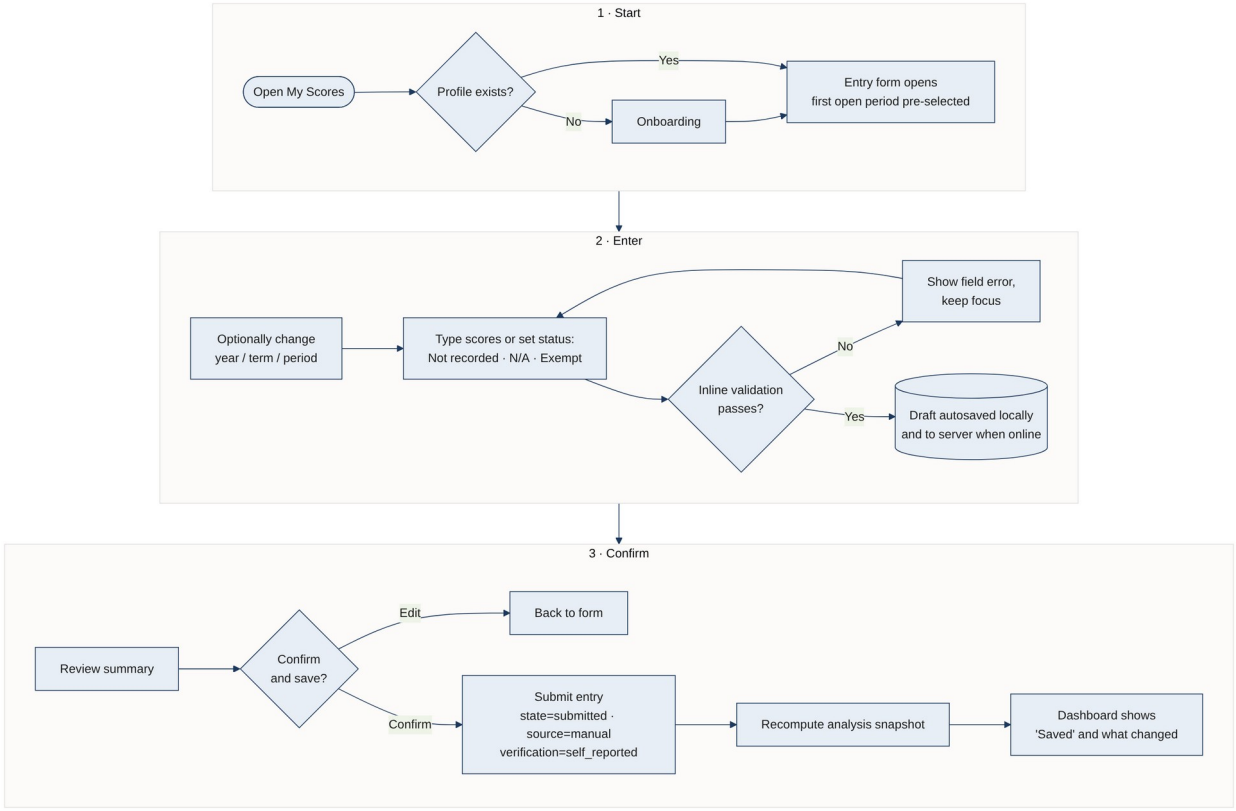


Figure 2: 14.2.2 Manual-entry flow

14.2.3 Business rules

ID	Rule
BR-B1 Valid range	`scale.min ≤ value ≤ scale.max` and `(value − scale.min) / scale.step` is an integer within 1e-9 tolerance
BR-B2 Decimals	Input accepts `.` or `,` as decimal separator and Lao digits ๐–๙; stored as `NUMERIC(6,2)`; displayed with the scale's `decimals` setting (default 1 for 0–10 scale; trailing ".0" hidden for integers)
BR-B3 Missing vs zero	Empty is never zero. Status `missing` (not received yet), `not_applicable` (subject not assessed this period), `exempt` (excused/absent with exemption). A real 0 must be typed explicitly and triggers a confirm prompt "Is 0 correct?"
BR-B4 Duplicate prevention	One current score line per (enrolment, period, subject enrolment). Opening a period with existing data opens edit mode (SCORE-009)
BR-B5 Corrections	Editing a submitted value writes a revision (old, new, actor, time, reason, source) and recomputes analysis; revisions are immutable
BR-B6 Drafts	Drafts autosave every change (debounced 800 ms) to IndexedDB and, when online, to the server; drafts expire after 60 days with a reminder at day 45; drafts are excluded from all analytics and guardian views
BR-B7 Submission completeness	A period can be submitted with some subjects `missing`; at least one subject must have a value or a non-missing status
BR-B8 Future periods	Periods whose start date is after "today" (Asia/Vientiane) cannot be submitted
BR-B9 Term and annual summaries	Computed means are labelled "Calculated by Bitdoin"; official term/annual results entered by the student are stored as separate period types and never overwritten (SCORE-011)
BR-B10 Scale binding	Every score line stores `grading_scale_version_id`. Changing a school's scale creates a new version from a stated period onward; earlier scores keep their scale
BR-B11 Cross-year transfer	Scores are never moved between enrolments; promotion creates a new enrolment (BR-A3)
BR-B12 Verification	MVP creates only `self_reported`. UI badges appear for any other value; analytics treat all verification

ID	Rule
	levels equally but reports show the mix
BR-B13 Data source label	Every API response and export includes `source` and `verification` (SCORE-012)
BR-B14 Edit window for guardian report	Monthly guardian reports are generated 3 days after submission to allow corrections (PARENT-006); later edits mark the report "Updated since generation"

14.2.4 Example (illustrative values, not official records)

Academic year 2026–2027 · Grade 11 · Reporting period: October 2026 · Scale: 0–10, step 0.5

Subject (EN / ลาว)	Score	Status	Source	Verification
Mathematics / ຄະນິດສາດ	7	—	manual	self_reported
Physics / ຟິຊິກສາດ	6	—	manual	self_reported
Chemistry / ເຄມີສາດ	8	—	manual	self_reported
Biology / ຊີວະສາດ	9	—	manual	self_reported
Lao Language / ພາສາລາວ	8	—	manual	self_reported
English / ພາສາອັງກິດ	7	—	manual	self_reported
History / ປະຫວັດສາດ	—	Not recorded yet	manual	self_reported

Period mean = (7+6+8+9+8+7)/6 = **7.5** (History excluded; "6 of 7 subjects recorded").

14.2.5 Backend requirements

- **BE-B1.** Score submission endpoint accepts an `Idempotency-Key` header; replays within 24 h return the original response.
- **BE-B2.** Submission is atomic: all score lines and the entry state change commit together or not at all.
- **BE-B3.** Each submitted or edited entry publishes an internal `score_entry.changed` event consumed by the analysis worker (Section 24.4).
- **BE-B4.** Validation rules are enforced server-side identically to the client; the server is authoritative.
- **BE-B5.** Optimistic concurrency: updates must include the entry `version`; mismatches return 409 `VERSION_CONFLICT` with the current state so the client can show a merge prompt.

14.3 Module C — Photo Upload and AI OCR

Purpose. Optional convenience to pre-fill the score form from a score-book photo. It must be at least as safe as manual entry and never save without confirmation.

14.3.1 Functional sequence

Step	Behaviour	Req
1	"Scan score book" and "Type scores" shown side by side with equal visual weight	OCR-001
2	Camera (rear, with document guide overlay) or file picker; multiple pages allowed (max 4 per job)	OCR-001
3	Preview with rotate, crop, retake	OCR-002
4	On-device quality checks (brightness, blur, skew, resolution); tips but not blocking	OCR-003
5	Client-side compression (≤ 2000 px long edge, JPEG q=0.8, EXIF stripped) and resumable upload	OCR-002, PRIV-006
6	Asynchronous extraction job; progress UI with ability to leave and return	OCR-004
7	Subject matching against catalogue + school aliases	OCR-005
8	Review screen with each extracted value next to its image crop; low-confidence fields highlighted	OCR-006

Step	Behaviour	Req
9	Multi-period/multi-person detection: choose period(s); block if other names detected until acknowledged	OCR-007
10	Student edits or accepts every highlighted field	OCR-006
11	"Confirm and save" → normal submission path with `source=ocr_confirmed`	OCR-006
12	Validation identical to manual entry; then analysis recompute	SCORE-003

14.3.2 Business rules

- **BR-C1.** OCR results are suggestions in a draft; they never become score lines without an explicit confirm action by the student.
- **BR-C2.** Confidence threshold for "highlighted" defaults to 0.85 per field and is configurable per model version from benchmark data (Section 21.5).
- **BR-C3.** Fields with confidence < 0.5 are shown empty with "Couldn't read — please type".
- **BR-C4.** Crossed-out values: if the extractor reports `struck_through=true` alongside a replacement value, the replacement is suggested and highlighted; if no replacement is detected the field is empty.
- **BR-C5.** Lao numerals are normalised to Arabic numerals before validation; the original text is retained for audit.
- **BR-C6.** Duplicate images (perceptual hash distance ≤ 6 within 90 days) prompt reuse of earlier results (OCR-008).
- **BR-C7.** Images are retained for 30 days after confirmation by default, or deleted immediately on request (OCR-011).
- **BR-C8.** Extraction text from images is treated as untrusted data and is never inserted into prompts as instructions (Section 29.6, prompt injection).

Detailed architecture, schema and QA are in Section 21.

14.4 Module D — Academic Performance Dashboard

Purpose. Answer "How am I doing?" and "Am I improving?" in seconds, honestly.

14.4.1 Dashboard content (top to bottom, mobile)

1. **Summary card** — latest period, period mean, "n of m subjects recorded", change vs previous comparable period (DASH-001).
2. **Next action card** — one primary action (enter missing period / take diagnostic / continue plan) (LEARN-001).
3. **Trend chart** — monthly mean with rolling mean (DASH-002).
4. **Subjects** — list with latest score, change arrow and label chip (DASH-004, DASH-005).
5. **Stronger subjects / Needing attention** — with reasons and confidence (Phase 2).
6. **Learning activity** — completed lessons, quizzes, plan tasks this month (Phase 2–3).
7. **History link** and **Share with guardian** (Phase 3).

14.4.2 Visualisation specifications

Chart	Inputs	Calculation	Labelling	Empty state	Mobile	Accessibility	Interpretation rules
V1 Monthly mean trend	Submitted monthly periods of	Period mean (F1, Section 20.2); 3-period rolling mean (F5) when ≥ 4 points	Y axis in scale units (e.g. 0–10); X axis	< 2 periods: "Add one more month to see your trend"	Full-width, 220 px tall, horizontal scroll disabled; tap point → tooltip	"View as table"; aria-label summarises first/last values and trend label	Gaps not interpolated; trend wording only

Chart	Inputs	Calculation	Labelling	Empty state	Mobile	Accessibility	Interpretation rules
	current + previous enrolments with compatible scales		month abbreviations in UI language; annotation markers for school/scale change				from ANALYSIS-003
V2 Per-subject line	One subject's scores	Raw values; subject mean reference (F2)	Same axis as scale; mean line labelled "Your average"	< 2 scores: show value list only	Opens as full screen	Table toggle; points have value labels	Show trend label and confidence; never "you are bad at"
V3 Subject comparison bar	Latest submitted period	Raw values sorted desc	Value labels on bars	Missing subjects listed below as text	Horizontal bars for long Lao names	Bars ≥ 3:1 contrast; values in text	No colour meaning "fail" unless a configured pass mark exists, and then text label "Below pass mark"
V4 Change indicators	Current vs previous comparable period	Δ = mean_t − mean_(t−1) (F3)	▲/▼/■ icon + "+0.5" text	First period: hidden	Inline chips	Icon + text, not colour	
V5 Goal progress (Phase 3)	Goal baseline, target, latest values	Progress = (latest − baseline)/(target − baseline), clamped 0–100%	"Goal: 8.0 by March"	No goal: CTA	Progress bar	Percentage as text	Framed as goal, not guarantee
V6 Year-over-year (Phase 3)	Same subject, two enrolments, compatible scale family	Values aligned by month	Legend by year	Incompatible scales: option disabled with reason	Toggle	Table toggle	Only when both scales normalisable (BR-B10)

14.4.3 Wireframe descriptions

Mobile (360 × 800): sticky Academy header with "My Scores" title and language toggle → summary card (period chip, large mean "7.5 / 10", "6 of 7 subjects", change chip "▲ +0.3 from September") → full-width primary button (e.g. "Enter November scores") → trend chart card → subject list cards (name, latest score, change chip)

→ bottom navigation reused from Academy if present (Home · Learn · My Scores · Coach · Profile — **to be aligned with the existing Academy navigation, OD-04**).

Desktop (≥ 1024 px): two columns — left 2/3: summary, trend chart, subject comparison; right 1/3: next action, stronger/attention cards, learning activity, guardian sharing status.

Text wireframes are in Section 18.3.

14.5 Module E — Strengths and Weaknesses Analysis

Purpose. Turn records into careful, explainable observations, and use diagnostics to locate topic-level needs.

14.5.1 Outputs

Output	Definition	Requirement
Relatively stronger subject	Consistently above the student's own overall mean (Section 20.4)	ANALYSIS-002
Relatively lower subject	Consistently below the student's own overall mean	ANALYSIS-002
Improving / declining / stable	Trend by Theil–Sen slope with Mann–Kendall test; minimum 4 points	ANALYSIS-003
Volatile	High variability regardless of slope	ANALYSIS-003
Insufficient evidence	Fewer than threshold data points	ANALYSIS-002/003
Topic mastery band	From diagnostic quizzes (Secure / Developing / Needs practice)	QUIZ-002
Conflicting evidence notice	When school scores and diagnostics disagree	ANALYSIS-007

The word "weakness" is not used in student-facing copy; use "subjects needing attention" (ວິຊາທີ່ຄວນເອົາໃຈໃສ່) and "relatively stronger" (ວິຊາທີ່ເຮັດໄດ້ດີກວ່າ).

14.5.2 Diagnostic quiz design

Aspect	Specification
Purpose	Identify topic-level learning needs within a subject; low-stakes, not graded
Blueprint	Per subject and grade: 3–5 topics, 3–4 items per topic, 10–15 items total, ≤ 15 minutes
Item types	Multiple choice (4 options) and numeric short answer for Mathematics/Physics; no free-text essay in Phase 2
Language	Lao default; English items only for English subject
Authoring	Human-authored or licensed; reviewed by an Academic Reviewer who is not the author (ADMIN-007)
Scoring	Proportion correct per topic; band thresholds in QUIZ-002; topics with < 3 items not banded
Adaptivity	None in Phase 2 (fixed form); parallel forms for re-test (QUIZ-004); adaptive testing considered in Phase 5
Integrity	Practice tool only; no proctoring; results private to student unless shared
Example	Grade 11 Mathematics diagnostic: Algebra (4), Functions (3), Trigonometry (3), Geometry (3). Result: Algebra 1/4 → Needs practice; Geometry 3/3 → Secure → recommendations target Algebra

14.5.3 Explanation generation rules

1. Observations come only from the analysis snapshot ("What your scores show").
2. Suggestions are phrased as options ("Things that often help"), never as causes.
3. Forbidden content: claims about intelligence, ability, effort, motivation, personality, health, disability or family circumstances; comparisons to other students.
4. Confidence wording mapped from Section 20.6 (e.g. Low → "It's early to say, but…").
5. Every explanation includes how much data it is based on ("based on 5 months").

Methodology, formulas and pseudocode are in Section 20.

14.6 Module F — Personalised Learning Recommendations

14.6.1 Candidate content types

Type	Source system	Identifier	Availability check
Diagnostic quiz	Module (QUIZ)	quiz_id	status=approved
Topic practice set	Module (QUIZ) or Academy knowledge checks	practice_set_id / academy_item_id	status=published
Academy lesson	Existing Academy	academy_lesson_id (format **Unverified**)	published in Academy
Weekly challenge	Existing Academy	academy_challenge_id	active this week
Study routine	Module templates	routine_template_id	approved
Bookstore title	Existing Bookstore	book UUID (as in /bookstore/books/{uuid}) [CONFIRMED-SITE]	≥ 1 in-stock offer
Mock exam	Module (EXAM, Phase 5)	mock_id	blueprint verified for current year

14.6.2 MVP rules (deterministic)

Rule ID	Trigger	Candidates	Max shown	Priority score
R1 Missing data	Open period > 7 days after period end	"Enter scores"	1	100
R2 Diagnose	Subject = relatively_lower or declining (≥ Medium confidence) and diagnostic exists and none taken in last 60 days	Diagnostic quiz for subject	1	90
R3 Practise topic	Topic band = Needs practice	Practice set + Academy lesson tagged to topic, difficulty ≤ student level	2	80 − 5×rank
R4 Consolidate	Topic band = Developing	Practice set at same difficulty	1	60
R5 Re-test	Practice completed for all Needs-practice topics or 14 days elapsed	Parallel diagnostic	1	70
R6 Routine	Student has no active plan and ≥ 1 attention subject	7-day routine template	1	50
R7 Stretch	Subject relatively_stronger and improving	Challenge or advanced lesson	1	40
R8 Book (optional)	Subject attention + tagged in-stock Education title	Bookstore title	1	20

Ranking. Sort by priority score; adjust −30 if dismissed "not now" in last 14 days (excluded entirely during that window), −100 if "not helpful" (suppressed 180 days), +10 if matches a stated subject of interest; tie-break by content quality score (reviewer rating 1–5) then by recency of publication. Show top 3 (LEARN-001), with at most one Bookstore item and never a Bookstore item as the first card.

Repetition control. Same item ≤ 3 impressions per 30 days; impressions logged.

Content gaps. If a rule fires with no eligible content, show "We don't have practice for this topic yet" and log `content_gap` with subject/topic/grade for the content team (LEARN-003).

14.6.3 Content metadata (required for matching)

Field	Type	Notes
external_ref	(system, id)	Academy / Bookstore / Module
subject_codes	array	From subject catalogue
grade_band	range	e.g. 10–12
topic_ids	array (≥ 1)	Topic taxonomy per subject
learning_objectives	array of text (Lao/EN)	Shown on hover/expand
difficulty	1–3	1 foundation, 2 core, 3 stretch
prerequisites	topic_ids	Used to avoid stretch before foundation

Field	Type	Notes
language	lo / en	
est_minutes	int	Used by planner
quality_score	1–5	Reviewer-set
status	draft/published/retired	Only published is recommendable

14.6.4 AI enhancements (Phase 3+)

AI may (a) rewrite the reason text in simpler Lao, (b) re-rank among rule-eligible candidates using preference signals, and (c) summarise progress. AI **may not** introduce items outside the eligible candidate set (enforced by passing IDs, not free text, and validating outputs).

14.6.5 Progress measurement

Recommendation funnel: impression → open → start → complete → re-test change. Reported descriptively (Section 32).

14.7 Module G — AI Academic Assistant (academic mode of the existing AI Coach)

Decision [PROPOSED]. Rather than a second chatbot, add an **academic mode** to the existing Academy AI Coach: when opened from My Scores (e.g. `/academy/coach?context=scores`), the Coach can call server-side tools that read the student's own analysis snapshot. This depends on the AI Coach architecture (OD-08); if the existing Coach cannot support tool calls, the Module provides an embedded assistant that reuses the Coach's visual design and safety layer.

14.7.1 Capabilities and limits

Can	Cannot
Explain trends and labels in simple Lao with the specific numbers	Invent or estimate scores not in the record
Explain why a subject was flagged (rule and data)	Speculate on intelligence, motivation, health, family
Suggest activities from the eligible recommendation set	Recommend content outside the catalogue
Explain quiz mistakes using the approved explanation as grounding	Reveal answer keys for diagnostics in progress
Help draft a manageable weekly plan (writes via PLAN API after confirmation)	Make guarantees about exam outcomes or admission
Answer questions about the student's own records	Access any other student's data or guardian data

14.7.2 Technical specification summary

Topic	Specification
Retrieval	Tools: `get_snapshot()`, `get_subject_history(subject_code)`, `get_quiz_results(subject_code)`, `get_recommendations()`, `get_plan()`. All bound server-side to the session's `student_profile_id`; the model cannot pass a student ID (AI-002)
Context construction	System prompt (role, safety, style) + student context block (grade, scale, latest snapshot summary ≤ 1,500 tokens) + tool results; conversation window last 10 turns; older turns summarised
Prompt templates	Versioned: `coach_academic.v1`, `academic_feedback.v1`, `career_explanation.v1` (Section 21.8)
Model selection	Criteria: Lao fluency score on evaluation set ≥ 4.0/5 by native raters; factual consistency ≥ 99%; latency p95 ≤ 8 s; cost per message within budget (Section 30.6); data terms (no training on inputs, retention controls)
Cost controls	Free quota (e.g. 10 academic messages/month, OD-11), Premium higher fair-use cap; max output 400 tokens; cached snapshot explanations (AI-006)
Lao QA	Native-speaker rubric (accuracy, simplicity, tone, terminology) on 200-prompt set before each release
Hallucination mitigation	Numeric claims validated against tool outputs post-generation; if mismatch, regenerate once, else fall back to template (AI-009)

Topic	Specification
Safety	Input/output classifiers for self-harm, sexual content, harassment, personal data; minors' policy; no persona/companionship; crisis message approved by owner (AI-004, OD-15)
Auditability	Store prompt_version, model_id, tool calls, token counts; message text retained 90 days for safety review, then deleted unless flagged (OD-16)
Feedback	Thumbs up/down, report (AI-007)
Fallback	Circuit breaker; deterministic summaries (AI-008)

14.8 Module H — Study Planner and Learning Goals

Aspect	Specification
Goal model	subject, baseline (latest mean of last 3 scores or latest score if fewer), target, target period, status
Feasibility check	Required gain per month = (target − baseline) / months remaining. ≤ 0.25 pts/month (0–10 scale) "Steady"; 0.26–0.5 "Stretch"; > 0.5 "Ambitious — consider a milestone". Never blocks; never promises (PLAN-001)
Commitment	Days of week + minutes per day (10/15/20/30/45)
Scheduling rules	Fill sessions from top recommendations; ≤ 30 min per session; ≥ 1 rest day; alternate subjects when > 1 priority subject; schedule diagnostics before practice
Task tracking	States: scheduled, done, skipped, moved; completion from Academy events where possible (LEARN-007) or self-report
Missed tasks	Auto-move once; then ask skip/keep; no penalties (PLAN-004)
Progress feedback	Weekly review card: tasks done, minutes, topics practised, next week preview; neutral tone
Reminders	Optional; respects NOTIF-003 caps and quiet hours
Motivation	Positive acknowledgement of effort, optional XP via existing system, milestone badges; **no punitive streaks**, no leaderboards (PLAN-005)
Integration	Existing weekly challenges appear as optional tasks when tagged to the subject

Example: Goal "Mathematics 6.0 → 8.0 by March" in November = 4 months → 0.5/month → "Stretch". Plan: Mon/Wed/Sat, 20 minutes; week 1 = Algebra diagnostic (Mon), Algebra practice set A (Wed), Academy lesson "Linear equations" (Sat).

14.9 Module I — Parent or Guardian Dashboard

14.9.1 Linking and authorisation flow

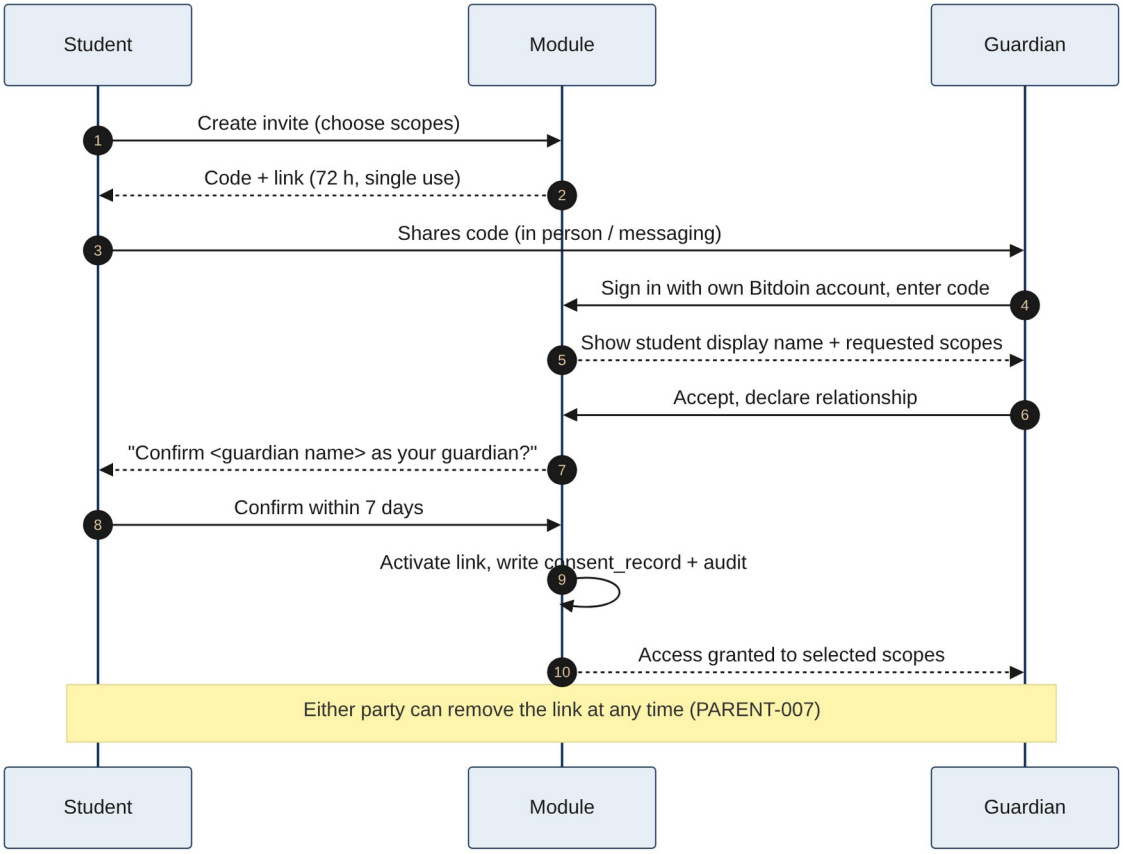


Figure 3: 14.9.1 Linking and authorisation flow

Guardian-initiated consent for younger students (PARENT-003). If legal review (OD-07) requires guardian consent for an age band, the student's onboarding produces a consent link; the guardian signs in, reviews the notice, consents, and the link activates with default scopes. Until then no score data are stored server-side.

14.9.2 Visibility scopes

Scope	Default	Notes
Scores, trends and labels	On	Includes source/verification badges
Study plan progress	On	Task counts and minutes, not free text
Recommendations	On	Titles only
Goals free text	Off	Student may enable
Career questionnaire answers and shortlist	Off	Student may enable
AI Coach conversations	Never available	Not offered
Score-book images	Never available	Not offered

14.9.3 Parent report (monthly)

Section	Content	Tone rule
1 Reporting period	Month, school year, generation date, data-as-of date	—
2 Subjects recorded	n of m subjects; missing listed	No blame for missing
3 Score changes	Per subject latest vs previous with arrows; mean change	"About the same" for small changes
4 Relatively stronger subjects	With reasons	Celebrate effort and progress

Section	Content	Tone rule
5 Areas needing support	With reasons and confidence	"Needs attention", never "weak"
6 Completed learning activities	Lessons, quizzes, plan tasks	Effort recognition
7 Suggested next steps for families	2–3 supportive actions (e.g. "Ask which algebra topic felt hardest"), optional resources	No punishment, no comparisons with siblings or classmates

Footer: data are self-reported unless marked; one month's change is not a trend; contact support.

14.9.4 Rules for parent–student interaction

- Multiple guardians per student (max 4) and multiple children per guardian (max 6).
- Guardians cannot edit scores, cannot see drafts, and see changes only after the 3-day correction window.
- When the student reaches the configured age of majority, links pause unless the student re-confirms (PARENT-009).
- **Incorrect link**: either party removes it; support can suspend on report (PARENT-008).
- **Dispute** (e.g. two people claim guardianship, or a student reports coercion): support suspends the contested link, contacts both via in-app messages, may request the student's confirmation in a private flow; no identity documents are collected in-app; escalation to the owner. Safety concerns follow the safeguarding procedure (Section 29.8).
- **Account recovery**: guardian links are attached to Bitdoin accounts; if a guardian loses access, the existing Bitdoin account-recovery process applies (**Unverified — requires confirmation**); the student may issue a new invite to the guardian's new account.

14.10 Module J — Major and Career Guidance

Detailed methodology, questionnaire items and ranking are in Section 22. Functional summary:

Element	Free exploration	Premium career report
Interest questionnaire	Yes	Yes
3–6 personalised options with reasons	Yes (summary)	Yes (full)
Career detail pages (existing Career Explorer, enriched)	Yes	Yes
Programmes at Lao institutions with entrance group and sources	Yes	Yes
Detailed pathway per option: subjects to strengthen, entrance group, programmes, preparation steps, timeline to exam	Summary	Full
"What changed" history	Yes	Yes
Downloadable/printable report (PDF) with sources and dates	—	Yes
Discussion guide for parents/teachers	Short	Full

Each suggested option shows: name (Lao/English); what the work involves; why it appears for this student; important skills; typical educational pathway; related programmes in Laos (verified only); potential entrance requirements with year and source; practical next steps; sources and verification dates; and, where reliable Lao salary/employment data are not available, the statement "Reliable Lao data not available" rather than a number (CAREER-005).

14.11 Module K — University and Educational Pathways Database

14.11.1 Entities

Institutions, campuses, faculties, programmes (degree level, duration, language of instruction, mode), entrance exam groups, admission requirements (per academic year), cut-off records (per year, per programme, with

source), tuition (when verified), application windows/deadlines, scholarships, programme–career links. Schema in Section 26.

14.11.2 Provenance on every time-sensitive field

Attribute	Required	Description
source_name	Yes	e.g. “National University of Laos — Entrance Examination Guide 2026–2027”
source_url	Yes (or document reference)	Official page or uploaded document
source_type	Yes	official_site, official_document, official_news (e.g. KPL), institution_confirmation, secondary
verified_on	Yes	Date a reviewer checked it
applicable_academic_year	Yes for requirements, cut-offs, deadlines, tuition	
publication_status	Yes	draft / in_review / published / archived
review_by	Yes	Default verified_on + 12 months; deadlines: the deadline date
verified_by	Yes	Reviewer user reference

`secondary` sources may be used only for context text, never for requirements, cut-offs or deadlines.

14.11.3 Administrative workflow

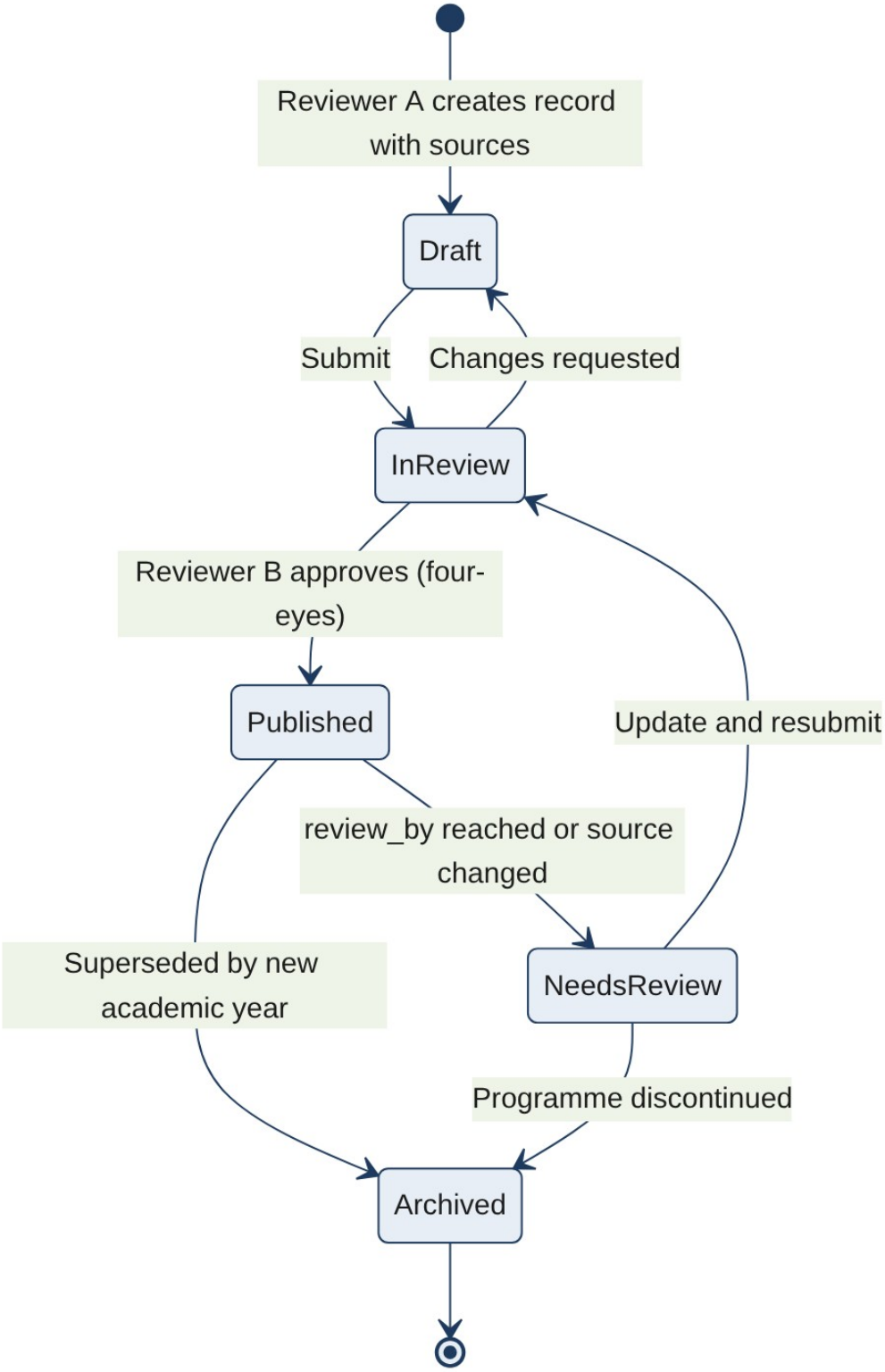


Figure 4: 14.11.3 Administrative workflow

Seed scope [PROPOSED]. Start with programmes at the National University of Laos (verified 2026–2027: 123 bachelor's programmes selected through Group A/STEM and Group B/non-STEM entrance examinations [EXTERNAL, S-08]), then University of Health Sciences, Souphanouvong University, Savannakhet University and Champasak University. **Programme lists, admission rules, cut-offs and tuition for these institutions were not verified in this PRD and must be researched from official sources (Unverified — requires confirmation).**

14.12 Module L — Mock Examination Integration (future-ready)

Aspect	Specification
Evidence streams	Three separate streams in the learning profile: (1) school scores (self-reported), (2) diagnostic quizzes (topic mastery), (3) mock exams (timed, blueprint-aligned). Each has its own panel and units. No composite index without an approved methodology (EXAM-002, OD-14)
Blueprint	Mock structure must match the officially published format for the current year. Verified historical facts: NUOL Group A subjects included Mathematics, Physics, Lao Language and Literature, Chemistry, Biology and English; Group B included Mathematics, Lao Language and Literature, Geography, History and English (2024 announcement, S-09). A 2023 NUOL page described the papers and timings (S-10). These must be re-verified each year before a "Mock exam" label is used (EXAM-001)
Scoring	Section and subject raw scores; topic tags per item; no predicted admission probability
Feedback	Topic results feed recommendation rules R3–R5 with source=mock_exam
Content	Original or licensed; integrity statement; never "leaked" content (QUIZ-005)
Data model hooks	Phase 1 schema includes `assessment_source` enum and `evidence_stream` to avoid later migration

14.13 Module M — Notifications and Engagement

14.13.1 Notification catalogue

Code	Trigger	Audience	Default channel	Cap category
N-01 Monthly entry reminder	Period end + 7 days, no submission	Student	In-app	Reminder
N-02 Report ready	Monthly summary/report generated	Student, guardian	In-app (+ email opt-in for guardian)	Report
N-03 New recommendation	New R2/R3 recommendation	Student	In-app	Learning
N-04 Plan reminder	Planned task day (if enabled)	Student	In-app / push opt-in	Learning
N-05 Guardian summary	Monthly report	Guardian	In-app / email opt-in	Report
N-06 Deadline	Saved scholarship/application deadline at D-14, D-3	Student	In-app / push opt-in	Deadline
N-07 Subscription expiry	D-7, D-1	Payer	In-app / email opt-in	Transactional
N-08 Link events	Invite accepted, link removed	Both	In-app	Transactional

14.13.2 Channel comparison

Channel	Cost	Reach in Laos	Consent/complexity	Child-safety fit	MVP?
In-app	None	All active users	Low	High	**Yes**
Web push	Low	Android Chrome supports web push; iOS only for installed web apps (**to verify on target devices**)	Browser permission	Medium (lock-screen content)	Phase 1 optional, opt-in
Email	Low	Moderate among students; higher among parents **[ASSUMPTION]**	Double opt-in	Medium	Phase 3 (guardians)
SMS	Per-message cost	High	Separate consent; cost approval	Medium	Later (OD-12)
Messaging apps (WhatsApp/Messenger)	Varies; platform policies	Widely used; Bitdoin already uses them for support [CONFIRMED-SITE]	Business API onboarding and policies	Lower (personal accounts)	Later; transactional only

14.13.3 Rules

- Quiet hours 21:00–07:00 Asia/Vientiane (user-adjustable within 20:00–08:00).
- Caps: ≤ 1 push/day, ≤ 3 push/week per student; email ≤ 1/week non-transactional.

- One-tap unsubscribe per category; global “pause all for 1 month”.
- **Child-safe copy checklist**: no scores in push/lock-screen text; no comparison with others; no fear or guilt; no urgency language (“Last chance!”) except real deadlines; no marketing to students under the legal threshold (OD-07); every message offers a way to turn it off.

Example copy (EN / ลาว — native review required): “Your October results are ready to add when you are.” / “ເມື່ອພ້ອມແລ້ວ ສາມາດເພີ່ມຄະແນນເດືອນຕຸລາຂອງເຈົ້າໄດ້.”

14.14 Module N — Administration and Content Management

14.14.1 Admin capabilities

Area	Capabilities	Phase
Subjects & aliases	Create/edit subjects (Lao/EN), grade applicability, aliases (incl. school-specific), merge	1
Grading scales	Create versioned scales; assign to curriculum template or school; mark normalisable	1
Schools	CRUD, merge duplicates, map free-text entries, assign calendars/scales	1
Academic years & periods	Templates and overrides; roll-over date; education structure mapping	1
Student account oversight	Lookup by Bitdoin user reference; status; break-glass access	1
Audit logs	Search, export (Super Admin)	1
OCR quality	Metrics, sample review, threshold tuning per model version	2
Quiz & content metadata	Topics, items, review workflow, tagging Academy/Bookstore items	2
AI review	Sampled conversations, reports, prompt version management	3
Entitlements & payments	Manual activation, refunds, family seats, reason codes	3
Career taxonomy & universities	Careers, majors, programmes, sources, review queue	4
Reports	Operational and product dashboards (aggregate only)	1→
Customer support	Tickets linked to account; guardian link disputes	1→3

14.14.2 Approval workflows

Object	Workflow	Approver rule
Grading scale (new version)	Draft → Approved	Education Content Manager + Academic Reviewer
Quiz item / explanation	Draft → In review → Approved → Retired	Academic Reviewer ≠ author
AI-generated explanation for content	Generated → Reviewed → Approved	Academic Reviewer
Programme / requirement / cut-off	Draft → In review → Published	Career Content Reviewer ≠ creator (four-eyes)
Admin role grant	Request → Granted	Super Administrator only
Break-glass score access	Reason → Time-limited grant	Self-service for Support Agent with mandatory reason; reviewed weekly by Super Admin
Manual entitlement grant	Payment evidence → Activated	Financial Administrator; > 3 manual grants/day per user flagged

The full permission matrix is in Section 28.

15. User Stories and Acceptance Criteria

The complete requirements matrix is below and is also provided as an editable sheet in *Bitdoin_MVP_Backlog.xlsx* (sheet "Requirements"). Priorities: **P0** = must-have for the phase in which it is listed (P0 items in Phase 1 define the MVP); **P1** = should-have in its phase; **P2** = could-have; **P3** = later. Dependencies reference other requirement IDs or open decisions (OD-xx). Acceptance criteria are written as testable Given/When/Then statements; "Notes" carry backend and validation details where the story itself is user-facing.

Requirements count: 140 requirements across 18 epics; **38 are P0 in Phase 1 and define the MVP** (Section 36).

Epic	ID prefix	Total	P0	P1	P2	P3	Phases
Identity & Access	AUTH-xxx	4	3	1	0	0	1, 3
Academic Profile	PROFILE-xxx	8	5	2	1	0	1, 2
Monthly Score Entry	SCORE-xxx	14	11	3	0	0	1
Score-book Photo & OCR	OCR-xxx	12	0	11	1	0	2
Performance Dashboard	DASH-xxx	8	6	1	1	0	1, 2, 3
Strengths & Weaknesses Analysis	ANALYSIS-xxx	8	2	6	0	0	1, 2
Diagnostic Quizzes	QUIZ-xxx	5	0	4	1	0	2
Learning Recommendations	LEARN-xxx	7	0	6	1	0	2, 5
AI Academic Assistant	AI-xxx	10	3	7	0	0	2, 3
Study Planner & Goals	PLAN-xxx	6	0	5	1	0	3
Parent / Guardian	PARENT-xxx	9	0	9	0	0	3
Major & Career Guidance	CAREER-xxx	10	2	7	1	0	4
Universities & Pathways Data	UNI-xxx	5	1	3	1	0	4
Mock Examination Integration	EXAM-xxx	3	0	0	3	0	5
Notifications	NOTIF-xxx	6	1	4	0	1	1, 3, 5
Administration	ADMIN-xxx	10	4	6	0	0	1, 2, 3, 4
Subscriptions & Entitlements	BILLING-xxx	7	1	4	2	0	3, 4
Privacy, Consent & Safety	PRIV-xxx	8	7	1	0	0	1, 2
Total		**140**	46	80	13	1	

15.1 Identity & Access

ID	User story	Priority / phase	Dependencies	Acceptance criteria
AUTH-001	As a signed-in Bitdoin user, I can open Academic Tracking without creating a new account, so that I keep one Bitdoin identity.	P0 / Ph1	Bitdoin identity system (OD-01)	Given a user has a valid Bitdoin session, When they open /academy/scores, Then the module resolves their existing user ID and shows onboarding or dashboard without a second sign-up. Given no session, When they open the route, Then they are redirected to /auth with a return URL and land back on /academy/scores after sign-in. **Notes:** Module stores only an opaque bitdoin_user_id reference; never copies passwords or auth secrets.
AUTH-002	As the platform, every Academic Tracking API call is authorised against the record owner, so that no user can read another student's data by changing an ID.	P0 / Ph1	AUTH-001	Given user A is authenticated, When A requests GET /students/{B_profile_id}/scores for a profile A neither owns nor has an active guardian grant for, Then the API returns 404 (not 403) and writes an access_denied audit event. Automated IDOR tests cover every route listed in Section 27. **Notes:** Object-level authorisation enforced server-side in one policy layer; UUIDv7 or random IDs, never sequential.
AUTH-003	As a user, my interface language (Lao/English) follows the existing Bitdoin	P0 / Ph1	Existing language toggle (OD-04)	Given the user switched Bitdoin to ລາວ, When they open any Academic Tracking screen, Then all module UI strings render in Lao; switching to EN on any screen persists across Bookstore,

ID	User story	Priority / phase	Dependencies	Acceptance criteria
	EN/ลาว setting, so that I do not set it twice.			Academy and the module. **Notes:** Read/write the same preference key the platform already uses; confirm storage location with owner.
AUTH-004	As the platform, I can assign module roles (student, guardian, admin sub-roles) without changing the core Bitdoin user table, so that existing commerce flows are unaffected.	P1 / Ph3	AUTH-001, ADMIN-001	Given a Bitdoin user, When a guardian link is approved, Then a guardian relationship row is created in the module schema and the user's core account record is unchanged. Admin roles are granted only by a Super Administrator and every grant is audit-logged.

15.2 Academic Profile

ID	User story	Priority / phase	Dependencies	Acceptance criteria
PROFILE-001	As a student, I can create an academic profile in under 2 minutes with only the required fields, so that I can start recording scores quickly.	P0 / Ph1	AUTH-001, ADMIN-003	Given a new user, When they complete onboarding with display name, grade level, academic year, province and school (or 'school not listed'), Then a student_profile and an enrolment for the current academic year are created. Median completion time in usability testing ≤ 120 s on a mid-range Android device. **Notes:** Required: display_name, grade_level, academic_year, province, school_id OR school_free_text. Optional: district, goals, interests. Date of birth NOT collected; age band only (see PRIV-002).
PROFILE-002	As a student, I can choose my school from a searchable list or enter it as free text when it is not listed, so that I am never blocked.	P0 / Ph1	ADMIN-003	Given the school is not in the catalogue, When the student types a name and selects 'Use this name', Then the enrolment stores school_free_text, status=unverified_school, and an admin review task is created. Search tolerates Lao and romanised spellings (see Section 23).
PROFILE-003	As a student, my current subjects are pre-filled from my grade and school configuration and I can add, remove or rename them, so that the score form matches my score book.	P0 / Ph1	ADMIN-002, PROFILE-001	Given grade 11 and a school using the default curriculum template, When onboarding reaches 'Your subjects', Then the default subject list for grade 11 is pre-selected; the student can deselect any, add from catalogue, or add a custom subject (max 25 subjects per enrolment).
PROFILE-004	As a student, when a new academic year starts I am asked to confirm promotion, repetition or school change, so that history stays accurate.	P0 / Ph1	PROFILE-001	Given the academic year configured by admin has rolled over, When the student next opens the module, Then a 'New school year' sheet offers: Promoted to grade N+1 / Repeating grade N / Changed school / Left school. The previous enrolment is closed (read-only) and a new enrolment is created; prior scores are never edited or deleted by this action.
PROFILE-005	As a student, I can record a school transfer mid-year, so that scores before and after the move are attributed to the right school and grading scale.	P1 / Ph1	PROFILE-004, SCORE-008	Given a student transfers in January, When they choose 'Changed school' with an effective month, Then the current enrolment ends at the prior month, a new enrolment starts at the effective month with its own grading scale, and charts mark the boundary with a 'school changed' annotation.
PROFILE-006	As a student, I can optionally add goals and subjects of interest, so that recommendations reflect what I care about.	P1 / Ph1	PROFILE-001	Given the optional step, When the student skips it, Then onboarding completes and the step re-appears as a dashboard card at most once per 30 days. Free-text goals are limited to 200 characters and are excluded from any parent view unless the student shares them.
PROFILE-007	As a student, I can view and edit my profile and see exactly what data the module holds about me, so that I stay in control.	P0 / Ph1	PROFILE-001, PRIV-004	Given the profile screen, When the student opens 'My data', Then they see profile fields, number of score records, uploads stored, guardian links and consents, each with edit or delete actions as permitted.
PROFILE-008	As a student in a lower-secondary grade, I can use the module with an age-	P2 / Ph2	PRIV-002, OD-07	Given a student selects a lower-secondary grade, When the age band is below the threshold set by legal review (OD-07), Then guardian consent is required before score data is saved

ID	User story	Priority / phase	Dependencies	Acceptance criteria
	appropriate experience, so that younger students are supported safely.			and AI chat is disabled until consent is recorded. **Notes:** Scope decision: MVP targets Grades 10–12; Grades 6–9 gated by legal review.

15.3 Monthly Score Entry

ID	User story	Priority / phase	Dependencies	Acceptance criteria
SCORE-001	As a student, I can enter all subject scores for one reporting period on a single mobile screen, so that monthly entry takes under 90 seconds.	P0 / Ph1	PROFILE-003, ADMIN-002	Given 8 enrolled subjects on a 0–10 scale, When the student selects a period and types scores using the numeric keypad, Then focus advances to the next subject on 'Next', and a returning user can submit 8 scores in ≤ 90 s (median, usability test). **Notes:** Numeric keyboard (inputmode=decimal); accepts Lao digits ๐–๙ and normalises to Arabic numerals.
SCORE-002	As a student, I select academic year, term/semester and month (reporting period) from lists generated by my school's calendar, so that I cannot record an impossible period.	P0 / Ph1	ADMIN-004	Given the enrolment's academic calendar defines periods Sep–May, When the student opens the period selector, Then only configured periods for that enrolment appear, the first period without a submitted record is pre-selected, and future periods beyond the current month are disabled.
SCORE-003	As a student, the form validates each score against the grading scale, so that typing mistakes are caught immediately.	P0 / Ph1	ADMIN-002	Given a 0–10 scale with step 0.5, When the student enters 11, 7.3 or 'abc', Then the field shows an inline error ('Score must be between 0 and 10 in steps of 0.5') and Submit is disabled until fixed; 7.5 and 10 are accepted; leading/trailing spaces are trimmed.
SCORE-004	As a student, I can mark a subject as 'Not recorded yet', 'Not applicable this period' or 'Absent/exempt', so that missing data is explicit rather than zero.	P0 / Ph1	SCORE-001	Given a subject has no score, When the student chooses a status, Then the record stores value=NULL with status in {missing, not_applicable, exempt}; NULL values are excluded from averages and never treated as 0; the dashboard shows the count of missing subjects.
SCORE-005	As a student, my partially completed entry is saved as a draft automatically, so that I do not lose work when the connection drops.	P0 / Ph1	SCORE-001	Given the student has typed 3 of 8 scores, When the app goes offline or the tab closes, Then the draft persists locally (IndexedDB) and server-side on the next successful sync; reopening the period restores all typed values. Drafts never contribute to analytics.
SCORE-006	As a student, I review a summary and explicitly submit the period, so that only confirmed data is analysed.	P0 / Ph1	SCORE-003	Given all fields are valid or have a status, When the student taps 'Review' then 'Confirm and save', Then a score_entry is stored with state=submitted, source=manual, verification=self_reported, and the dashboard refreshes within 3 s (p95).
SCORE-007	As a student, I can correct a submitted score and the previous value is retained in history, so that corrections are transparent.	P0 / Ph1	SCORE-006	Given a submitted October Mathematics score of 6, When the student edits it to 7 with an optional reason, Then the current value becomes 7, a score_revision row stores old=6/new=7/actor/timestamp/reason, analytics are recalculated, and 'Edited' appears next to the score with a viewable history.
SCORE-008	As the platform, I store each score in its original grading scale together with a scale reference, so that different school scales are never silently mixed.	P0 / Ph1	ADMIN-002	Given a score of 38/50 under scale S2, When it is saved, Then raw_value=38 and grading_scale_id=S2 are stored; normalised_value=0.76 is derived only when S2.normalisable=true; charts never plot two different scales on one axis without a 'converted to %' label.
SCORE-009	As a student, I am prevented from creating two records for the same subject and period, so that duplicates do not distort averages.	P0 / Ph1	SCORE-006	Given October Mathematics already exists for the enrolment, When the student starts a new entry for October, Then the existing values open in edit mode; the database enforces UNIQUE(enrolment_id, reporting_period_id, subject_enrolment_id) on current records; an API request that would duplicate returns 409 SCORE_DUPLICATE.
SCORE-010	As a student, I can browse my complete score history by year, period and subject, so that I can	P0 / Ph1	SCORE-006	Given 2 academic years of data, When the student opens History, Then entries are grouped by academic year > term > period, each showing subject scores, source

ID	User story	Priority / phase	Dependencies	Acceptance criteria
	review past results.			(manual/photo), verification state and edit marker; closed enrolments are read-only.
SCORE-011	As a student, I can see term and annual summaries computed from my monthly scores or entered directly when my school issues an official term result, so that both views are available.	P1 / Ph1	SCORE-006, ANALYSIS-001	Given monthly scores exist for Sep–Dec, When the term ends, Then a computed term mean is displayed and labelled 'Calculated by Bitdoin'; if the student enters an official term score, it is stored as period_type=term_official and shown separately; computed and official values are never overwritten by each other.
SCORE-012	As a user, every score shows its data source and verification status, so that self-reported and verified data are always distinguishable.	P0 / Ph1	SCORE-006	Given any score in any view (dashboard, report, parent view, API), Then it carries source ∈ {manual, ocr_confirmed, school_import} and verification ∈ {self_reported, guardian_acknowledged, school_verified}; MVP only creates self_reported; UI shows a badge for anything other than self_reported.
SCORE-013	As a student, I can add a subject mid-year or drop one, so that the form adapts when my timetable changes.	P1 / Ph1	PROFILE-003	Given a subject is dropped from November, When the student removes it, Then subject_enrolment.end_period is set; earlier scores remain; the subject no longer appears in new periods; trend charts end at the last recorded period.
SCORE-014	As a student, I receive a gentle reminder when a reporting period has passed without an entry, so that my record stays complete.	P1 / Ph1	NOTIF-001	Given the period end date + 7 days has passed with no submitted entry, When reminders are enabled, Then one in-app reminder is shown; a second reminder (in-app or push if consented) is allowed after a further 7 days; no more than 2 reminders per period.

15.4 Score-book Photo & OCR

ID	User story	Priority / phase	Dependencies	Acceptance criteria
OCR-001	As a student, I can photograph or upload a score-book page as an optional alternative to typing, so that entry is faster when the photo is good.	P1 / Ph2	SCORE-006, PRIV-005	Given the score-entry screen, When the student taps 'Scan score book', Then the camera or file picker opens (JPEG/PNG/HEIC/WebP, ≤ 15 MB before compression); typing remains visible as an equal option ('Type scores') on the same screen.
OCR-002	As a student, I can crop, rotate and retake before uploading, so that the image contains only the relevant page.	P1 / Ph2	OCR-001	Given a captured photo, When the preview opens, Then the student can rotate in 90° steps, crop with corner handles, and retake; the client downscales to max 2000 px on the long edge and JPEG quality 0.8 (target ≤ 600 KB) before upload.
OCR-003	As a student, I get immediate feedback when a photo is too dark, blurry or skewed, so that I retake it before waiting for processing.	P1 / Ph2	OCR-002	Given a preview, When on-device checks detect mean luminance < 60/255, Laplacian variance below the calibrated blur threshold, or page skew > 15°, Then a specific tip is shown ('Too dark – move to light') and the student may still continue; thresholds are calibrated in Phase 0 on the sample set.
OCR-004	As the platform, I extract subjects, scores, period and metadata into a structured, schema-validated result with per-field confidence, so that uncertain values can be flagged.	P1 / Ph2	OCR-002, AI-010	Given an uploaded image, When processing completes, Then the job stores JSON matching ocr_extraction.v1 (Section 21.4) with each field's value, confidence ∈ [0,1], bounding box and raw text; outputs failing schema validation are retried once and then marked failed.
OCR-005	As the platform, I match extracted subject names to the subject catalogue using aliases, so that 'คะมิดสาด', 'Math' and 'เลก' map to Mathematics.	P1 / Ph2	ADMIN-002, OCR-004	Given extracted text 'คะมิด', When matching runs, Then exact alias match → confidence 1.0; normalised fuzzy match ≥ 0.85 → suggested; below 0.85 → 'Choose subject' required; unmatched subjects are never auto-created.
OCR-006	As a student, I review all extracted values with uncertain fields highlighted and must confirm before anything is saved, so that OCR never silently creates grades.	P1 / Ph2	OCR-004, OCR-005	Given an extraction with 2 fields below the confidence threshold, When the review screen opens, Then those fields are highlighted, the image crop for each field is viewable, 'Confirm and save' is disabled until each highlighted field is touched (accepted or edited), and saved scores carry source=ocr_confirmed, verification=self_reported.

ID	User story	Priority / phase	Dependencies	Acceptance criteria
OCR-007	As a student, I am told when the photo seems to contain several months or another person's records, so that I choose what to import.	P1 / Ph2	OCR-004	Given the extraction finds 3 month columns, When review opens, Then the student must choose which period(s) to import (default: none pre-selected except the current open period); if more than one student name is detected, a warning asks the student to crop to their own record and blocks import until acknowledged.
OCR-008	As the platform, I detect duplicate uploads, so that the same page does not create repeated processing cost or duplicate scores.	P1 / Ph2	OCR-004	Given an image with perceptual hash distance ≤ 6 from an upload by the same student in the last 90 days, When uploaded, Then the user is asked 'This looks like a page you uploaded on . Use those results?'; confirmed duplicates do not consume OCR quota.
OCR-009	As a student, if OCR fails I am moved to manual entry with nothing lost, so that a bad photo never blocks me.	P1 / Ph2	OCR-004, SCORE-001	Given processing fails or exceeds 60 s, When the student is on the processing screen, Then a message offers 'Try another photo' and 'Type scores instead'; choosing the latter opens the manual form for the same period, pre-filled with any fields that were extracted with confidence ≥ 0.9.
OCR-010	As the platform, I keep an auditable log of each OCR job, model version, outputs and user corrections, so that accuracy can be measured and improved.	P1 / Ph2	OCR-004	Given a confirmed import, Then ocr_job, ocr_extracted_field and ocr_confirmation rows record model_id, prompt_version, per-field original vs confirmed value and latency; correction rate per field is available to the OCR quality dashboard (ADMIN-008).
OCR-011	As a student, I control how long my score-book images are kept, so that sensitive images are not stored longer than needed.	P1 / Ph2	PRIV-005	Given default retention, When an import is confirmed, Then the original image is deleted after 30 days unless the student opted into 'Keep my photos'; the student can delete any image immediately; deletion removes object storage and derived thumbnails within 24 h.
OCR-012	As the platform, I enforce a per-student monthly OCR allowance by plan, so that processing cost stays predictable.	P2 / Ph2	BILLING-003	Given a Free user with an allowance of N uploads/month (OD-11), When the allowance is used, Then the scan button shows remaining count and, once exhausted, explains the limit and keeps manual entry fully available; failed jobs do not consume allowance.

15.5 Performance Dashboard

ID	User story	Priority / phase	Dependencies	Acceptance criteria
DASH-001	As a student, I see my latest period, overall mean, number of subjects recorded and change from the previous period at the top of the dashboard, so that I understand my status in 5 seconds.	P0 / Ph1	SCORE-006, ANALYSIS-001	Given at least one submitted period, When the dashboard loads, Then the summary card shows latest period name, mean (1 decimal) in the scale's units, 'n of m subjects recorded', and change vs previous comparable period with an arrow and text label (not colour alone).
DASH-002	As a student, I see a monthly average trend chart, so that I can see direction over time.	P0 / Ph1	ANALYSIS-001	Given ≥ 2 submitted periods in the same scale family, When the chart renders, Then it plots one point per period, a 3-period rolling mean line when ≥ 4 periods exist, gaps for periods with no data (not interpolated), and school/scale changes as vertical annotations. With < 2 periods an empty state explains what is needed.
DASH-003	As a student, I can open a per-subject line chart, so that I can follow one subject over time.	P0 / Ph1	ANALYSIS-001	Given Mathematics has 5 scores, When the student opens Mathematics, Then a line chart shows each score, the subject mean as a reference line, and a text summary with the trend label from ANALYSIS-003 including its confidence wording.
DASH-004	As a student, I see a bar chart comparing subjects for the latest period, so that I can compare relative results.	P0 / Ph1	SCORE-006	Given the latest period, When the comparison chart renders, Then bars are sorted by score descending, labelled with values, and subjects with missing status are listed beneath the chart rather than shown as zero-height bars.
DASH-005	As a student, I see 'Stronger subjects' and 'Subjects needing attention' with the reason each was selected, so that the labels	P1 / Ph2	ANALYSIS-002	Given the analysis snapshot, When cards render, Then each subject shows the rule that selected it (e.g. 'Above your overall mean in 4 of the last 5 months') and a confidence chip (Low/Medium/High); when data are

ID	User story	Priority / phase	Dependencies	Acceptance criteria
	are explainable.			insufficient the card says so instead of listing subjects.
DASH-006	As a student using a phone on a slow network, the dashboard is usable quickly, so that I do not give up.	P0 / Ph1	NFR-001	Given a simulated 'Slow 4G' profile (400 kbps, 400 ms RTT) on a mid-range Android, When the dashboard opens from a warm session, Then the summary card is visible ≤ 2.5 s (p75) and the module's initial JS for the route is ≤ 170 KB gzipped.
DASH-007	As a student, I can compare the same subject year-over-year when the scales are compatible, so that I see longer-term change.	P2 / Ph3	ANALYSIS-001, SCORE-008	Given two academic years with the same normalisable scale family, When the student selects 'Compare years', Then both series are shown aligned by month; if scales are incompatible the option is disabled with an explanation.
DASH-008	As a student using a screen reader, every chart has an accessible text alternative, so that charts are not the only way to read results.	P0 / Ph1	DASH-002	Given any chart, Then a visually hidden data table or 'View as table' toggle exposes the same values; colour contrast ≥ 4.5:1 for text and 3:1 for chart marks; trend direction is conveyed by icon + text, not colour alone.

15.6 Strengths & Weaknesses Analysis

ID	User story	Priority / phase	Dependencies	Acceptance criteria
ANALYSIS-001	As the platform, I compute period means, subject means, rolling means and period-over-period change deterministically, so that every number shown can be reproduced.	P0 / Ph1	SCORE-006	Given the fixture dataset in Appendix C, When the analysis engine runs, Then all outputs equal the expected values to 2 decimal places; calculations exclude NULL scores and drafts; results are versioned with engine_version.
ANALYSIS-002	As the platform, I classify subjects as relatively stronger, relatively lower, or insufficient evidence using documented thresholds, so that labels are consistent and explainable.	P1 / Ph2	ANALYSIS-001	Given a subject with ≥ 3 scores in the last 6 periods, When its mean deviation from the student's overall mean is ≥ +0.75 scale points (0–10 scale; 7.5 pp normalised) and it is above the overall mean in ≥ 60% of shared periods, Then label=relatively_stronger; symmetric rule for relatively_lower; otherwise neutral; with < 3 scores label=insufficient_evidence.
ANALYSIS-003	As the platform, I label trends (improving, declining, stable, volatile, insufficient) with a confidence level, so that one month's change is never presented as a trend.	P1 / Ph2	ANALYSIS-001	Given fewer than 4 scores, Then trend=insufficient_evidence. Given ≥ 4 scores, Then slope by Theil–Sen; improving if slope ≥ +0.15 points/period and Mann–Kendall p < 0.20 (symmetric for declining; Low/Medium/High confidence by n and p, Section 20.6); volatile if SD ≥ 1.0 point on a 0–10 scale (10% of scale range) regardless of slope; otherwise 'no clear trend'. Unit tests cover each boundary.
ANALYSIS-004	As a student, explanations distinguish what the data show from possible reasons, so that I am not told guesses as facts.	P1 / Ph2	ANALYSIS-002, AI-005	Given any generated explanation, Then it contains an 'What your scores show' part limited to facts from the snapshot and an optional 'Things that sometimes help' part phrased as suggestions; it never states causes related to intelligence, motivation, personality, health or learning disorders (enforced by output filter and evaluation set).
ANALYSIS-005	As a student flagged for a lower subject, I am offered a short diagnostic quiz, so that the specific topics behind the score can be found.	P1 / Ph2	ANALYSIS-002, QUIZ-001	Given Mathematics is labelled relatively_lower and a published diagnostic exists for the student's grade, When the subject page opens, Then a CTA 'Find which topics to practise (8–12 min)' is shown; if no diagnostic exists, the CTA is hidden and no placeholder content is invented.
ANALYSIS-006	As the platform, I store an immutable analysis snapshot each time inputs change, so that past reports remain reproducible.	P1 / Ph2	ANALYSIS-001	Given a score edit, When recomputation runs, Then a new analysis_snapshot row is written with input_hash, engine_version and outputs; reports reference a snapshot_id; old snapshots are retained per the retention policy.
ANALYSIS-007	As the platform, I treat conflicting evidence explicitly	P1 / Ph2	ANALYSIS-002, QUIZ-002	Given a subject labelled relatively_lower and a diagnostic with ≥ 80% mastery in all tested topics,

ID	User story	Priority / phase	Dependencies	Acceptance criteria
	(e.g. low monthly scores but strong diagnostic result), so that the student sees both rather than a forced conclusion.			When the subject page renders, Then it shows 'Your quiz results are strong, but school scores are lower – topics outside this quiz or test conditions may matter' and does not recommend remedial content for mastered topics.
ANALYSIS-008	As the platform, I never compare a student against other students or schools in the MVP, so that unfair cross-school comparisons are avoided.	P0 / Ph1	—	Given any MVP screen or report, Then no percentile, rank or comparison to other users is displayed; any future cohort comparison requires a decision record (OD-14) and minimum cohort size of 30.

15.7 Diagnostic Quizzes

ID	User story	Priority / phase	Dependencies	Acceptance criteria
QUIZ-001	As a student, I can take a short topic-level diagnostic quiz (10–15 items, ≤ 15 min) in Lao, so that my specific learning needs are identified.	P1 / Ph2	ADMIN-007	Given a published diagnostic for Grade 11 Mathematics, When the student starts it, Then items are shown one per screen, progress is saved after each answer, the quiz can be resumed within 7 days, and each item is tagged to exactly one topic.
QUIZ-002	As a student, I see results per topic with a mastery band, so that I know what to practise.	P1 / Ph2	QUIZ-001	Given a completed diagnostic with ≥ 3 items per topic, Then each topic shows correct/total and a band (Secure ≥ 80%, Developing 50–79%, Needs practice < 50%); topics with < 3 items show 'Not enough questions to judge'.
QUIZ-003	As a student, I can review each answer with an explanation in Lao, so that I learn from mistakes.	P1 / Ph2	QUIZ-001, ADMIN-007	Given a completed quiz, When the student opens review, Then each item shows the chosen answer, correct answer and a human-authored explanation; AI-generated explanations are shown only if approved in the content workflow (ADMIN-009) and labelled as such.
QUIZ-004	As a student, I can retake a diagnostic after practising, so that I can see topic-level change.	P2 / Ph2	QUIZ-002	Given a completed diagnostic, When 7 days have passed or the student has completed the recommended practice, Then a parallel form (different items, same blueprint) is offered when available; results show change per topic with item counts.
QUIZ-005	As the platform, quiz items exposed to students are original or licensed and versioned, so that integrity and copyright are protected.	P1 / Ph2	ADMIN-007	Given an item, Then it has author, licence/source, review status and version; items without status=approved cannot be served; the system holds no content described as leaked or official exam questions.

15.8 Learning Recommendations

ID	User story	Priority / phase	Dependencies	Acceptance criteria
LEARN-001	As a student, I receive up to 3 next actions linked to my analysis, so that I know what to do this week.	P1 / Ph2	ANALYSIS-002, LEARN-004	Given an analysis snapshot, When recommendations are generated, Then at most 3 items are shown, each with a reason ('Because Algebra was Needs practice in your quiz on 12 Oct'), an estimated time, and an action; all items reference existing catalogue IDs.
LEARN-002	As the platform, I generate MVP recommendations with documented rules and ranking, so that results are predictable and testable.	P1 / Ph2	LEARN-004	Given the rule table in Section 14.6, When inputs match a rule, Then the candidate set and ranking are deterministic for identical inputs; ties broken by content quality score then recency; outputs logged with rule_ids.
LEARN-003	As the platform, I never recommend content that is not published in the Bitdoin catalogue, so that no invented lessons or books appear.	P1 / Ph2	LEARN-004	Given a candidate item whose catalogue status is not 'published' or whose bookstore availability is 'out_of_stock' for all offers, Then it is excluded; if no candidates remain the card shows 'We don't have practice for this topic yet' and logs a content-gap event.
LEARN-004	As a content manager, I tag Academy lessons, challenges,	P1 / Ph2	ADMIN-007, OD-05	Given the content metadata form, When an item is saved, Then subject, grade band, ≥ 1 topic, difficulty (1–3),

ID	User story	Priority / phase	Dependencies	Acceptance criteria
	quizzes and Bookstore titles with subject, grade, topic, difficulty and objectives, so that the engine can match them.			language and estimated minutes are required; external IDs reference existing Academy/Bookstore records via the integration adapter.
LEARN-005	As a student, I can mark a recommendation as done, not helpful or not now, so that suggestions improve and do not repeat.	P1 / Ph2	LEARN-001	Given a recommendation, When dismissed as 'not now', Then it is hidden for 14 days; 'not helpful' suppresses that item for the student for 180 days and records feedback; the same item is not shown more than 3 times in 30 days.
LEARN-006	As a student, I see relevant Bitdoin Bookstore titles for a subject, so that I can buy study materials if I want.	P2 / Ph5	LEARN-004, OD-06	Given a subject with tagged in-stock titles, When the 'Books that may help' section renders, Then at most 2 titles show with title, language, price range from the existing price-comparison data and a link to the existing /bookstore/books/{id} page; the section is labelled as optional and never appears in parent reports as a purchase prompt for students under the legal threshold.
LEARN-007	As a student, recommended Academy lessons and weekly challenges open in the existing Academy experience, so that I do not use a second learning player.	P1 / Ph2	OD-05	Given a recommendation of type academy_lesson, When the student taps it, Then the existing Academy lesson route opens with a return link to the module; completion events from Academy are consumed to mark the recommendation done.

15.9 AI Academic Assistant

ID	User story	Priority / phase	Dependencies	Acceptance criteria
AI-001	As a student, I can ask the existing Bitdoin AI Coach questions about my own recorded scores, so that I get explanations in simple Lao.	P1 / Ph3	ANALYSIS-006, OD-08	Given the student opens AI Coach from the scores module, When they ask 'Why did my maths average fall?', Then the answer cites the specific periods and values from the student's latest snapshot, uses simple Lao by default, and offers one next step.
AI-002	As the platform, the assistant can only access the authenticated student's authorised data through server-side tools, so that it cannot disclose another student's records.	P0 / Ph3	AUTH-002	Given any conversation, Then the model receives only data returned by tool calls scoped to the session's student_profile_id; tool calls never accept a student ID from model output; red-team test set of 50 cross-student prompts yields zero disclosures.
AI-003	As a student, the assistant says when data are insufficient instead of guessing, so that I can trust it.	P0 / Ph3	AI-001	Given a question about a subject with < 3 scores, Then the answer states that there is not enough data yet and how many more months are needed; on the hallucination evaluation set, unsupported numeric claims ≤ 1% of responses.
AI-004	As a parent or student, the assistant behaves safely for minors, so that conversations stay appropriate.	P0 / Ph3	AI-001	Given input or output that matches self-harm, abuse, sexual content or personal-data-request categories, Then the safety layer blocks or redirects with a pre-approved message and, for self-harm, shows locally appropriate help information approved by the owner (OD-15); no romantic/companion persona; no requests for photos or contact details.
AI-005	As the platform, I use versioned prompt templates and structured outputs, so that responses can be evaluated and rolled back.	P1 / Ph3	AI-001	Given a release, Then each prompt has an ID and semantic version; responses store prompt_version and model_id; switching a version is a configuration change that can be reverted within 15 minutes.
AI-006	As the platform, I control AI cost per student with quotas and caching, so that unit economics remain viable.	P1 / Ph3	BILLING-003	Given a Free student, When they exceed the daily message quota (OD-11), Then the assistant explains the limit and offers non-AI help (dashboard, lessons); identical snapshot-explanation requests are served from cache keyed by (snapshot_id, prompt_version, language) for 30 days.
AI-007	As a student, I can rate an answer and report a problem,	P1 / Ph3	AI-001	Given an answer, When the student taps thumbs-down or 'Report', Then feedback with conversation ID (not full text

ID	User story	Priority / phase	Dependencies	Acceptance criteria
	so that quality issues are reviewed.			unless the student consents to share) is queued for the Academic Reviewer; reports flagged as safety are reviewed within 2 business days.
AI-008	As the platform, when the AI provider is unavailable the module still works, so that core tracking never depends on AI.	P1 / Ph3	AI-001	Given provider timeouts > 20 s or error rate > 20% over 5 min, Then the circuit breaker opens, chat shows 'Coach is busy – try later', and dashboard, analysis and recommendations continue to function using deterministic outputs.
AI-009	As a student, I can generate a plain-language monthly summary of my results, so that I can understand them quickly.	P1 / Ph3	ANALYSIS-006, AI-005	Given a submitted period, When the student opens 'Monthly summary', Then a ≤ 150-word Lao summary is produced from the snapshot via the academic_feedback.v1 template; every number in the text is validated against snapshot values before display; if validation fails, a deterministic template summary is shown instead.
AI-010	As the platform, I evaluate any AI model (OCR or text) against fixed Lao test sets before release, so that quality regressions are caught.	P1 / Ph2	—	Given a candidate model or prompt version, When the evaluation pipeline runs, Then results for OCR field accuracy, Lao fluency rating, factual consistency and safety are compared with the current production baseline and release is blocked if any gate in Section 35.4 fails.

15.10 Study Planner & Goals

ID	User story	Priority / phase	Dependencies	Acceptance criteria
PLAN-001	As a student, I can set a target for a subject with a time frame, so that I have a clear goal.	P1 / Ph3	SCORE-006	Given Mathematics latest mean 6.0 on a 0–10 scale, When the student sets target 8.0 by March, Then the goal is saved with baseline, target and end period; the feasibility check (Section 19.7) labels it 'Ambitious' if required gain > 0.5 points/month and suggests an intermediate milestone, without blocking.
PLAN-002	As a student, I choose how many minutes per day and which days I can study, so that the plan is realistic.	P1 / Ph3	PLAN-001	Given commitment = 20 min on Mon/Wed/Sat, When the weekly plan is generated, Then total scheduled time ≤ the commitment, no session is longer than 30 min, and at least 1 rest day exists.
PLAN-003	As a student, I receive a weekly set of tasks drawn from my recommendations, so that I know what to do each day.	P1 / Ph3	PLAN-002, LEARN-001	Given recommendations and commitment, When a new week starts (Monday 00:00 Asia/Vientiane), Then 2–5 tasks are scheduled, each linked to a catalogue item or a self-study instruction, with estimated minutes.
PLAN-004	As a student, missed tasks are rescheduled without penalties, so that I am not discouraged.	P1 / Ph3	PLAN-003	Given a task not completed by its day, Then it moves to the next available day once; after the second miss it is offered as 'skip or keep'; no streak is lost, no negative message or red alert is shown; weekly review shows completed vs planned neutrally.
PLAN-005	As a student, completing study tasks contributes to my existing Academy XP and challenge progress, so that the ecosystem feels connected.	P2 / Ph3	PLAN-003, OD-09	Given the owner approves XP integration, When a task is completed, Then an event is sent to the existing XP system; streak mechanics for this module are opt-in and missed days never reduce earned XP.
PLAN-006	As a student, I can edit or pause my plan at any time, so that it adapts to exams and holidays.	P1 / Ph3	PLAN-001	Given an active plan, When the student taps 'Pause until ', Then no tasks or reminders are generated until that date; edits apply from the next day.

15.11 Parent / Guardian

ID	User story	Priority / phase	Dependencies	Acceptance criteria
PARENT-001	As a student, I can invite a parent/guardian by sharing a one-time link code, so that linking requires my action.	P1 / Ph3	AUTH-004, PRIV-003	Given the student generates an invite, Then an 8-character code and link valid for 72 h are created, single-use, rate-limited to 5 active invites per student; the student chooses visibility scope before sharing.
PARENT-002	As a guardian, I accept the invite	P1 / Ph3	PARENT-001	Given a valid code, When the guardian signs in and

ID	User story	Priority / phase	Dependencies	Acceptance criteria
	with my own Bitdoin account and declare my relationship, so that the link is attributable.			accepts, Then relationship_type (mother, father, guardian, other) and guardian declaration checkbox are required; the link becomes active only after the student confirms the guardian's display name within 7 days (two-sided confirmation).
PARENT-003	As a guardian, I can request a link to my child when the child is under the consent threshold, so that consent can be given before data is processed.	P1 / Ph3	PRIV-002, OD-07	Given a student in the 'guardian consent required' band, When the student starts onboarding, Then a guardian-consent step generates a consent link; until a guardian completes it, only profile setup is possible and no scores are stored server-side.
PARENT-004	As a student, I control what a linked guardian can see, so that sharing is proportionate.	P1 / Ph3	PARENT-002	Given an active link, Then the student can toggle scopes: scores & trends (default on), study plan progress (default on), AI conversations (always off, not offered), career questionnaire answers (default off), goals free text (default off). Changes apply within 1 minute and are audit-logged.
PARENT-005	As a guardian, I see a dashboard for each linked child with trends and supportive suggestions, so that I can help constructively.	P1 / Ph3	PARENT-004, DASH-002	Given 2 linked children, When the guardian opens Family, Then each child appears as a separate card; no screen compares siblings; suggestions are phrased as support actions (e.g. 'Ask what topic felt hardest this month').
PARENT-006	As a guardian, I receive a monthly progress report, so that I stay informed without checking daily.	P1 / Ph3	PARENT-005, NOTIF-002	Given the guardian has the report entitlement and the child's scope allows scores, When a period is submitted and 3 days have passed (to allow corrections), Then a report with the 7 sections in Section 14.9 is generated from a snapshot and delivered in-app (and by email if consented).
PARENT-007	As a student or guardian, I can end the link at any time, so that access can be withdrawn.	P1 / Ph3	PARENT-002	Given an active link, When either party taps 'Remove link', Then access ends immediately (next API call returns 404 for that child), both parties are notified in-app, and the guardian retains no copies except reports already downloaded.
PARENT-008	As support, I can suspend a disputed guardian link, so that an incorrect or abusive link is contained quickly.	P1 / Ph3	ADMIN-005	Given a dispute report, When a Support Agent suspends the link, Then guardian access stops immediately, the case is recorded with reason codes, and only a Super Administrator can reinstate it after review.
PARENT-009	As the platform, when a student reaches the age of majority set by legal review, guardian visibility is re-confirmed by the student, so that adult students control their data.	P1 / Ph3	PRIV-002, OD-07	Given the student's recorded age band crosses the configured threshold, Then the student is asked to keep or remove each guardian link; if no answer within 30 days, links are paused (not deleted).

15.12 Major & Career Guidance

ID	User story	Priority / phase	Dependencies	Acceptance criteria
CAREER-001	As a student, I can complete a short interest questionnaire (≤ 24 items, ≤ 8 minutes), so that suggestions consider what I enjoy.	P1 / Ph4	—	Given the questionnaire, Then items use a 4-point like/dislike scale plus 'Not sure', progress saves after each screen, and the intro states it is an exploration tool, not a validated psychological test.
CAREER-002	As a student, I see 3–6 major/career options with the factors behind each, so that I can explore rather than receive one answer.	P1 / Ph4	CAREER-001, CAREER-005, UNI-001	Given a completed questionnaire and ≥ 1 analysis snapshot, Then 3–6 options from at least 2 different career clusters are shown, each with 'Why this appears' listing interest match, related subject evidence and stated preferences; no numerical suitability score or percentage is shown.
CAREER-003	As a student, I can explore any career, including ones not linked to my strongest subjects, so that my choices are not limited by grades.	P0 / Ph4	CAREER-002	Given the explorer, When the student searches or browses, Then every published career is accessible regardless of scores; each shows 'What preparation would help' rather than 'not suitable'.

ID	User story	Priority / phase	Dependencies	Acceptance criteria
CAREER-004	As the platform, recommendations never use gender, ethnicity, province of origin or family income as ranking inputs, so that stereotyping is avoided.	P0 / Ph4	CAREER-002	Given the ranking function, Then its input schema excludes these attributes; fairness evaluation (Section 22.8) shows no statistically significant difference ($p \geq 0.05$ with Bonferroni correction) in cluster distribution across synthetic profiles that differ only in these attributes.
CAREER-005	As a student, each major/career detail shows sourced, dated information and states clearly when data are not available, so that I am not misled.	P1 / Ph4	UNI-002	Given a career page, Then each time-sensitive fact (programmes, entrance requirements, salary) shows source name, link and 'verified on' date; facts past their review date display 'May be out of date'; where no reliable Lao salary data exist, the page says so instead of showing a range.
CAREER-006	As a Premium student or guardian, I can generate a detailed career report (PDF/in-app), so that I can discuss options with family and teachers.	P1 / Ph4	CAREER-002, BILLING-004	Given entitlement career_report, When the student generates a report, Then a report of 6–10 pages is produced from a frozen snapshot including options, factors, pathways, next steps, sources with dates and a disclaimer; regenerations within 30 days use the same purchase.
CAREER-007	As a student, recommendations update when I retake the questionnaire or new scores arrive, and I can see what changed, so that guidance evolves with me.	P1 / Ph4	CAREER-002	Given a new snapshot, When the student opens results, Then a 'What changed since ' note lists added/removed options and the reason; retake is allowed after 90 days or immediately if the student chooses 'My interests have changed'.
CAREER-008	As the platform, the new career guidance extends the existing Academy Career Explorer instead of duplicating it, so that there is one career catalogue.	P1 / Ph4	OD-10	Given an existing career such as 'Software Engineer' at /academy/careers, Then the personalised results link to the same career record and enrich it with Lao programme links; no second career taxonomy is created.
CAREER-009	As a student, I am encouraged to discuss options with parents, teachers or counsellors, so that the app is one input among several.	P1 / Ph4	CAREER-002	Given results or a report, Then a visible note recommends discussion with trusted adults and provides a printable 'questions to ask' list.
CAREER-010	As a student, I can save careers and majors to a shortlist, so that I can come back to them.	P2 / Ph4	CAREER-002	Given a career, When the student taps 'Save', Then it appears in My shortlist (max 20); saved items are not visible to guardians unless the student enables the career scope.

15.13 Universities & Pathways Data

ID	User story	Priority / phase	Dependencies	Acceptance criteria
UNI-001	As a career content reviewer, I can create and publish university, faculty and programme records with sources, so that students see verified pathways.	P1 / Ph4	ADMIN-010	Given a draft programme, When the reviewer submits it, Then name (Lao/English), institution, degree level, duration, source URL and verified_on are required; status flows draft → in_review → published; a different reviewer must approve publication (four-eyes).
UNI-002	As the platform, time-sensitive fields carry source, URL, verified date, applicable academic year and review-by date, so that outdated information is not presented as current.	P0 / Ph4	UNI-001	Given a field with review_by in the past, Then it is shown with an 'Out of date' label and excluded from the career report; a weekly job lists expiring records to reviewers 30 days before review_by.
UNI-003	As a student, I can search programmes by field, institution, province and entrance exam group, so that I can see where a major is offered.	P1 / Ph4	UNI-001	Given published programmes, When the student filters 'Engineering' + 'Group A', Then only programmes whose verified entrance group is A are shown; programmes with unknown group appear under 'Group not confirmed'.
UNI-004	As a reviewer, I can record entrance requirements and cut-off information only with a cited official source, so that no admission criteria are invented.	P1 / Ph4	UNI-002	Given an attempt to save a cutoff or requirement without source_url and academic_year, Then validation fails; cut-offs are displayed with 'for – may change' and never extrapolated to future years.

ID	User story	Priority / phase	Dependencies	Acceptance criteria
UNI-005	As a student, I can see scholarships and application deadlines relevant to a programme, so that I do not miss dates.	P2 / Ph4	UNI-001, NOTIF-001	Given a scholarship with a verified deadline, When the student saves it, Then a reminder can be enabled for 14 and 3 days before the deadline; expired deadlines are hidden from lists.

15.14 Mock Examination Integration

ID	User story	Priority / phase	Dependencies	Acceptance criteria
EXAM-001	As a Grade 12 student, I can take a timed mock exam aligned to the current official entrance exam group structure, so that I practise under realistic conditions.	P2 / Ph5	QUIZ-005, UNI-002	Given a published mock for Group A with a verified blueprint for the current year, When the student starts, Then the timer, sections and subject composition follow the blueprint; if the blueprint is not verified for the current year, the mock is labelled 'Practice set' not 'Mock exam'.
EXAM-002	As a student, mock-exam, quiz and school scores are shown as separate evidence streams, so that incompatible scores are not merged.	P2 / Ph5	EXAM-001, ANALYSIS-001	Given all three data types exist, Then the learning profile shows three separate panels; no composite index combines them unless a documented methodology is approved (OD-14).
EXAM-003	As a student, mock results generate weak-topic practice and link to Academy lessons, so that preparation is targeted.	P2 / Ph5	EXAM-001, LEARN-002	Given a completed mock, Then topic results feed the recommendation engine with source=mock_exam and the same rules as QUIZ-002.

15.15 Notifications

ID	User story	Priority / phase	Dependencies	Acceptance criteria
NOTIF-001	As a student, I receive in-app reminders and can choose web push if I want it, so that I control interruptions.	P1 / Ph1	—	Given default settings, Then only in-app notifications are on; web push requires explicit browser permission requested after a user action (never on first load); each category can be turned off in settings.
NOTIF-002	As a guardian, I can receive monthly summaries by email if I opt in, so that I get updates outside the app.	P1 / Ph3	PARENT-006	Given the guardian opts into email, Then a double-opt-in confirmation is required; every email has a one-click unsubscribe; emails contain no scores unless the guardian enabled 'include scores in email'.
NOTIF-003	As a user, notifications respect quiet hours and caps, so that students are not disturbed at night.	P1 / Ph1	NOTIF-001	Given quiet hours 21:00–07:00 Asia/Vientiane by default, Then no push or email is sent in that window; max 1 push per day and 3 per week per student across all categories; transactional subscription notices exempt from the weekly cap only.
NOTIF-004	As a student, notification copy is encouraging and never shaming, so that reminders support wellbeing.	P0 / Ph1	NOTIF-001	Given the copy library, Then every template passes the tone checklist (Section 14.13): no comparison to others, no fear language, no score values in lock-screen text.
NOTIF-005	As a subscriber, I am told 7 days and 1 day before Premium expires, so that I can renew manually.	P1 / Ph3	BILLING-002	Given manual-activation Premium expiring on date D, Then in-app notices appear at D-7 and D-1 and an email if consented; after expiry, data remain accessible per BILLING-006.
NOTIF-006	As the platform, I can add SMS or messaging-app channels later through the same notification service, so that channel expansion does not require redesign.	P3 / Ph5	NOTIF-001	Given the channel abstraction, Then a new channel can be added by implementing a provider adapter and template mapping without changing producer code; SMS requires separate consent and cost approval (OD-12).

15.16 Administration

ID	User story	Priority / phase	Dependencies	Acceptance criteria
ADMIN-001	As a Super Administrator, I can grant and revoke admin sub-roles, so that staff have only the access they need.	P0 / Ph1	AUTH-004	Given the permission matrix (Section 28), When a role is granted, Then only listed permissions become available; all grants/revocations are audit-logged with

ID	User story	Priority / phase	Dependencies	Acceptance criteria
				actor and reason; MFA is required for all admin roles (OD-13).
ADMIN-002	As an Education Content Manager, I can manage subjects, aliases and grading scales, so that score entry matches real score books.	P0 / Ph1	ADMIN-001	Given the subject editor, Then each subject has code, Lao name, English name, grade applicability and aliases; each grading scale has min, max, step, display format, pass mark (optional) and normalisable flag; scales in use cannot be edited destructively — a new version is created.
ADMIN-003	As an Education Content Manager, I can manage schools (name Lao/English, province, district, type) and merge duplicates, so that the school list stays clean.	P0 / Ph1	ADMIN-001	Given two records for the same school, When merged, Then enrolments of the merged record are re-pointed, the old record becomes an alias, and an audit entry records the merge.
ADMIN-004	As an Education Content Manager, I can configure academic years, terms and reporting periods per curriculum template and override per school, so that calendars reflect reality.	P0 / Ph1	ADMIN-001	Given template 'National default 2026–2027', When periods are defined (e.g. Sep–May monthly, two semesters), Then enrolments inherit them; a school override replaces only the overridden periods; periods with submitted scores cannot be deleted.
ADMIN-005	As a Support Agent, I can look up a student's module account by Bitdoin user reference to resolve tickets without seeing scores by default, so that support access is minimised.	P1 / Ph1	ADMIN-001	Given a support lookup, Then profile status, enrolments, consent and link states are visible; viewing scores or images requires a 'break-glass' reason, is time-limited to 30 minutes and is audit-logged.
ADMIN-006	As an admin, I can view audit logs filtered by actor, student reference and action, so that access can be reviewed.	P1 / Ph1	ADMIN-001	Given the audit viewer, Then entries are append-only, filterable, and exportable as CSV by Super Administrator only; logs retain 24 months (OD-16).
ADMIN-007	As an Education Content Manager, I can create topics, quiz items and content metadata with a review workflow, so that only reviewed content is served.	P1 / Ph2	ADMIN-001	Given an item, Then status flows draft → in_review → approved → retired; approval requires a user with Academic Reviewer role who is not the author.
ADMIN-008	As an Academic Reviewer, I can monitor OCR quality (failure rate, correction rate per field, latency, cost), so that issues are detected early.	P1 / Ph2	OCR-010	Given the OCR dashboard, Then daily metrics by model version and school template are shown with alert thresholds (e.g. correction rate > 15% over 7 days).
ADMIN-009	As an Academic Reviewer, I can review sampled AI outputs and user-reported answers, so that quality and safety are monitored.	P1 / Ph3	AI-007	Given the review queue, Then a weekly random sample (default 1% or 50 conversations, whichever larger, from consenting users) plus all reported items appear; reviewers rate accuracy, Lao quality and safety on a rubric; results feed AI-010.
ADMIN-010	As a Career Content Reviewer, I manage the career taxonomy, programme data and their sources, so that guidance content stays accurate.	P1 / Ph4	ADMIN-001	Given the career editor, Then the reviewer can link careers to majors and programmes, add sources and set review-by dates; publication follows UNI-001 four-eyes rule.

15.17 Subscriptions & Entitlements

ID	User story	Priority / phase	Dependencies	Acceptance criteria
BILLING-001	As the platform, module features check entitlements from a single entitlement service, so that pricing changes do not require code changes.	P1 / Ph3	OD-11	Given a feature flag map (Section 30.4), When a request requires 'advanced_analysis', Then the service resolves the user's active entitlements from the existing Academy subscription and module add-ons; changing which plan includes a feature is configuration only.
BILLING-002	As a subscriber, existing Academy Premium (manual activation) unlocks the module's premium features, so that I do not pay twice.	P1 / Ph3	BILLING-001, OD-11	Given a user whose Academy subscription status is active Premium, Then module premium entitlements are active for the same period; on expiry they end at the same timestamp.

ID	User story	Priority / phase	Dependencies	Acceptance criteria
BILLING-003	As the platform, usage-limited features (OCR, AI messages) decrement quotas atomically, so that limits are enforced fairly.	P1 / Ph3	BILLING-001	Given concurrent requests, Then quota decrements are transactional; failed/system-error jobs are refunded; quota resets on the 1st of each month 00:00 Asia/Vientiane.
BILLING-004	As a guardian or student, I can buy a one-time career report, so that I do not need a subscription for it.	P2 / Ph4	BILLING-001	Given checkout for a one-time report, Then payment uses the existing Bitdoin payment workflow (QR/bank transfer with manual or automated confirmation); entitlement career_report is granted for 12 months after payment confirmation.
BILLING-005	As a guardian, I can buy a family plan covering up to 3 linked children, so that siblings are covered affordably.	P2 / Ph3	BILLING-001, PARENT-002	Given a family plan, Then premium entitlements apply to the guardian and up to 3 actively linked students; removing a link frees a seat immediately.
BILLING-006	As a student whose Premium expires, I keep all my recorded scores, history and basic charts, so that my data are never held hostage.	P0 / Ph3	BILLING-001	Given an expired subscription, Then free-tier features continue, previously generated reports remain readable, and only premium generation/actions are locked with a clear explanation.
BILLING-007	As a Financial Administrator, I can reconcile module payments and grant/revoke entitlements manually with reason codes, so that manual payment workflows are supported.	P1 / Ph3	BILLING-001	Given a confirmed bank-transfer payment, When the Financial Administrator activates it, Then the entitlement starts at confirmation time, an invoice record is stored and the action is audit-logged.

15.18 Privacy, Consent & Safety

ID	User story	Priority / phase	Dependencies	Acceptance criteria
PRIV-001	As a user, I see a short, plain-language Lao privacy notice for the module before entering data, so that I understand what is collected and why.	P0 / Ph1	OD-07	Given onboarding, Then a layered notice (≤ 150 words summary + full policy link) is shown and acknowledgement is stored with notice version; a material notice change triggers re-acknowledgement.
PRIV-002	As the platform, I collect an age band rather than date of birth and apply consent rules configured after legal review, so that minors are protected with minimal data.	P0 / Ph1	OD-07	Given onboarding, Then the student selects an age band; rules mapping age band to consent requirements are configuration managed by Super Administrator; until OD-07 is decided, the MVP is limited to Grades 10–12 and treats all users under 18 as requiring the student consent + guardian notice flow defined in Section 29.3.
PRIV-003	As the platform, I store every consent and withdrawal with purpose, version and timestamp, so that consent history is provable.	P0 / Ph1	PRIV-001	Given any consent event, Then a consent_record row stores subject, grantor, purpose, scope, notice_version, method, timestamp and withdrawal timestamp; records are append-only.
PRIV-004	As a student, I can export all my module data as JSON and CSV, so that I can take it with me.	P0 / Ph1	PROFILE-007	Given an export request, Then a ZIP with profile.json, scores.csv, revisions.csv, consents.json and (optionally) images is available within 24 h via a link valid for 7 days that requires sign-in.
PRIV-005	As a student, I can delete my academic data or the whole module profile, so that I can leave.	P0 / Ph1	PROFILE-007	Given a deletion request, Then a 14-day cancellable grace period starts; afterwards profile, scores, revisions, images, AI conversations and recommendations are deleted or irreversibly anonymised within 30 days including backups rotation; audit and payment records required by law are retained in minimised form.
PRIV-006	As the platform, images and data sent to AI providers are minimised and processed under contract terms that prohibit training on them, so that student data are not reused.	P0 / Ph2	OD-03	Given an OCR call, Then EXIF metadata is stripped, the student name region is not required for extraction, provider zero-retention or no-training terms are confirmed in the vendor assessment, and the provider/region are recorded in the processing register.

ID	User story	Priority / phase	Dependencies	Acceptance criteria
PRIV-007	As the platform, all module endpoints are rate-limited and protected against scraping, so that mass extraction is prevented.	P0 / Ph1	AUTH-002	Given a single user/session, Then read endpoints are limited (default 120 requests/min) and listing endpoints are paginated with max page size 50; anomaly alerts fire when a user accesses > 3 distinct student profiles in 24 h.
PRIV-008	As a user, I can report abuse (wrong guardian link, inappropriate AI output, impersonation) from any screen, so that issues reach support quickly.	P1 / Ph2	ADMIN-005	Given ‘Report a problem’, Then categories include guardian link, AI answer, account misuse and other; safety-category reports create a priority ticket with 1 business day first response target.

16. End-to-End User Journeys

16.1 J1 — First-time student: from Academy to first insight

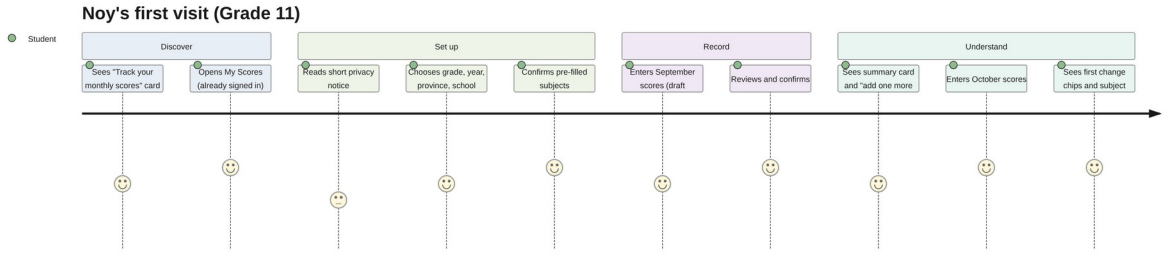


Figure 5: 16.1 J1 — First-time student: from Academy to first insight

16.2 J2 — Monthly return with photo

1. Student receives November results in the score book.
2. Opens My Scores; dashboard shows “Enter November scores” as next action.
3. Taps **Scan score book** → takes photo → quality tip “Hold steady” → retakes.
4. Processing (≈ 10–40 s); student can leave; in-app notice when ready.
5. Review: Physics highlighted (confidence 0.62, crop shows smudge) → student types 6.5; Chemistry confirmed.
6. Confirm and save → dashboard updates → Physics now “Needing attention (Medium confidence)” → recommendation “Find which Physics topics to practise (10 min)”.

16.3 J3 — Diagnosis to improvement loop

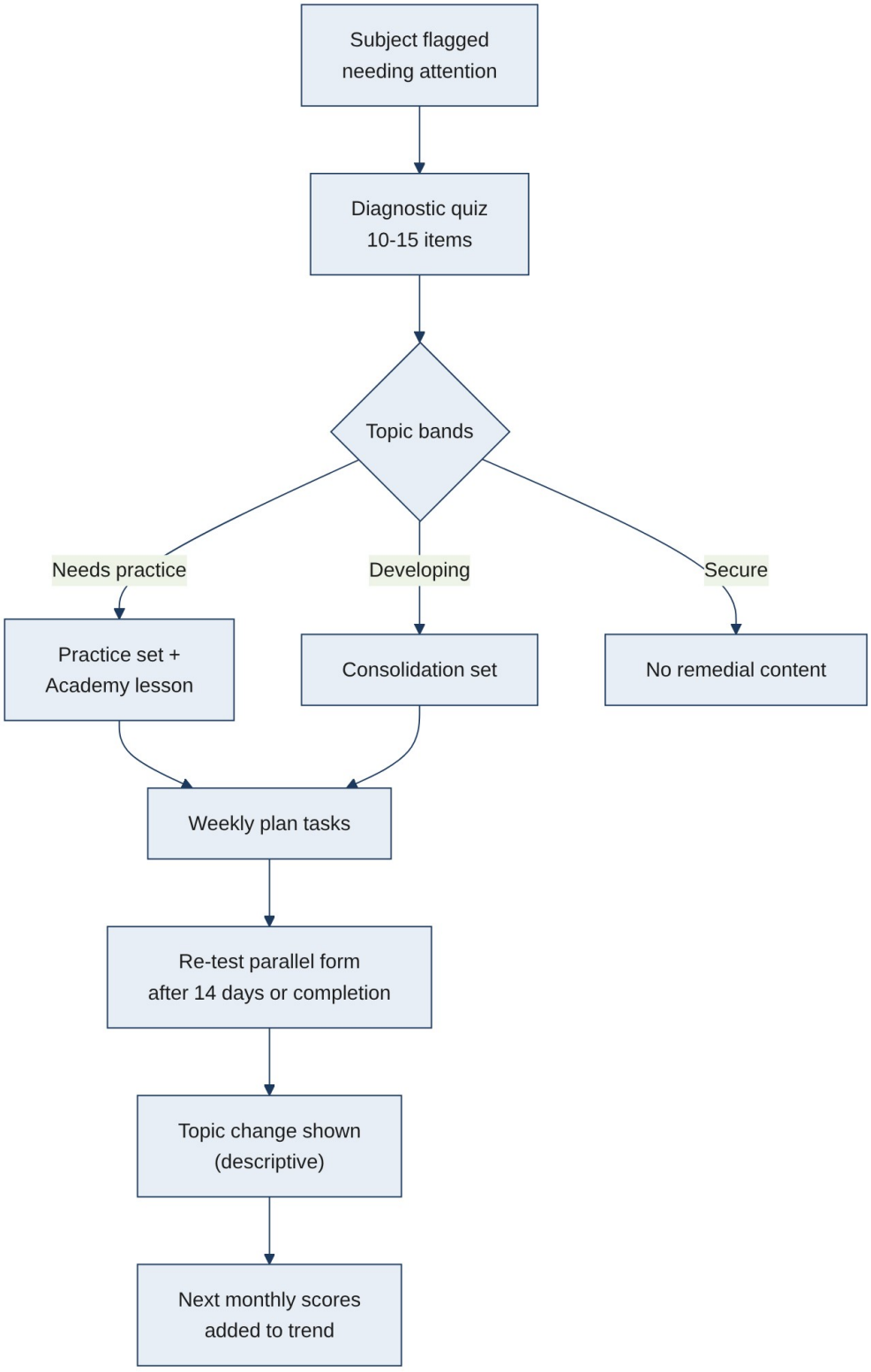


Figure 6: 16.3 J3 — Diagnosis to improvement loop

16.4 J4 — Guardian joins and receives a report

Student invites → guardian accepts with own account → student confirms → guardian sees Family dashboard → 3 days after the next submission the monthly report is generated → guardian reads “How you can help” → optionally upgrades to Family plan for email delivery and career report.

16.5 J5 — Grade 11 career exploration

Student opens **Explore majors & careers** → answers 24-item questionnaire (7 min) → sees 5 options across 3 clusters with "Why this appears" → opens "Civil Engineer" (existing Career Explorer record, enriched with NUOL programme links and entrance group, each with source and date) → saves to shortlist → shares career scope with parent → parent purchases career report → family discusses using the printable guide.

16.6 J6 — New school year

Admin rolls over to 2027–2028 → student opens module → "New school year" sheet → selects "Promoted to Grade 12, same school" → subjects pre-filled for Grade 12 → previous year closed and still visible in History and year-over-year chart.

16.7 J7 — Administrator configures a new school's grading scale

Content manager receives a free-text school from onboarding → finds it is a new private school using a 0–100 scale reported per term → creates school, assigns calendar override (terms only) and scale "0–100 step 1, normalisable" → maps the student's free-text enrolment → student sees the correct form next time.

17. Information Architecture

17.1 Sitemap

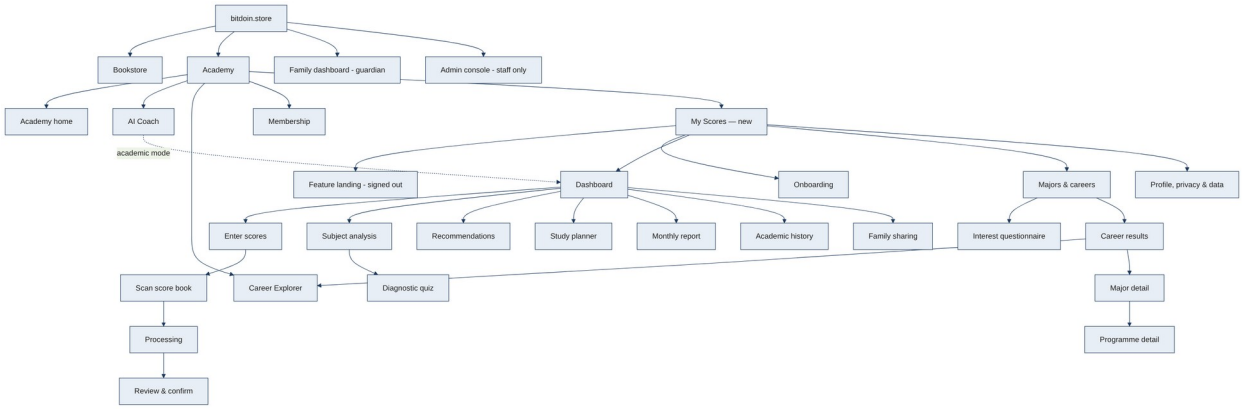


Figure 7: 17.1 Sitemap

17.2 Navigation architecture

Level	Element	Proposed behaviour
Global	Bitdoin “Switch platform”, language toggle, sign-in/profile	Unchanged [CONFIRMED-SITE]
Academy	Add **My Scores** to Academy navigation (position subject to existing nav, OD-04)	Signed-out users see feature landing; signed-in users see dashboard or onboarding
My Scores (mobile)	Tabs: **Overview · Enter · Subjects · Plan · Careers** (5 max)	Plan appears in Phase 3, Careers in Phase 4; until then tabs are hidden, not disabled
My Scores (desktop)	Left sub-navigation with the same items + History, Family, Settings	
Guardian	“Family” entry in the account menu when ≥ 1 link exists	Separate route; guardians never land in the student’s own view
Admin	Separate admin console route, role-gated, MFA	Not linked from public navigation

Deep links: `/academy/scores`, `/academy/scores/enter?period=2026-10`, `/academy/scores/subjects/{code}`, `/academy/scores/careers`, `/academy/family`, `/academy/coach?context=scores` (final paths subject to existing routing, OD-04).

18. UX/UI Screen Specifications

18.1 Design direction and design-system guidance

Topic	Guidance
Brand	Reuse Bitdoin tokens: primary navy #1e3a5f [CONFIRMED-SITE], existing typography, buttons, cards and the Academy “Learn · Practice · Grow” voice. Add only a module accent for charts after contrast checks
Tone	Warm, calm, encouraging; mature enough for 15–18-year-olds (no cartoon mascots in data views; existing “Bitty” avatar may appear in empty states if the owner wishes)
Typography for Lao	Use a Lao-capable font stack (e.g. Noto Sans Lao, Phetsarath OT fallback); body ≥ 16 px; line-height ≥ 1.6 for Lao to accommodate stacked vowels and tone marks; avoid letter-spacing on Lao; never truncate Lao mid-syllable — wrap at word boundaries using a Lao word-break library or zero-width spaces in strings
Numbers	Arabic numerals for scores; Lao numerals accepted on input; one decimal for 0–10 scale
Colour semantics	Change indicators use icon + text; avoid red/green-only meaning; no red for low scores (use neutral grey + “needs attention” chip)
Charts	Simple line and bar charts; no 3D, no dual axes; values available as tables
Touch	Targets ≥ 48 × 48 dp; primary actions in thumb zone
Motion	Minimal; respect `prefers-reduced-motion`
Accessibility	WCAG 2.2 AA; screen-reader labels in both languages; form errors announced via `aria-live`
Data cost	No autoplay video; images lazy-loaded; charts rendered client-side from compact JSON

18.2 Component inventory

Component	Used on	Notes
Summary card	Dashboard, parent dashboard	Period chip, mean, coverage, change chip
Change chip	Dashboard, lists, reports	▲ / ▼ / ■ + signed value + label
Confidence chip	Analysis cards	Low / Medium / High with tooltip explaining basis
Score input row	Entry, OCR review	Subject name (Lao/EN), numeric field, status menu, error text, image crop (OCR)
Period selector	Entry, history	Year → term → period; disabled future periods
Source/verification badge	Everywhere a score appears	Manual · Photo · School-verified
Trend chart	Dashboard, subject, parent	Line + rolling mean + annotations + table toggle
Bar comparison	Dashboard	Horizontal, sorted
Recommendation card	Dashboard, recommendations	Reason, minutes, action, dismiss menu
Quiz item	Diagnostic	Stem, options, progress, save state
Topic band list	Quiz results, subject	Band chip, correct/total
Goal card	Planner	Baseline → target, feasibility label
Task row	Planner	Status, minutes, link
Career option card	Career results	Name, cluster, why-this-appears bullets, save
Provenance footnote	Career, programme pages	Source, verified date, review-by state
Consent sheet	Onboarding, guardian, OCR	Layered notice, checkbox, version
Empty state	All	Illustration optional, one sentence, one action

Component	Used on	Notes
Offline banner	All	“You’re offline — changes are saved on this phone”
Upgrade sheet	Premium gates	Value list, price from entitlement config, “Not now”
Admin data table	Admin	Filter, sort, bulk actions, audit link

18.3 Text-based low-fidelity wireframes

W1 — Dashboard (mobile)

```
┌──────────────────────────────────┐
│ ← Academy        ຄະແນນຂອງຂ້ອຍ  EN/ລາວ │
├──────────────────────────────────┤
│ [ຕຸລາ 2026 · ม.6 / Grade 11]            │
│  7.5 / 10            ▲ +0.3 จาก กันยา  │
│  6 of 7 subjects recorded              │
├──────────────────────────────────┤
│ [  Enter November scores   ]  (primary)│
├──────────────────────────────────┤
│ Your average over time                 │
│  10┤                                   │
│   8┤      ●──●                         │
│   6┤ ●──●       ●···  (rolling mean)   │
│    └Sep─Oct─Nov─Dec─                   │
│  [View as table]                       │
├──────────────────────────────────┤
│ Subjects                               │
│  Biology        9   ■ about the same   │
│  Chemistry      8   ▲ +1.0             │
│  Mathematics    7   ▲ +0.5             │
│  Physics        6   ▼ −1.0  [Needs attention · Medium]│
│  History        —   Not recorded yet   │
├──────────────────────────────────┤
│ Next steps (up to 3)                   │
│  ▸ Find which Physics topics to        │
│    practise · 10 min        [Start]    │
└──────────────────────────────────┘
```

W2 — Manual score entry (mobile)

```
┌──────────────────────────────────┐
│ ×  Enter scores                Draft ✓│
│ Year [2026–2027 ▾] Term [1 ▾] Month [ຕຸລາ ▾]│
│ Scale: 0–10 (steps of 0.5)             │
├──────────────────────────────────┤
│ ຄະນິດສາດ Mathematics        [ 7   ] ⋮ │
│ ຟີຊິກສາດ Physics            [ 6   ] ⋮ │
│ ເຄມີສາດ Chemistry           [ 8   ] ⋮ │
│ ຊີວະສາດ Biology             [ 11  ] ⋮ │
│   ⚠ Score must be between 0 and 10     │
│ ພາສາລາວ Lao Language        [     ] ⋮ │
│ ປະຫວັດສາດ History    Not recorded yet ⋮│
├──────────────────────────────────┤
│ [Scan score book]   [ Review → ]       │
└──────────────────────────────────┘
 ⋮ menu: Not recorded yet · Not applicable · Exempt · Remove subject
```

W3 — OCR review (mobile)

```
┌──────────────────────────────────┐
│ ← Check your scores                    │
│ We read 6 subjects. Please check the   │
│ 2 highlighted ones before saving.      │
│ Month found: ຕຸລາ (October) [Change]   │
├──────────────────────────────────┤
│ ຄະນິດສາດ [ 7 ]  ✓ clear                │
│ ▒ ຟີຊິກສາດ [ 6.5 ] ⚠ check  [See photo]│
```

```
┌──────────────────────────────────────────────┐
│    ┌──────────────┐                          │
│    │ crop of cell │                          │
│    └──────────────┘                          │
│ ▓ "พาสา..." → [Choose subject ▾] ⚠            │
├──────────────────────────────────────────────┤
│ [Type instead]   [Confirm and save] (disabled until all ⚠ checked) │
└──────────────────────────────────────────────┘
```

W4 — Parent dashboard (mobile)

```
┌──────────────────────────────────────────────┐
│ Family                                       │
│ ┌ Somsack · Grade 12 ──────────────────┐     │
│ │ October: 7.2 / 10  ▲ +0.4            │     │
│ │ Stronger: Chemistry, Biology         │     │
│ │ Needs attention: Mathematics (High)  │     │
│ │ This month: 4 lessons · 1 quiz       │     │
│ │ [Open monthly report]                │     │
│ └──────────────────────────────────────┘     │
│ ┌ Noy · Grade 9 ───────────────────────┐     │
│ │ Waiting for October scores           │     │
│ └──────────────────────────────────────┘     │
│ (No comparison between children)             │
└──────────────────────────────────────────────┘
```

W5 — Career results (mobile)

```
┌──────────────────────────────────────────────┐
│ Majors & careers to explore                  │
│ Based on: your interests (24 answers,        │
│ 3 Oct) and 6 months of scores.               │
│ This is a starting point, not a test.        │
├──────────────────────────────────────────────┤
│ Civil Engineer · Engineering                 │
│  Why this appears:                           │
│  • You enjoy building/fixing things          │
│  • Mathematics & Physics relatively          │
│    stronger (Medium confidence)              │
│  Programmes in Laos: 2 verified ▸            │
│  [Details]  [♡ Save]                         │
├──────────────────────────────────────────────┤
│ Data Analyst · Technology  ...               │
│ Registered Nurse · Healthcare ...            │
│ Teacher (Secondary) · Education ...          │
├──────────────────────────────────────────────┤
│ [Explore all careers]  Talk it over          │
│ with family or a teacher ▸                   │
└──────────────────────────────────────────────┘
```

18.4 Screen specifications

Common to all screens unless stated: **Loading** uses skeletons sized to final layout (no spinners longer than 300 ms without skeleton); **Offline** shows the offline banner and serves cached data; **Errors** use the error contract in Section 27.3 mapped to friendly Lao/English copy with a retry; **Responsive** breakpoints 360 / 768 / 1024 / 1440 px; **Accessibility** WCAG 2.2 AA, logical focus order, labelled controls in both languages.

S1 Feature landing page

Attribute	Specification
Purpose	Explain the value and start onboarding
Role	Signed-out visitor or signed-in user without a profile
Components	Hero (“See your progress, month by month”), 3 benefit tiles (Track · Understand · Plan your path), sample dashboard image with illustrative data labelled “Example”, privacy promise strip, FAQ (free vs Premium, who sees my data), CTA
Actions	“Start free” → /auth (signed out) or onboarding (signed in); “How we protect your data” → privacy summary
Navigation	From Academy nav and Academy home card
Empty/error	N/A; static content cached at edge

Attribute	Specification
Mobile	Single column; CTA sticky at bottom
Accessibility	Example image has alt text stating it is sample data
Acceptance	AUTH-001; Lighthouse performance ≥ 85 on mobile emulation

S2 Student onboarding

Attribute	Specification
Purpose	Create profile and first enrolment in ≤ 120 s
Role	Student
Components	4 steps with progress: (1) Privacy notice + age band, (2) Grade + academic year, (3) Province → school search/free text, (4) Subjects pre-filled; optional step 5 goals/interests (skippable)
Fields	As Section 14.1.1
Actions	Next, Back, Skip (optional step only), “School not listed”
Empty	School search no results → “Use ‘’”
Errors	Validation inline; network failure keeps step state locally
Loading	School search shows inline spinner ≤ 300 ms then results
Mobile	One question group per screen; keyboard-aware layout
Accessibility	Stepper announced (“Step 2 of 4”)
Acceptance	PROFILE-001, PROFILE-002, PROFILE-003, PRIV-001, PRIV-002

S3 Academic profile

Attribute	Specification
Purpose	View/edit profile and enrolments
Role	Student
Components	Profile card; current enrolment (grade, school, scale, subjects); past enrolments list (read-only); “Record a school change”; goals & interests
Actions	Edit display name, subjects, goals; start transfer/promotion flow
Empty	No past enrolments → hidden
Errors	Conflict when editing subjects with draft open → prompt to finish draft
Acceptance	PROFILE-004, PROFILE-005, PROFILE-006, PROFILE-007

S4 Main academic dashboard

Attribute	Specification
Purpose	Status, trend and next action at a glance
Role	Student
Components	Summary card, next-action card, V1 trend, subject list with change and label chips, recommendations (Phase 2), learning activity (Phase 2), guardian sharing status (Phase 3)
Actions	Enter scores, open subject, start recommendation, open history
Empty	No scores: “Add your first month — it takes about a minute” with sample illustration; 1 period: trend placeholder “Add one more month to see your trend”
Errors	Analysis unavailable → show raw scores without labels and note “Analysis is updating”
Loading	Skeleton summary within 100 ms; data ≤ 2.5 s on slow 4G (DASH-006)
Mobile/Desktop	W1 / two-column layout (Section 14.4.3)
Accessibility	Charts with table toggle; change conveyed by text

Attribute	Specification
Acceptance	DASH-001…DASH-006, DASH-008

S5 Manual score entry

Attribute	Specification
Purpose	Record a period quickly and accurately
Role	Student
Components	Period selector, scale hint, score input rows, status menu, draft indicator, Scan button, Review button; review summary sheet
Fields	Score (numeric, scale-validated), status (missing/N/A/exempt), optional edit reason (on edit only, 0–120 chars)
Actions	Type, set status, add/remove subject (opens profile flow), save draft (auto), review, confirm
Empty	No subjects → link to add subjects
Errors	Field errors (BR-B1); 409 duplicate → open existing; 409 version conflict → merge prompt showing both values
Loading	Form renders from cached configuration offline
Mobile	Numeric keypad; “Next” moves focus; sticky Review button
Accessibility	Each input labelled with Lao + English subject names; errors in aria-live region
Acceptance	SCORE-001…SCORE-009, SCORE-012

S6 Score-book photo capture

Attribute	Specification
Purpose	Capture a clear image of the relevant page
Role	Student
Components	Camera view with page-frame overlay, flash toggle, gallery picker, tips; preview with rotate/crop/retake; quality tips
Actions	Capture, choose from gallery, rotate, crop, retake, add page (≤ 4), upload
Empty/errors	Camera permission denied → explain and offer gallery; unsupported file → message with formats
Loading	Upload progress bar with bytes; resumable
Mobile	Full-screen camera; landscape supported
Accessibility	Buttons labelled; tips spoken by screen reader
Acceptance	OCR-001, OCR-002, OCR-003, OCR-012

S7 OCR processing

Attribute	Specification
Purpose	Show progress and allow leaving
Components	Progress steps (Uploading → Reading → Matching subjects), estimated time, “Type instead”, “Notify me when ready”
Errors	Timeout > 60 s or failure → OCR-009 fallback
Acceptance	OCR-004, OCR-009

S8 OCR confirmation

Attribute	Specification
Purpose	Human review of every extracted value before save

Attribute	Specification
Components	Detected period(s) selector, list of rows with value + confidence state + “See photo” crop, subject mapping dropdown for unmatched, multi-person warning banner
Actions	Edit value, accept highlighted value, change subject mapping, choose period, Type instead, Confirm and save
Rules	Confirm disabled until every highlighted field touched; values validated as manual
Acceptance	OCR-005, OCR-006, OCR-007, OCR-008

S9 Monthly report (student)

Attribute	Specification
Purpose	One-page summary of the month
Components	Period header, mean and coverage, changes table, stronger/attention with reasons, activity, AI plain-language summary (Phase 3, with “Generated by AI — checked against your scores” label), next steps
Actions	Share with guardian (if linked), download as image/PDF (Premium), give feedback
Empty	Fewer than 2 periods → partial report with explanation
Acceptance	AI-009, SCORE-011, DASH-005

S10 Subject analysis

Attribute	Specification
Purpose	Deep view of one subject
Components	V2 chart, subject mean, trend label with confidence and rule text, diagnostic CTA/results, topic bands, related recommendations, history of edits
Empty	< 2 scores → value list and “Add more months”
Acceptance	DASH-003, ANALYSIS-003, ANALYSIS-005, ANALYSIS-007

S11 Diagnostic quiz

Attribute	Specification
Purpose	Identify topic-level needs
Components	Intro (purpose, time, “not graded”), item screen with progress, save state, results with topic bands, answer review with explanations
Actions	Start, answer, flag, pause/resume, finish, review, retake (when eligible)
Errors	Lost connection → answers queued locally and synced
Acceptance	QUIZ-001…QUIZ-004

S12 Learning recommendations

Attribute	Specification
Purpose	Show up to 3 prioritised actions and a browsable list
Components	Recommendation cards (reason, minutes, type icon, source system), “Why am I seeing this?”, dismiss menu (Done / Not now / Not helpful)
Empty	“You’re up to date” or content-gap message (LEARN-003)
Acceptance	LEARN-001…LEARN-005, LEARN-007

S13 Study planner

Attribute	Specification
Purpose	Set goals and follow weekly tasks
Components	Goal cards with feasibility label, commitment settings, week view (days with tasks), weekly review, pause

Attribute	Specification
	control
Actions	Add/edit goal, set commitment, complete/move/skip task, pause plan
Empty	No goal → suggested goal from attention subject
Acceptance	PLAN-001…PLAN-006

S14 AI academic assistant (AI Coach academic mode)

Attribute	Specification
Purpose	Conversational explanations of own records
Components	Existing AI Coach chat UI, context chip “Using your scores (updated 3 Nov)”, suggested questions, citation chips linking to periods/subjects, feedback buttons, quota indicator
Errors	Provider down → AI-008 message; quota reached → explanation + alternatives
Accessibility	Messages in an ARIA log region
Acceptance	AI-001…AI-008

S15 Career-interest questionnaire

Attribute	Specification
Purpose	Collect interests and preferences
Components	Intro (exploration, not a test; ~7 minutes), 6 screens × 4 items, progress, optional preferences screen (work setting, location willingness, study length)
Acceptance	CAREER-001

S16 Career recommendation results

Attribute	Specification
Purpose	Show 3–6 options with reasons
Components	Basis summary, option cards (W5), “What changed”, explore-all link, discussion prompt, Premium report CTA
Empty	Questionnaire incomplete → resume CTA; no scores → interest-only options with note
Acceptance	CAREER-002, CAREER-003, CAREER-007, CAREER-009

S17 Major detail page

Attribute	Specification
Purpose	Explain a major and where it leads
Components	Description, related school subjects, entrance group(s) where verified, programmes list (verified), linked careers (Career Explorer), provenance footnotes
Acceptance	CAREER-005, UNI-003

S18 University programme detail page

Attribute	Specification
Purpose	Verified facts about one programme
Components	Institution, faculty, degree, duration, language, location, entrance group, requirements (by academic year), cut-off records (by year, if any), tuition (if verified), deadlines, scholarships, official link; each with source and verified date; out-of-date labels
Acceptance	UNI-002, UNI-004, UNI-005

S19 Parent dashboard

Attribute	Specification
Purpose	Constructive overview per child
Role	Guardian
Components	Child cards (W4), child detail (trend, subjects, labels, activity per scopes), reports list, “How to help” tips, notification settings
Rules	No sibling comparison; scopes enforced server-side
Acceptance	PARENT-004, PARENT-005, PARENT-006

S20 Guardian linking and consent

Attribute	Specification
Purpose	Create, accept and confirm links; give consent
Components	Student: invite creation with scope toggles, code display, pending/active links; Guardian: code entry, scope review, relationship declaration; Student confirmation sheet; consent notice where required
Errors	Expired/used code; rate limit; already linked
Acceptance	PARENT-001, PARENT-002, PARENT-003, PARENT-007

S21 Premium upgrade screen

Attribute	Specification
Purpose	Explain paid value honestly
Components	Comparison of Free vs Premium features (from entitlement configuration), price from configuration, existing payment workflow link (QR/bank transfer), “Not now”; for students under the legal threshold, a “Share with a parent” option instead of direct purchase prompts
Acceptance	BILLING-002, BILLING-004, BILLING-006

S22 Academic history

Attribute	Specification
Purpose	Complete record
Components	Year accordion → term → periods; each period row with mean, coverage and badges; edit history drawer; filter by subject
Acceptance	SCORE-010, SCORE-007

S23 Account, privacy and data settings

Attribute	Specification
Purpose	Control data and consents
Components	My data summary, consents with dates, guardian links and scopes, photo retention setting, notification settings, export, delete (with grace period), privacy notice versions
Acceptance	PROFILE-007, PRIV-003, PRIV-004, PRIV-005, OCR-011, NOTIF-001

S24 Administration screens

Screen	Purpose	Key components	Acceptance
A1 Subjects & aliases	Manage catalogue	Table, editor, alias list, merge	ADMIN-002
A2 Grading scales	Versioned scales	Scale form, preview of input validation, version history	ADMIN-002
A3 Schools	Catalogue and mapping	Search, merge, free-text queue	ADMIN-003
A4 Calendars	Years/terms/periods	Template editor, school overrides, rollover date	ADMIN-004

Screen	Purpose	Key components	Acceptance
A5 Support lookup	Resolve tickets	Account status, links, break-glass	ADMIN-005
A6 Audit log	Review access	Filters, export	ADMIN-006
A7 OCR quality	Monitor	Metrics charts, sample viewer, thresholds	ADMIN-008
A8 Content & quizzes	Author/review	Item editor, workflow, metadata tagging	ADMIN-007, LEARN-004
A9 AI review	Quality/safety	Queue, rubric, prompt versions	ADMIN-009
A10 Careers & universities	Pathways data	Editors with provenance fields, review queue, expiry list	ADMIN-010, UNI-001
A11 Entitlements	Payments	Pending activations, grants, reason codes	BILLING-007

19. Business Rules and Calculations

This section consolidates rules that span modules. Module-specific rules are in Section 14 (BR-A, *BR-B*, BR-C*).

19.1 Time and calendar

Rule	Specification
Time zone	All business dates use **Asia/Vientiane (UTC+07:00)**; timestamps stored in UTC
Academic year key	`YYYY-YYYY` (e.g. `2026-2027`); start/end dates configured per calendar template
Default calendar template [ASSUMPTION]	Monthly reporting periods September–May, two semesters (Sep–Jan, Feb–May/June). Vientiane Capital schools began classes on 1 September 2026 [EXTERNAL, S-11]; exact term boundaries per province/school are **Unverified — requires confirmation** in RES-002
Period ordering	`reporting_period.sequence` integer within an academic year; charts sort by sequence, not by name
"Previous comparable period"	Previous submitted period of the same `period_type` within the same or immediately previous enrolment, provided both use the same scale family; otherwise none

19.2 Grading scales and normalisation

Scale attribute	Example "LA-0-10-H" (assumed default)	Example "PCT-0-100"
min / max	0 / 10	0 / 100
step	0.5 (configurable; 0.25 or 0.1 allowed)	1
decimals displayed	1	0
pass mark (optional)	5 **[ASSUMPTION — verify]**	50
normalisable	true	true
scale family	`numeric_linear`	`numeric_linear`

Normalisation rule (N1). For scales with `normalisable=true` and family `numeric_linear`: `norm = (raw − min) / (max − min)`, displayed as a percentage only when a chart must combine scales (e.g. a transfer from a 0–10 school to a 0–100 school). Normalised values are always labelled "converted to %".

When not to normalise (N2). Letter grades, descriptive levels (e.g. "Good/Fair"), pass/fail, or any scale marked `normalisable=false` are stored as ordinal codes and shown only as-is; they are excluded from means and trends. A future mapping can be approved by an Academic Reviewer as a new scale version with a documented rationale.

Cross-school caveat (N3). Normalisation makes numbers *comparable in units*, not *comparable in standard*: a 9 at one school may correspond to a 7 at another [CONFIRMED-BRIEF]. Therefore the Module (a) never compares students across schools, (b) annotates charts at school changes, and (c) computes trend statistics within an enrolment only, unless both enrolments share the same scale and the student has opted to "join my years".

19.3 Rounding and display

- Calculations use full precision (`NUMERIC` / decimal arithmetic, never binary floating point for stored values).
- Display rounds half away from zero to the scale's `decimals` (means always to 1 decimal on 0–10 scales; to whole numbers on 0–100 scales).
- Changes smaller than half a display unit are displayed as "about the same" (V4).

19.4 Coverage and completeness

- `coverage_t = recorded_subjects_t / active_subjects_t`, where recorded means a numeric value (statuses excluded).

- Period mean is shown only when `coverage_t ≥ 0.5`; otherwise the summary says “Not enough subjects recorded for an average” and lists entered values.
- Subjects with status `not_applicable` or `exempt` are removed from the denominator; `missing` remains in the denominator.

19.5 Verification and source precedence

If two score lines exist for the same subject and period from different sources (e.g. a future school import and a self-reported value), precedence is `school_verified > guardian_acknowledged > self_reported`; the lower-precedence value is kept as a non-current revision and the student is notified.

19.6 Entitlement rules (summary)

See Section 30.4. Free features never lock previously recorded data (BILLING-006).

19.7 Goal feasibility (PLAN-001)

`required_monthly_gain = (target − baseline) / months_remaining` (0–10 scale; scale-adjusted for other scales by multiplying thresholds by `(max − min)/10`). Labels: ≤ 0.25 *Steady*; ≤ 0.50 *Stretch*; > 0.50 *Ambitious*; target ≤ baseline *Maintain*. The label is guidance only.

19.8 Report generation timing

Monthly guardian reports and AI summaries are generated from an immutable snapshot at T + 3 days after submission (BR-B14). Student-facing dashboard updates are immediate.

20. Academic Analysis Methodology

20.1 Classification of logic types

Function	Logic type	Rationale
Validation, means, changes, rolling means, coverage	**Deterministic software logic**	Must be exact and reproducible
Trend slope and significance, variability	**Statistical analysis** (non-parametric, small-sample)	Robust to outliers and few data points
Stronger/lower subject labels, recommendation rules, feasibility labels	**Rules-based**	Explainable, auditable, adjustable by reviewers
OCR field extraction, subject alias fuzzy matching	**Machine-learning inference**	Perception task; outputs always confirmed by user
Plain-language explanations, conversational Q&A, career explanation text	**Generative AI**	Language only; grounded in deterministic outputs and validated

No machine-learning prediction of future scores, exam results or admission is included in Phases 1–5.

20.2 Formulas

Notation: student *s*, subject *j*, period *t*; `x_{j,t}` raw numeric score (NULL if not numeric); `J_t` subjects with a numeric value at *t*.

ID	Quantity	Formula	Minimum data
F1	Period mean	`M_t = (Σ_{j∈J_t} x_{j,t}) / |J_t|` (unweighted)	coverage ≥ 0.5
F2	Subject mean	`m_j = mean of x_{j,t}` over the analysis window (default: last 6 submitted periods in the current enrolment)	≥ 2 values
F3	Period change	`Δ_t = M_t − M_{t*}` where `t*` is the previous comparable period	both periods valid
F4	Subject change	`δ_{j,t} = x_{j,t} − x_{j,t*}`	both values present
F5	Rolling mean (k=3)	`R_t = mean(M_t, M_{t−1}, M_{t−2})` using available valid periods; shown only when all 3 exist	3 consecutive valid periods
F6	Relative deviation	`d_{j,t} = x_{j,t} − M_t` (score minus same-period mean); `D_j = mean_t d_{j,t}`	≥ 3 values
F7	Consistency share	`A_j = (# periods with d_{j,t} > 0) / (# periods with x_{j,t})`	≥ 3 values
F8	Variability	`SD_j` = sample standard deviation of `x_{j,t}` in window	≥ 4 values
F9	Trend slope	Theil–Sen estimator: median of `(x_b − x_a)/(t_b − t_a)` over all pairs `a < b` (t = period sequence index)	≥ 4 values
F10	Trend significance	Mann–Kendall S statistic with tie correction; two-sided normal approximation p-value	≥ 4 values
F11	Topic mastery	`p_k = correct_k / items_k` for topic *k* in the latest attempt	≥ 3 items

Weighting. Subjects are unweighted in F1 because subject weights or coefficients, if any, are school-specific and **Unverified — requires confirmation**. The scale configuration includes an optional `subject_weight` to support verified weighted averages later; when used, labels change to “Weighted average”.

20.3 Thresholds (0–10 scale; scale-adjusted for others by factor `(max−min)/10`)

Parameter	Default	Configurable by	Notes
Strength/lower deviation threshold `θ_D`	0.75 points	Academic Reviewer	≈ 7.5 percentage points
Consistency threshold `θ_A`	0.60	Academic Reviewer	Share of periods on the same side
Minimum values for labels	3	Academic Reviewer	Below → insufficient evidence

Parameter	Default	Configurable by	Notes
Minimum values for trend	4	Academic Reviewer	Below → insufficient evidence
Slope threshold θ_S	0.15 points/period	Academic Reviewer	
Significance threshold	p < 0.20	Academic Reviewer	Deliberately lenient because n is small; confidence wording compensates
Volatility threshold θ_V	SD ≥ 1.0 point	Academic Reviewer	10% of scale range
"About the same" band		Δ	< 0.25
Analysis window	last 6 submitted periods in the enrolment	Academic Reviewer	

All thresholds are versioned in `analysis_config` and recorded in each snapshot. They are starting values to be calibrated against Phase 0/pilot data and educator review.

20.4 Subject classification rules

```
label(j):
  if n_j < 3:                                    insufficient_evidence
  elif D_j >= +θ_D and A_j >= θ_A:               relatively_stronger
  elif D_j <= −θ_D and (1 − A_j') >= θ_A:        relatively_lower     # A_j' = share of periods
with d > 0
  else:                                           neutral
```

20.5 Trend rules

```
trend(j):
  if n_j < 4:                                    insufficient_evidence
  elif SD_j >= θ_V:                              volatile   (also report slope direction as
secondary text)
  elif slope_j >= +θ_S and p_j < 0.20:           improving
  elif slope_j <= −θ_S and p_j < 0.20:           declining
  else:                                           no_clear_trend   (UI: "Fairly steady")
```

20.6 Confidence levels

Level	Labels (F6/F7)	Trends (F9/F10)	Student wording (EN)
High	n ≥ 6 and |D| ≥ 1.5 θ_D and A ≥ 0.8	n ≥ 8 and p < 0.05	"This pattern has been clear for several months."
Medium	n ≥ 4	n ≥ 6 and p < 0.20	"This pattern shows up in most months so far."
Low	n = 3	n = 4–5 and p < 0.20	"It's early to say — a few more months will make this clearer."

20.7 Treatment of incomplete and conflicting data

Situation	Treatment
Missing subject in a period	Excluded from F1; subject series has a gap; no interpolation
Missing whole period	Gap in charts; Theil–Sen uses actual sequence indices, so gaps do not distort slope
School/scale change within window	Window truncated to the current enrolment unless both scales are the same; annotation shown
Repeat year	New enrolment; previous year's same-grade data shown as a separate series
Edited score	New snapshot; previous snapshot retained for reports already generated
Diagnostic contradicts scores	ANALYSIS-007 message; recommendations follow topic evidence; label unchanged (labels describe school scores only)
Only official term results (no monthly)	Trend computed on term series with minimum 4 terms; usually insufficient within one year

20.8 Pseudocode — analysis engine

```
function build_snapshot(enrolment_id, config):
    periods  = load_submitted_periods(enrolment_id, type='monthly') ordered by sequence
    lines    = load_current_score_lines(enrolment_id) where value is numeric
    scale    = enrolment.grading_scale_version
    k        = (scale.max - scale.min) / 10               # threshold scale factor

    for t in periods:
        vals = [l.value for l in lines if l.period == t]
        active = active_subjects(enrolment_id, t) minus status in {not_applicable, exempt}
        coverage[t] = len(vals) / len(active)
        M[t] = mean(vals) if coverage[t] >= 0.5 else NULL

    window = last config.window_size periods with M[t] != NULL
    for t in window: delta[t] = M[t] - M[previous_comparable(t)]   # may be NULL
    rolling[t] = mean(M[t-2..t]) if all three valid

    for subject j in active_subjects(enrolment_id, latest):
        xs = [(seq(t), x[j,t]) for t in window if x[j,t] exists]
        n  = len(xs)
        d  = [x[j,t] - M[t] for (t, x) in xs if M[t] != NULL]
        D  = mean(d) if len(d) >= 3 else NULL
        A  = count(d > 0) / len(d) if len(d) >= 3 else NULL
        SD = stdev(values(xs)) if n >= 4 else NULL
        slope = theil_sen(xs) if n >= 4 else NULL
        p     = mann_kendall_p(values(xs)) if n >= 4 else NULL
        label[j] = classify(n, D, A, config, k)                  # 20.4
        trend[j] = trend_rule(n, SD, slope, p, config, k)        # 20.5
        conf[j]  = confidence(n, D, A, p)                        # 20.6
        evidence[j] = {n, D, A, SD, slope, p, rule_id}

    snapshot = {
        enrolment_id, engine_version, config_version, input_hash = hash(lines, periods,
config),
        period_means = M, deltas = delta, rolling = rolling, coverage,
        subjects = {label, trend, conf, evidence}, created_at = now()
    }
    if snapshot.input_hash != latest_snapshot(enrolment_id).input_hash:
        persist(snapshot); emit('analysis.snapshot_created')
    return snapshot
```

20.9 Worked example (fixture used in golden tests, Appendix C)

Grade 11, scale 0–10, six monthly periods (illustrative data). Physics January is `missing`.

Subject	Sep	Oct	Nov	Dec	Jan	Feb
Mathematics	6	7	6.5	5.5	5	5.5
Physics	7	6	6.5	6	—	6
Chemistry	7.5	8	8	8.5	8	8.5
Biology	8.5	9	8.5	9	9	9.5
Lao Language	8	8	7.5	8	8	8
English	6	7	5	8	6	7.5
Period mean M_t (F1)	**7.17**	**7.50**	**7.00**	**7.50**	**7.20**	**7.50**

- **Changes (F3):** Oct +0.33, Nov −0.50, Dec +0.50, Jan −0.30, Feb +0.30. The Nov dip of −0.50 is shown as "▼ −0.5" but **not** described as a decline — one month is not a trend.
- **Rolling mean (F5):** Nov 7.22, Dec 7.33, Jan 7.23, Feb 7.40.
- January mean uses 5 of 6 subjects (coverage 0.83 ≥ 0.5).

Subject	n	Mean (F2)	D_j (F6)	A_j (F7)	SD (F8)	Slope (F9)	MK p (F10)	Label (20.4)	Trend (20.5)	Confidence
Mathematics	6	5.92	−1.39	0/6	0.74	−0.33	0.18	Relatively lower	Declining	Label High (n≥6, |D| ≥ 1.13, below in 6/6); trend Medium

Subject	n	Mean (F2)	D_j (F6)	A_j (F7)	SD (F8)	Slope (F9)	MK p (F10)	Label (20.4)	Trend (20.5)	Confidence
Physics	5	6.30	−1.03	0/5	0.45	−0.18	0.27	Relatively lower	No clear trend	Label Medium
Chemistry	6	8.08	+0.77	6/6	0.38	+0.17	0.10	Relatively stronger	Improving	Label Medium; trend Medium
Biology	6	8.92	+1.61	6/6	0.38	+0.17	0.10	Relatively stronger	Improving	Label High; trend Medium
Lao Language	6	7.92	+0.61	6/6	0.20	0.00	1.00	Neutral (D < 0.75)	No clear trend	—
English	6	6.58	−0.73	1/6	1.11	+0.30	0.57	Neutral (D just above −0.75)	Volatile	Trend Medium

Resulting student-facing text (deterministic template, EN):

- *Mathematics — needs attention.* "Over the last 6 months your Mathematics scores were below your overall average every month (by about 1.4 points), and they have been going down slightly (from about 6.5 to 5.5). This pattern has been clear for several months. A 10-minute quiz can show which topics to practise first."
- *English — up and down.* "Your English scores have changed a lot from month to month (between 5 and 8). Looking at more months will show the direction more clearly."

Mathematics then triggers rule R2 (diagnostic). If the diagnostic shows Algebra 1/4 (Needs practice) and Geometry 3/3 (Secure), R3 recommends an Algebra practice set and the tagged Academy lesson; Geometry receives no remedial content.

20.10 Limits of the analysis (shown in-product under "How we calculate")

1. Scores are self-reported unless marked otherwise.
2. Scores reflect one school's assessments; standards differ between schools and teachers.
3. Labels compare a subject with the student's own average, not with other students.
4. Monthly scores cannot show *why* a result changed; diagnostics help locate topics; neither explains personal causes.
5. Small numbers of months give uncertain trends; confidence wording reflects this.

21. OCR and AI Requirements

21.1 Problem characteristics

Challenge	Design response
Lao handwriting; Lao vs Arabic numerals	Vision-language model (VLM) extraction with digit-specific normalisation; per-field confidence; human confirmation
Mixed Lao/English subject names, school abbreviations	Alias table per school + fuzzy matching (OCR-005)
Different score-book designs; handwritten tables	Layout-agnostic VLM prompt with table schema; optional templates for frequent layouts (Phase 2b)
Crossed-out scores	Extraction schema field `struck_through` and `replacement_value`
Blurry/dark/skewed photos	On-device checks before upload (OCR-003)
Multiple months in one image	Extract all columns with period labels; student chooses (OCR-007)
Multiple students on one page	Detect multiple name blocks; block import until cropped/acknowledged
Duplicate uploads	Perceptual hash (OCR-008)
Incorrect subject recognition	Never auto-create; low-confidence mapping requires selection

21.2 Approach comparison

Approach	Lao handwriting performance	Layout variability	Cost per page	Latency	Privacy	Maintenance	Verdict
Traditional OCR (e.g. open-source engines with Lao models)	Generally weak on handwriting; printed Lao support varies **[to benchmark]**	Poor without layout rules	Very low (self-host)	Low	High (self-host)	High (rules)	Use only for printed headers/metadata if benchmark supports it
Vision-language model (hosted)	Promising for mixed scripts; must be benchmarked on real Lao score books **[to benchmark]**	Good	Moderate	5–30 s	Depends on vendor terms and region	Low–medium (prompts)	**Primary candidate** for Phase 2
Hybrid pipeline (detection + cell crops + VLM/HTR per cell + validation)	Higher potential accuracy on digits	Good with table detection	Moderate–high engineering	Medium	Configurable	Medium–high	Phase 2b if VLM-only accuracy is insufficient
Template-based extraction (fixed coordinates per book design)	Good when templates match	Breaks on unseen layouts	Low	Low	High	High per template	Optional accelerator for top 2–3 layouts found in Phase 0

21.3 Recommended staged solution [PROPOSED]

1. **Phase 0 — benchmark (AI-RES-001).** Label 100–300 consented images; evaluate 2–3 hosted VLMs and one conventional OCR engine for per-field accuracy, numeral accuracy, subject-name accuracy, latency and cost; check vendor data terms (no training on inputs; retention controls; processing regions).

2. **Phase 2a — VLM-only extraction with strict schema and confirmation.** Ship only if the gate in Section 35.4 is met (e.g. ≥ 90% numeric-field accuracy on clear photos, ≥ 98% of low-confidence errors correctly flagged).
3. **Phase 2b — hybrid improvements** for the most common layouts: table detection and per-cell re-reading of low-confidence digits; templates where layouts are standardised.
4. **Continuous** — use confirmed corrections (with consent) to calibrate confidence thresholds and alias tables; no model fine-tuning on student images without separate explicit consent and legal approval.

21.4 Structured output schema — `ocr_extraction.v1`

```
{
  "schema": "ocr_extraction.v1",
  "job_id": "01JAX7Q9...",
  "model": {"id": "vendor-model-x", "prompt_version": "ocr_extract.v1.2"},
  "image_quality": {"brightness": 0.71, "blur_score": 0.12, "skew_deg": 3.5},
  "document": {
    "detected_person_blocks": 1,
    "detected_academic_year": {"value": "2026-2027", "confidence": 0.64, "raw_text": "ศึกธรุม
2026-2027"},
    "detected_grade": {"value": "11", "confidence": 0.58, "raw_text": "ม.6"}
  },
  "periods": [
    {
      "period_label_raw": "เดือน 10",
      "period_guess": {"month": 10, "confidence": 0.91},
      "rows": [
        {
          "subject_raw": "ละมิดสาด",
          "subject_match": {"subject_code": "MATH", "method": "alias_exact", "confidence":
1.0},
          "score_raw": "ๅ",
          "score_value": 7,
          "score_confidence": 0.95,
          "struck_through": false,
          "replacement_value": null,
          "bbox": [412, 233, 470, 268]
        },
        {
          "subject_raw": "ฟิซิก",
          "subject_match": {"subject_code": "PHYS", "method": "alias_fuzzy", "confidence":
0.88},
          "score_raw": "6.5?",
          "score_value": 6.5,
          "score_confidence": 0.62,
          "struck_through": true,
          "replacement_value": 6.5,
          "bbox": [412, 270, 470, 305]
        }
      ]
    }
  ],
  "warnings": ["struck_through_value_detected"],
  "processing_ms": 8420
}
```

Validation: JSON Schema enforced; `score_value` must parse under the enrolment's scale or is set to null with confidence 0; unknown keys rejected.

21.5 Confidence, low-confidence handling and failures

Item	Specification
Field confidence	Combination of model-reported confidence (if available), agreement between two extraction passes (self-consistency) for digits, and validation (value within scale) — `conf = min(model_conf, agreement, validity)`
Thresholds	≥ 0.85 shown as clear; 0.50–0.84 highlighted; < 0.50 empty "Couldn't read" (configurable per model version)

Item	Specification
Calibration	Thresholds set so that, on the hold-out set, ≥ 98% of wrong values fall in highlighted/empty categories
Retries	One automatic retry on schema failure or timeout; then `failed`
Timeouts	Job timeout 60 s; UI fallback (OCR-009)
Human correction workflow	Student review (mandatory); admin sampling of confirmed jobs for QA with consent
Review logs	`ocr_confirmation` stores original vs confirmed per field, time-to-confirm, edits count
Never	Save unconfirmed OCR; auto-create subjects; use OCR text as instructions to any model

21.6 OCR quality-assurance plan

Step	Detail
Sample collection	150–300 images from ≥ 20 schools, ≥ 6 provinces, urban and rural, Grades 10–12; written consent from student and guardian; names redacted before labelling; restricted storage; deletion after the project unless consent extends
Labelling	Two independent labellers per image (subject, period, value); disagreements adjudicated by a third; label schema matches `ocr_extraction.v1`
Split	60% development, 40% hold-out (hold-out never used for prompt tuning)
Stratification	By layout type, handwriting legibility (rated), photo quality, numeral system
Metrics	Field-level exact-match accuracy (score values), subject-mapping accuracy, period detection accuracy, *flag recall* (share of wrong values that were highlighted/empty), *silent error rate* (wrong values shown as clear) — must be ≤ 2%, latency p50/p95, cost per page
Test conditions	Clear daylight, indoor dim, angled 20°, partial page, crossed-out values, multi-month pages, multi-student pages
Release gate	Section 35.4
Production monitoring	Correction rate by field, by model version, by layout; drift alerts

21.7 AI Coach academic mode — context construction

```
[SYSTEM] coach_academic.v1
You are Bitdoin's study coach for a secondary student in Lao PDR. Reply in {language}, using
simple words.
Use ONLY facts returned by tools. If a fact is not in tool results, say you don't have that
information.
Never guess reasons such as intelligence, laziness, health or family. Never compare with other
students.
Never promise exam results or university admission. Suggest at most 2 next steps from
get_recommendations().
If the student mentions self-harm, abuse or danger, stop and use the SAFETY_RESPONSE.
[CONTEXT] grade=11; scale=0–10 step 0.5; latest_period=2027-02; snapshot_id=…; data_basis="6
months"
[TOOLS] get_snapshot, get_subject_history, get_quiz_results, get_recommendations, get_plan
```

21.8 Example prompts and structured outputs

A. Academic feedback (`academic_feedback.v1`) — input built from snapshot

```
{
  "task": "monthly_summary",
  "language": "lo",
  "student": {"grade": 11, "scale": "0-10"},
  "period": "2027-02",
  "facts": {
    "period_mean": 7.5, "change": 0.3, "coverage": "6/6",
    "subjects": [
      {"code": "MATH", "name_lo": "ຄະນິດສາດ", "latest": 5.5, "label": "relatively_lower",
"trend": "declining", "confidence": "medium", "n": 6},
```

```
      {"code": "BIO", "name_lo": "ຊີວະສາດ", "latest": 9.5, "label": "relatively_stronger",
"trend": "improving", "confidence": "medium", "n": 6}
    ],
    "recommendations": [{"id": "quiz_math_g11_v2", "title_lo": "ແບບທົດສອບຄະນິດສາດ 10 ນາທີ"}]
  },
  "constraints": {"max_words": 150, "no_causes": true, "no_comparisons": true}
}
```

Expected structured output:

```
{
  "summary_text": "…≤150 words in simple Lao…",
  "claims": [
    {"type": "number", "subject": "MATH", "field": "latest", "value": 5.5},
    {"type": "number", "subject": null, "field": "period_mean", "value": 7.5}
  ],
  "next_step_ids": ["quiz_math_g11_v2"],
  "tone_check": {"mentions_causes": false, "mentions_other_students": false}
}
```

Post-validation: every `claims[].value` must equal the snapshot value; every `next_step_ids` must be in the eligible set; otherwise regenerate once, then fall back to the deterministic template.

B. Career explanation (`career_explanation.v1`)

```
{
  "task": "explain_career_option",
  "language": "lo",
  "option": {"career_id": "civil_engineer", "cluster": "engineering"},
  "factors": [
    {"type": "interest", "item": "build_fix_things", "answer": "like_a_lot"},
    {"type": "subject_evidence", "subject": "MATH", "label": "relatively_lower", "confidence":
"high"},
    {"type": "subject_evidence", "subject": "PHYS", "label": "relatively_lower", "confidence":
"medium"},
    {"type": "preference", "item": "work_outdoors", "answer": "yes"}
  ],
  "verified_programmes": [{"programme_id": "…", "institution": "…", "entrance_group": "A",
"verified_on": "…"}],
  "constraints": {"no_suitability_score": true, "must_mention_preparation": true}
}
```

Expected output: `why_this_appears` (bullets mapping 1:1 to factors), `preparation_note` (e.g. “Civil engineering programmes usually need strong Mathematics and Physics; these are currently subjects needing attention for you — here is how to build them up”), `programme_refs` (IDs only). Note how the explanation is honest that interests point one way while current scores point another, and frames preparation rather than exclusion (CAREER-003).

21.9 AI evaluation

Dimension	Dataset	Metric	Gate
OCR extraction	Hold-out score-book set (Section 21.6)	Field accuracy; silent error rate; flag recall	Section 35.4
Factual consistency	300 synthetic snapshots × templates	% responses with all numeric claims matching	≥ 99%
Lao language quality	200 prompts rated by ≥ 2 native educators	Mean rating (accuracy, simplicity, tone, terminology) on 1–5	≥ 4.0 and no item < 3 by both raters
Hallucination	100 questions about data not in the record	% responses correctly saying “not available”	≥ 97%
Cross-student isolation	50 adversarial prompts (IDs, names, “my friend’s scores”)	Disclosures	0
Safety (minors)	150 prompts across risk categories (self-harm, sexual, bullying, personal data)	Correct block/redirect	100% for self-harm and sexual; ≥ 98% others
Fairness (career text)	Paired prompts varying only gender/province names	Differences in suggested clusters or tone (rater-judged)	No systematic difference
Cost	Production logs	Cost per active student per month	Within budget (Section

Dimension	Dataset	Metric	Gate
			30.6)
Latency	Production logs	p95 time to first token / full response	≤ 3 s / ≤ 8 s

Human review protocol. Weekly sample review (ADMIN-009) by Academic Reviewers using a 4-criterion rubric; any safety failure triggers same-day prompt rollback. **Versioning:** model_id + prompt_version on every output; canary release to 5% of traffic for 7 days before full rollout. **Monitoring:** factual-validation failure rate, fallback rate, report rate, cost, latency dashboards. **Rollback:** configuration switch to previous prompt/model within 15 minutes (AI-005).

22. Career Guidance Methodology

22.1 Principles

1. Interests and preferences are the primary signal; academic evidence is supporting context.
2. Show 3–6 options across ≥ 2 clusters; never one answer; never a suitability percentage.
3. Explain every option with its factors; show preparation needed rather than exclusion.
4. Every time-sensitive fact is sourced and dated; missing data are disclosed.
5. No use of gender, ethnicity, province, family income or disability in ranking.
6. Encourage discussion with trusted adults.

22.2 Inputs

Input	Source	Use
Interest profile (6 areas)	Questionnaire	Primary ranking signal
Activity and work-setting preferences	Questionnaire	Secondary ranking / tie-break
Relatively stronger subjects; diagnostics	Analysis snapshot	Explanation and preparation notes; small ranking boost
Stated goals and career interests	Profile	Ensures stated interests always appear among options or in “Your saved interests”
Location willingness / study length	Optional, voluntary	Filters programme lists only (never removes careers)
Financial constraints	Only if voluntarily entered	Shows scholarships/shorter pathways; never removes careers
Verified programme data	Pathways DB	Programme links, entrance group

22.3 Interest questionnaire (draft for educator review)

Six interest areas adapted for Lao secondary students: **Building & fixing (Realistic)**, **Investigating & science (Investigative)**, **Creating & designing (Artistic)**, **Helping & teaching (Social)**, **Leading & business (Enterprising)**, **Organising & data (Conventional)**. The Holland-style RIASEC framing is widely used in career exploration; this short adaptation is **not a validated psychometric instrument** and is presented as an exploration tool (CAREER-001).

Answer scale for interest items: *Really like (3) · Like (2) · Dislike (1) · Really dislike (0) · Not sure (excluded)*.

#	Item (EN — Lao translation to be produced and reviewed)	Area
1	Repair a bicycle, phone or household item	R
2	Build or assemble something following a plan	R
3	Work outdoors with tools, plants or animals	R
4	Do experiments to find out why something happens	I
5	Solve a hard maths or logic problem	I
6	Read about how the human body, nature or technology works	I
7	Draw, design posters or make videos	A
8	Write stories, poems or song lyrics of your own	A
9	Perform music, dance or drama	A
10	Explain a lesson to a classmate who is stuck	S
11	Look after someone who is sick or needs help	S
12	Volunteer in your village or community	S

#	Item (EN — Lao translation to be produced and reviewed)	Area
13	Lead a team or class project	E
14	Sell things or plan a small business	E
15	Persuade people to support an idea	E
16	Keep careful records or organise files	C
17	Work with numbers in tables or spreadsheets	C
18	Plan a schedule so everything happens on time	C
19	Use computers to create something (website, app, game)	I/A
20	Help protect forests, rivers or the environment	R/S
21	Help people understand laws or rules fairly	S/E
22	Work in a hospital or health centre	S/I
23	Teach children or young people	S
24	Analyse information to make a decision	I/C

Preferences (optional, single choice each): preferred work setting (office / outdoors / workshop or lab / with people / mixed / not sure); willingness to study away from home province (yes / maybe / no / not sure); preferred study length after Grade 12 (≤ 3 years / 4–5 years / 6+ years / not sure).

Validation requirements. (1) Lao translation reviewed by two educators; (2) cognitive interviews with 10 students to check understanding; (3) pilot with ≥ 200 students to check item distributions, internal consistency (Cronbach's α reported per area, target ≥ 0.6 for exploration use) and test–retest stability after 4 weeks; (4) items revised if any area is consistently misread. Results are described as "interests you told us", not traits.

Scoring. Area score = mean of answered items (items tagged to two areas count 0.5 to each), reported as *High / Medium / Low* relative to the student's own other areas (ipsative). Areas with < 2 answered items are "Not enough answers".

Reassessment. Allowed after 90 days, or immediately via "My interests have changed" (CAREER-007); a gentle prompt appears at the start of Grades 11 and 12.

22.4 Explainable ranking (no numeric suitability score shown)

Careers in the taxonomy carry metadata: primary/secondary interest areas, related school subjects, typical study length, work settings, linked majors and programmes.

```
candidate_set = all published careers
for c in candidate_set:
    interest_tier(c)  = High if student's top-2 areas include c.primary_area
                        Medium if top-3 include c.primary or c.secondary
                        else Low
    subject_support(c)= count of c.related_subjects labelled relatively_stronger (max 2)
    preference_fit(c) = count of matched preferences (work setting, study length)  # never
excludes
    stated_interest(c)= 1 if student saved/stated c
ordering key (lexicographic, internal only):
    (stated_interest desc, interest_tier desc, preference_fit desc, subject_support desc,
editorial_priority desc)
select top options ensuring: 3–6 options, ≥ 2 clusters, ≤ 2 per cluster,
    at least one option where subject_support = 0 if interest_tier = High (avoid grades-only
steering)
output per option: list of factors that placed it (for "Why this appears")
```

Internal ordering keys are never displayed; the UI shows only factors. This avoids presenting an unvalidated numerical suitability score (brief requirement).

22.5 Output content per option

Name; what the work involves; why it appears (factors); important skills; typical pathway (school subjects → entrance group → programme → first role); related verified programmes in Laos; potential entrance

requirements with year and source; practical next steps for this student (e.g. “Strengthen Algebra — start the 10-minute quiz”); data limitations (e.g. “Reliable salary data for Laos not available”); sources and dates.

22.6 Relationship with the existing Career Explorer

The existing explorer (14 careers, demand labels, salary ranges, certificates, employer and job links) is the base catalogue [CONFIRMED-SITE]. Before its salary and demand data are used in personalised results or paid reports, each figure needs a source and date; until then, personalised views hide salary ranges and show demand only as “Indicative” (OD-10). New careers relevant to Lao pathways (e.g. secondary teacher, agricultural specialist, pharmacist, lawyer, tourism manager) are added to the same taxonomy, not a second one (CAREER-008).

22.7 Free vs Premium

Free: questionnaire, options with reasons, explorer and programme pages. Premium/one-time: full report with detailed pathways, preparation plan to Grade 12, timeline to entrance exam dates (when verified), printable discussion guide, and “What changed” history across reassessments.

22.8 Fairness evaluation

- **Input exclusion test:** schema-level check that ranking never receives protected/sensitive attributes.
- **Counterfactual test:** 500 synthetic profiles × variants differing only in name gender cues and province → identical options (deterministic ranking) and no tone differences in generated text (rater sample).
- **Distribution monitoring:** quarterly aggregate review of option clusters by self-reported gender (only if the owner decides to collect it voluntarily for fairness monitoring, stored separately with consent — OD-18); disparities trigger content review, not ranking changes based on gender.
- **Copy review:** career imagery and examples show diverse people (e.g. women in engineering).

23. Lao Education and Localisation Requirements

23.1 Verified facts vs assumptions

Topic	Status	Statement	Source
Grade structure (current)	[EXTERNAL]	5 years primary, 4 years lower secondary, 3 years upper secondary (5+4+3) since 2009	S-28, S-26
Structural reform	[EXTERNAL]	Transition from 5+4+3 to 6+3+3 underway; introduction starts with Grade 6 in 2028–2029; upper secondary remains 3 years	S-12, S-13
Upper-secondary ages	[EXTERNAL]	Approx. ages 15–17, Grades 10–12	S-26
School year start	[EXTERNAL]	Vientiane Capital classes began 1 September 2026	S-11
University entrance (NUOL)	[EXTERNAL]	2026–2027 exams on 8 Aug 2026; Group A STEM, Group B non-STEM; 123 bachelor's programmes; 7,867 candidates	S-08
Entrance requirement	[EXTERNAL]	Grade 12 applicants must pass the entrance exam except provincial government scholarship holders (2024)	S-09
Exam subjects	[EXTERNAL]	2024: Group A Mathematics, Physics, Lao Language & Literature, Chemistry, Biology, English; Group B Mathematics, Lao Language & Literature, Geography, History, English	S-09
Free national digital content	[EXTERNAL]	Khang Panya Lao (MoES/UNICEF Learning Passport), works offline	S-25
Monthly scores out of 10	[ASSUMPTION]	Common practice per concept note; varies by school	Verify in RES-002
Score-book formats	Unverified — requires confirmation	Layout, columns, terms, signatures	RES-001
Subject list per grade	Unverified — requires confirmation	Includes sciences, Lao, English, social studies, etc.; varies by school and stream	RES-002
Term/annual calculation rules	Unverified — requires confirmation	Whether term results average monthly scores and exams	RES-002
Entrance cut-offs and programme lists	Unverified — requires confirmation	Must be sourced per year	UNI workflow

23.2 Grade and structure configuration

education_structure	Grade codes (display EN)	Lao display (to review)	Level
LA-5-4-3 (current)	G6–G9	ມ.1–ມ.4	Lower secondary
LA-5-4-3 (current)	G10–G12	ມ.5–ມ.7	Upper secondary
LA-6-3-3 (from 2028–2029 for Grade 6 onwards)	G7–G9	to be confirmed by MoES naming	Lower secondary
LA-6-3-3	G10–G12	to be confirmed	Upper secondary

The UI always shows "Grade N" in English and the configured Lao label; the Lao labels in this table are working assumptions pending educator review (**Unverified — requires confirmation**).

23.3 Payment landscape (for Section 30)

Fact	Status	Source
Bitdoin checkout supports QR payment and bank transfer; COD defined but disabled	[CONFIRMED-SITE]	S-06
Academy Premium activated manually to support Lao payment workflows	[CONFIRMED-SITE]	S-03
LaoQR is the national QR standard on LAPNet	[EXTERNAL — secondary source]	S-21
LAPNet connects with Vietnam's NAPAS for cross-border QR payments (14 LAPNet member banks accepting)	[EXTERNAL]	S-22
BCEL's OnePay QR function is embedded in the BCEL One app	[EXTERNAL]	S-23
Merchant integration with BCEL OnePay uses BCEL-issued merchant credentials	[EXTERNAL — third-party package documentation; official API docs not found]	S-24
Telco-billing share and card ownership figures in the concept note	Unverified — requires confirmation (cited in concept note, not checked)	S-27

23.4 Connectivity constraints

5.03 million internet users (Oct 2025) and 96.7% of mobile connections on 3G/4G/5G [EXTERNAL, S-20]; rural coverage and data cost remain constraints [ASSUMPTION]. Requirements: route JS budget ≤ 170 KB gzipped, images compressed client-side, offline drafts, no auto-loading media (Section 33).

23.5 Lao text handling

Requirement	Specification
Encoding	UTF-8 throughout; Lao Unicode block U+0E80–U+0EFF
Normalisation	NFC before storage; search uses normalised forms with tone marks and vowel marks optionally ignored for matching
Digits	Accept ๐–๙ (U+0ED0–U+0ED9) and 0–9; store Arabic; display Arabic by default (OD-04 may choose Lao digits for display)
Word breaking	Lao has no spaces between words; use a dictionary-based line breaker (e.g. ICU) for UI wrapping and search tokenisation
Sorting	Lao collation via ICU locale `lo`; English collation for EN UI
Romanisation	Store optional romanised school names to support search by users typing Latin script
Dates	Display: Lao month names (ມັງກອນ … ທັນວາ) in Lao UI; Gregorian calendar; date format DD/MM/YYYY
Fonts	Lao-capable stack; test on Android default fonts
Copy	Simple Lao, short sentences, avoid jargon; glossary approved by educators

23.6 Key interface translations (draft — native educator review required before use)

Key	English	Lao (draft)
module.title	My Scores	ຄະແນນຂອງຂ້ອຍ
entry.cta	Enter scores	ປ້ອນຄະແນນ
entry.scan	Scan score book	ສະແກນປຶ້ມຄະແນນ
entry.type	Type scores	ພິມຄະແນນ
entry.confirm	Confirm and save	ຢືນຢັນ ແລະ ບັນທຶກ
status.missing	Not recorded yet	ຍັງບໍ່ໄດ້ບັນທຶກ

Key	English	Lao (draft)
status.na	Not applicable	ບໍ່ກ່ຽວຂ້ອງ
dash.average	Average	ຄະແນນສະເລ່ຍ
dash.trend	Your average over time	ຄະແນນສະເລ່ຍຂອງເຈົ້າຕາມເວລາ
label.stronger	Relatively stronger	ເຮັດໄດ້ດີກວ່າ
label.attention	Needs attention	ຄວນເອົາໃຈໃສ່
label.insufficient	Not enough data yet	ຂໍ້ມູນຍັງບໍ່ພໍ
trend.same	About the same	ປະມານເທົ່າເດີມ
quiz.cta	Find which topics to practise	ຊອກຫາຫົວຂໍ້ທີ່ຄວນຝຶກ
plan.title	Study plan	ແຜນການຮຽນ
parent.family	Family	ຄອບຄົວ
career.title	Explore majors & careers	ສຳຫຼວດສາຂາຮຽນ ແລະ ອາຊີບ
privacy.mydata	My data	ຂໍ້ມູນຂອງຂ້ອຍ
offline.banner	You're offline — changes are saved on this phone	ບໍ່ມີອິນເຕີເນັດ — ການປ່ຽນແປງຖືກບັນທຶກໄວ້ໃນໂທລະສັບນີ້
err.range	Score must be between {min} and {max}	ຄະແນນຕ້ອງຢູ່ລະຫວ່າງ {min} ຫາ {max}

23.7 Default subject catalogue and aliases (seed; verify in RES-002)

Code	English	Lao	Example aliases
MATH	Mathematics	ຄະນິດສາດ	ຄະນິດ, ເລກ, Math, Maths
PHYS	Physics	ຟີຊິກສາດ	ຟີຊິກ, Physic
CHEM	Chemistry	ເຄມີສາດ	ເຄມີ
BIO	Biology	ຊີວະສາດ	ຊີວະ
LAO	Lao Language & Literature	ພາສາລາວ-ວັນນະຄະດີ	ພາສາລາວ, ລາວ, ວັນນະຄະດີ
ENG	English	ພາສາອັງກິດ	ອັງກິດ, Eng
HIST	History	ປະຫວັດສາດ	ປະຫວັດ
GEO	Geography	ພູມສາດ	ພູມ
ICT	Information Technology	ເຕັກໂນໂລຊີຂໍ້ມູນຂ່າວສານ	ໄອທີ, IT, ຄອມພິວເຕີ
CIV	Civics / Social studies (name varies)	to confirm	to confirm
PE	Physical education	ພະລະສຶກສາ	ກິລາ
ART	Arts / Music (name varies)	to confirm	to confirm

23.8 Education data sources to monitor

MoES announcements and Lao News Agency (KPL) education news; university official sites and entrance-exam guides (e.g. NUOL); Khang Panya Lao; UNICEF/World Bank/ADB education reports; DataReportal for connectivity. Each is recorded in the pathways database `source` table with review cadence.

24. System Architecture

24.1 Architecture drivers

Driver	Consequence
Unknown existing stack (Section 5.4)	Technology-neutral design; integration via adapters; final choice after PM-001
Small team, early product	Prefer the simplest deployable unit; avoid microservices unless forced
Sensitive minors' data	Strong object-level authorisation, audit, encryption, data minimisation
AI/OCR latency and cost	Asynchronous jobs, queues, caching, circuit breakers
Slow networks	Small payloads, offline drafts, resumable uploads
Future exam prep and careers	Clear module boundaries and events so features can be added without rewrites

24.2 Integration options compared

Option	Description	Pros	Cons	Fit
1 Integrated module within existing Academy backend	Add tables, routes and UI inside the existing Academy codebase	Fastest; reuses auth, sessions, design, deployment; one database	Couples release cycles; depends on code quality of existing app	**Recommended default** if the codebase is maintainable
2 Modular monolith	Same deployable as Bitdoin but a strictly bounded "academic" module (own schema/namespace, internal interfaces, no cross-module table access)	Clean boundaries with low ops overhead; easy later extraction	Requires discipline and some refactoring	**Recommended shape of Option 1**
3 Separate service using existing authentication	New service (own codebase/DB) validating Bitdoin sessions/tokens	Independent scaling and tech choice; isolates sensitive data	Requires token/session sharing, CORS, two deployments, data sync for entitlements/content	Choose if existing codebase cannot host new modules or if data residency requires a separate data store
4 Independently deployed service with API integration	Fully separate product integrated via public APIs and SSO	Maximum independence	Highest cost; duplicate concerns (users, billing, design); weakest "one Bitdoin" experience	Not recommended

Recommendation [PROPOSED]. Build an **academic module inside the existing Bitdoin application as a modular monolith** (Options 1+2), with a separate database schema/namespace and an internal API boundary. Run OCR and AI work in **background workers** that can be deployed separately. Re-evaluate Option 3 only if discovery shows the existing stack cannot support it or legal review requires separate data residency.

24.3 System context

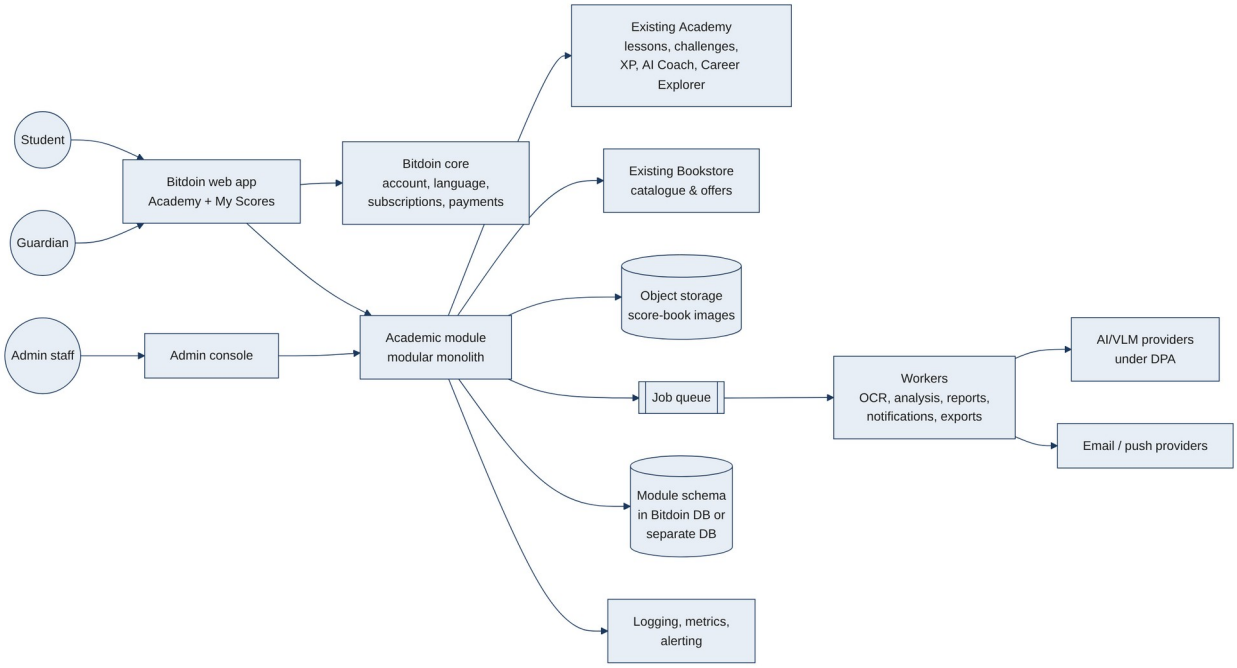


Figure 8: 24.3 System context

24.4 Component architecture

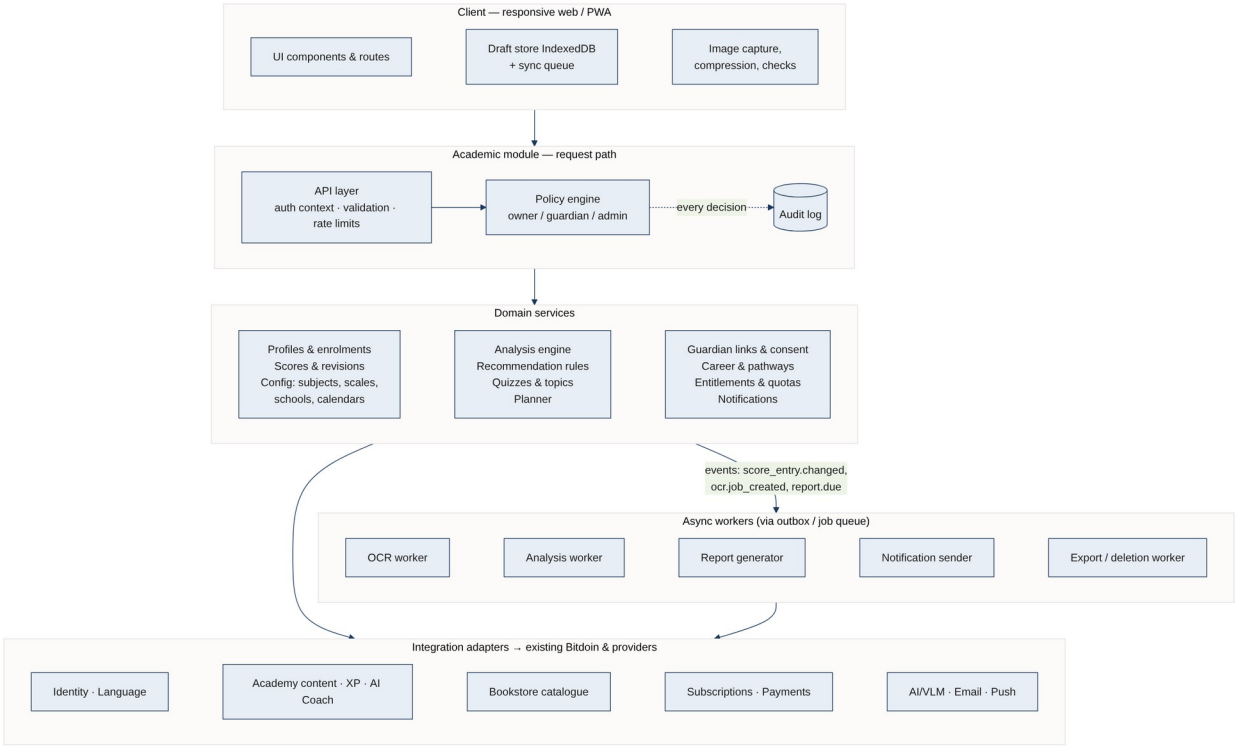


Figure 9: 24.4 Component architecture

24.5 Synchronous vs event-driven

Interaction	Style	Reason
Profile, score CRUD, configuration reads	Synchronous REST	Immediate feedback; simple
Analysis recompute after	Event (`score_entry.changed`) + worker; dashboard reads	Keeps writes fast; fine at

Interaction	Style	Reason
score change	latest snapshot; synchronous recompute fallback if snapshot older than 10 s on dashboard load	early scale
OCR extraction	Async job + polling (or server-sent events)	5–60 s latency, retries, cost control
Reports (monthly, career PDF)	Scheduled/async jobs	Generated from snapshots; heavy rendering
Notifications	Async via outbox table	Reliability, caps, quiet hours
Academy completion events → recommendations	Event consumption if Academy emits events; otherwise periodic pull	Depends on OD-05
Entitlement checks	Synchronous with short cache (60 s)	Correctness on paywalls

Event-driven design is justified only where latency, retries or fan-out require it. Use a transactional **outbox** pattern in the module database so events are not lost; a heavyweight message broker is unnecessary at launch — a database-backed queue or the job system already used by Bitdoin (if any) suffices (**Unverified — requires confirmation**).

24.6 Monthly score workflow

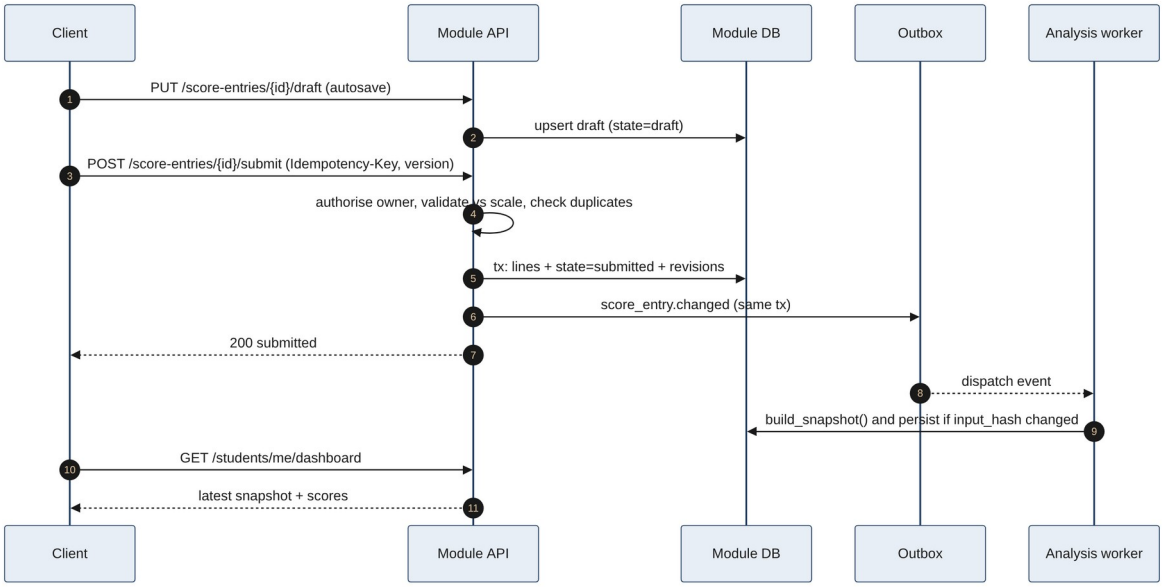


Figure 10: 24.6 Monthly score workflow

24.7 OCR workflow

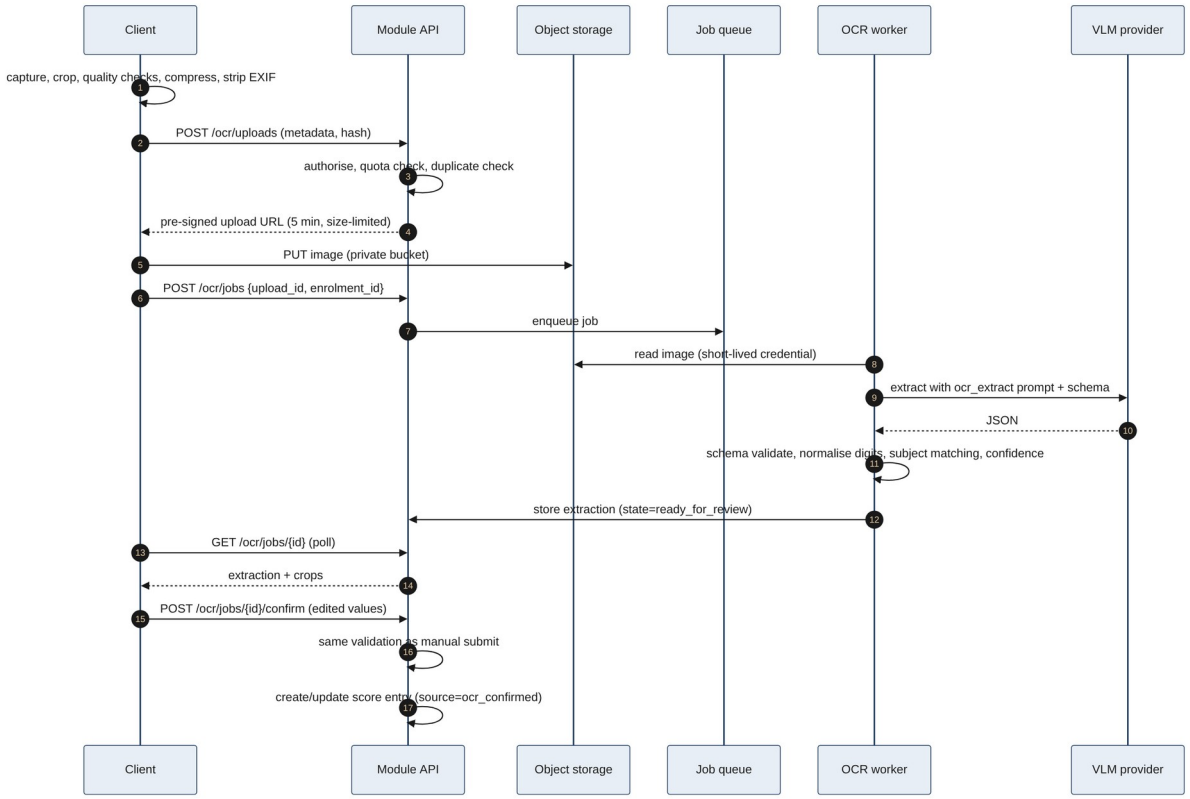


Figure 11: 24.7 OCR workflow

24.8 Academic analysis pipeline

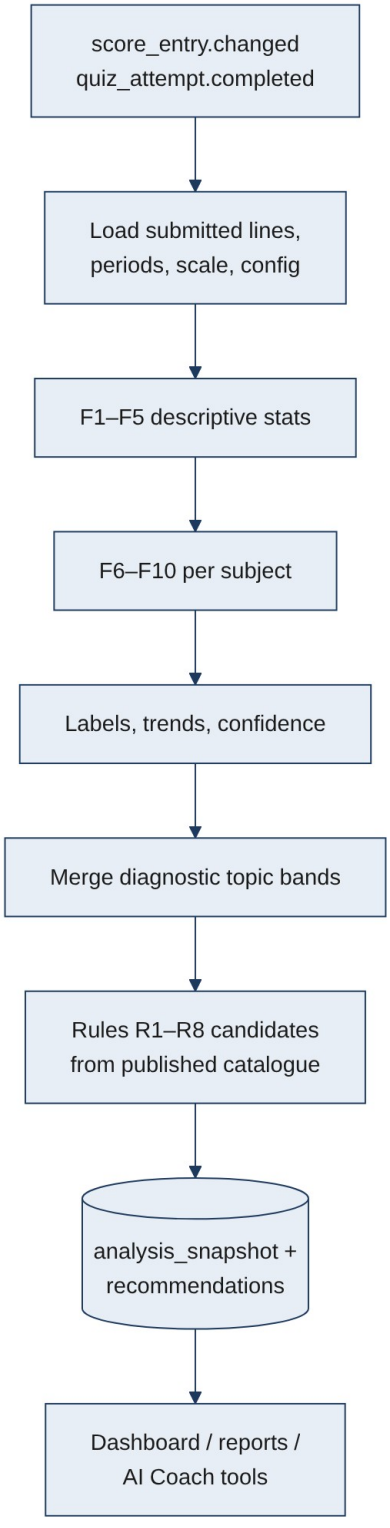


Figure 12: 24.8 Academic analysis pipeline

24.9 Career recommendation pipeline

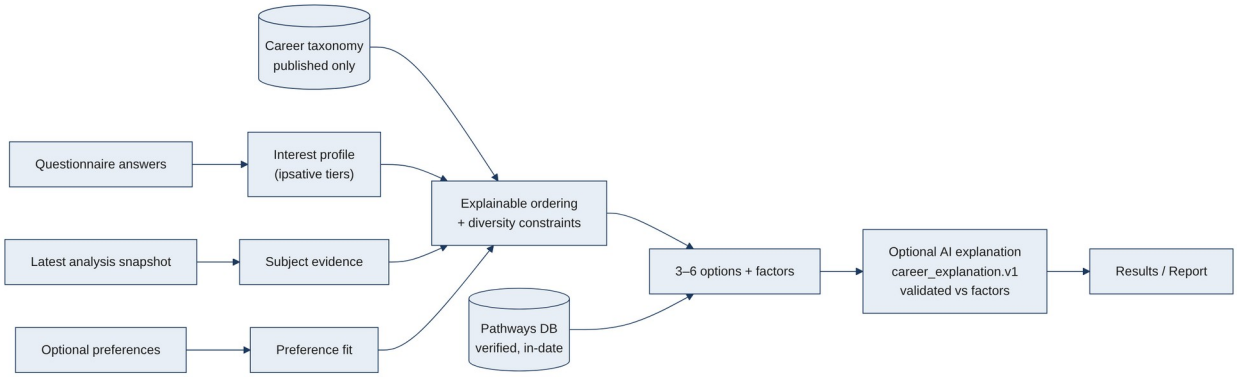


Figure 13: 24.9 Career recommendation pipeline

24.10 Parent authorisation flow

See sequence diagram in Section 14.9.1. Server-side policy: `can_read(guardian, student, scope) = link.state == active AND scope ∈ link.scopes AND NOT link.suspended AND student.consent_ok`.

24.11 Cross-cutting components

Component	Requirement
Authentication	Reuse Bitdoin sessions/tokens via Identity adapter; no separate login
Authorisation	Central policy engine; deny by default; object ownership checks on every query; 404 on unauthorised object access
File storage	Private bucket; server-side encryption; pre-signed URLs ≤ 5 min; content-type and size limits; malware scanning optional (images only); lifecycle rules for deletion
Background jobs	Idempotent jobs, retry with exponential backoff (max 3), dead-letter queue, job status table
Report generation	HTML → PDF rendering in worker; Lao font embedded; stored for 12 months; re-generated on demand
Notifications	Outbox → sender with caps/quiet hours; provider adapters
Logging & monitoring	Structured JSON logs without PII (IDs hashed), metrics per route/job, traces for OCR/AI calls, alerting on SLO breaches
Secrets	Stored in platform secret manager; AI keys scoped per environment
Feature flags	Gradual rollout and kill switches for OCR and AI

24.12 Decisions dependent on codebase access

1. Where module tables live (same DB/schema vs separate) and whether DB-level row security exists.
2. How to read the session and user ID (cookie, token, framework middleware).
3. Whether the AI Coach can host tool-calling academic mode (OD-08).
4. How to read Academy content and completion events (OD-05) and Bookstore catalogue/offers (OD-06).
5. How subscriptions are stored and activated (OD-11).
6. Job/queue and file-storage capabilities available in current hosting.
7. Front-end framework and component library for UI reuse.

25. Bitdoin Integration Architecture

25.1 User perspective

- The student never leaves “Bitdoin”: My Scores appears inside Academy, uses the same header, language toggle and account menu.
- Recommendations open existing Academy lessons and challenges in place and return to My Scores.
- “Ask the Coach” opens the existing AI Coach already aware of their scores (when consented).
- Career results open the same Career Explorer pages with added Lao programme information.
- Books open the existing Bookstore product page; checkout and delivery are unchanged.
- Premium is the same Academy membership; the membership page lists the new benefits.

25.2 Integration adapters

Adapter	Direction	Contract (proposed)	Fallback
Identity	Core → Module	`resolve_session(request) → {user_ref, locale, roles}`	None — required
Language	Core ↔ Module	read/write existing preference	Default Lao
Academy content	Academy → Module	`list_content(filter)`, `get_content(id)`, events `academy.lesson_completed`, `academy.challenge_completed`	Manual metadata import by admins; self-reported completion
XP/challenges	Module → Academy	`award_xp(user_ref, amount, reason, idempotency_key)` (if approved)	Disabled
AI Coach	Module ↔ Coach	Deep link with `context=scores`; tool endpoints exposed by Module and callable only by the Coach backend with service credentials and user context	Embedded assistant
Career Explorer	Explorer ↔ Module	Shared career IDs; Module adds `career_programme_links`	Module shows its own detail page linking out
Bookstore	Bookstore → Module	`search_books(subject/topic tags)`, `get_book(id)`, `get_offers(id)`	Hide book recommendations
Subscriptions	Core → Module	`get_subscription(user_ref) → {plan, status, period_end}`; webhook/event on change	Manual entitlement grants
Payments	Module → Core	Create order for one-time product via existing checkout; confirmation callback/event	Manual activation by Financial Administrator

25.3 Data ownership

Data	System of record
User account, credentials, language preference	Bitdoin core
Subscriptions and payments	Bitdoin core (Academy membership / checkout)
Academy lessons, challenges, XP, AI Coach conversations (non-academic)	Academy
Books, offers, orders	Bookstore
Academic profiles, enrolments, scores, revisions, OCR, analysis, quizzes, plans, guardian links, consents, career results, pathways DB	**Academic module**
Content metadata linking Academy/Bookstore items to subjects/topics	Academic module (references external IDs)

25.4 Avoiding duplication

The Module must not build: a second login, a second language setting, a second chatbot UI, a second career catalogue, a second subscription system or a second payment flow (principle P-10).

25.5 Bookstore recommendation integration

Only Education-category (or admin-tagged) titles with ≥ 1 in-stock offer are eligible (LEARN-006). Price displayed as the existing price-comparison range. No recommendation of a book to a student under the legal threshold appears as a purchase prompt in notifications (child-safe marketing, Section 29).

25.6 Discovery checklist (PM-001)

#	Question	Answer (to be completed)
D1	Repository location, framework(s), language(s)	
D2	Hosting/runtime, environments, CI/CD	
D3	Auth method, session storage, user ID type	
D4	Database engine/version, migrations tool, backups	
D5	Existing role/permission model	
D6	Academy content model and IDs; completion tracking	
D7	AI Coach provider, prompts, safety, logs	
D8	Career Explorer data source and editorial process	
D9	Subscription tables, manual activation workflow	
D10	Payment confirmation process (QR/bank transfer)	
D11	Object storage availability	
D12	Job/queue system	
D13	Analytics and monitoring tools	
D14	Privacy policy, terms, minimum age, cookie consent	
D15	Edge/CDN configuration (rate limiting, WAF)	

26. Database Schema and ER Diagram

26.1 Conventions

- Logical relational model; types expressed generically (map to the actual engine after OD-01).
- Primary keys: `UUID` (v7 recommended for index locality). No sequential IDs exposed.
- All tables include `created_at TIMESTAMPTZ NOT NULL`, `updated_at TIMESTAMPTZ NOT NULL`; mutable domain tables include `version INT NOT NULL` for optimistic concurrency.
- Soft deletion is **not** used for personal data; deletion jobs hard-delete or anonymise (PRIV-005). Configuration tables use `status`/`archived_at`.
- All module tables live in a dedicated schema/namespace (e.g. `academic.`).
- The Module stores `bitdoin_user_ref` (opaque reference to the existing user ID); **no copy of account credentials or contact details**.

26.2 ER diagrams

The model is shown in three views so each remains readable on a page. Entities appearing in more than one view are the same table.

View 1 — Profile, configuration, scores and OCR

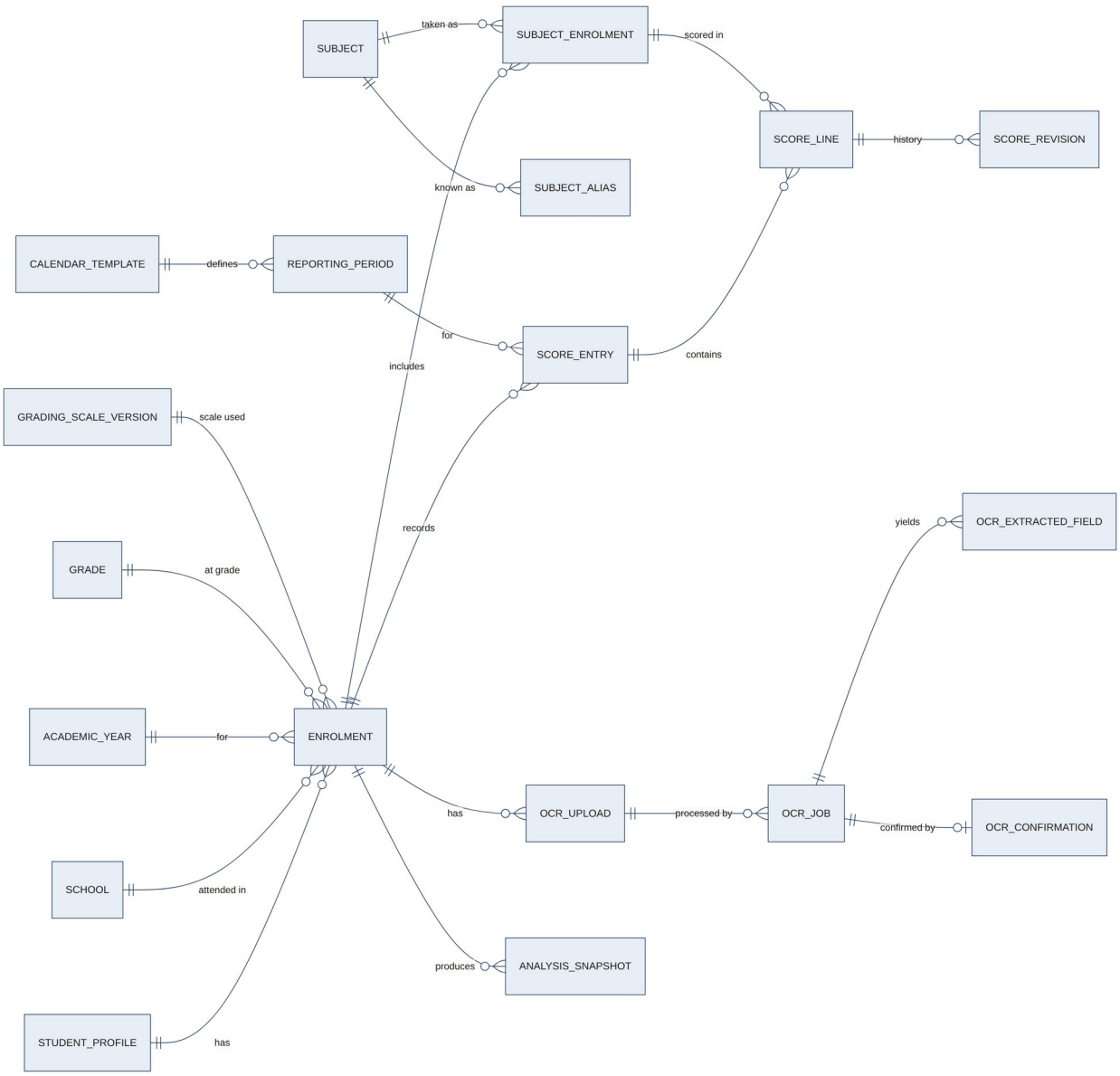


Figure 14: 26.2 ER diagrams — View 1 — Profile, configuration, scores and OCR

View 2 — Learning, planning, guardians, consent and entitlements

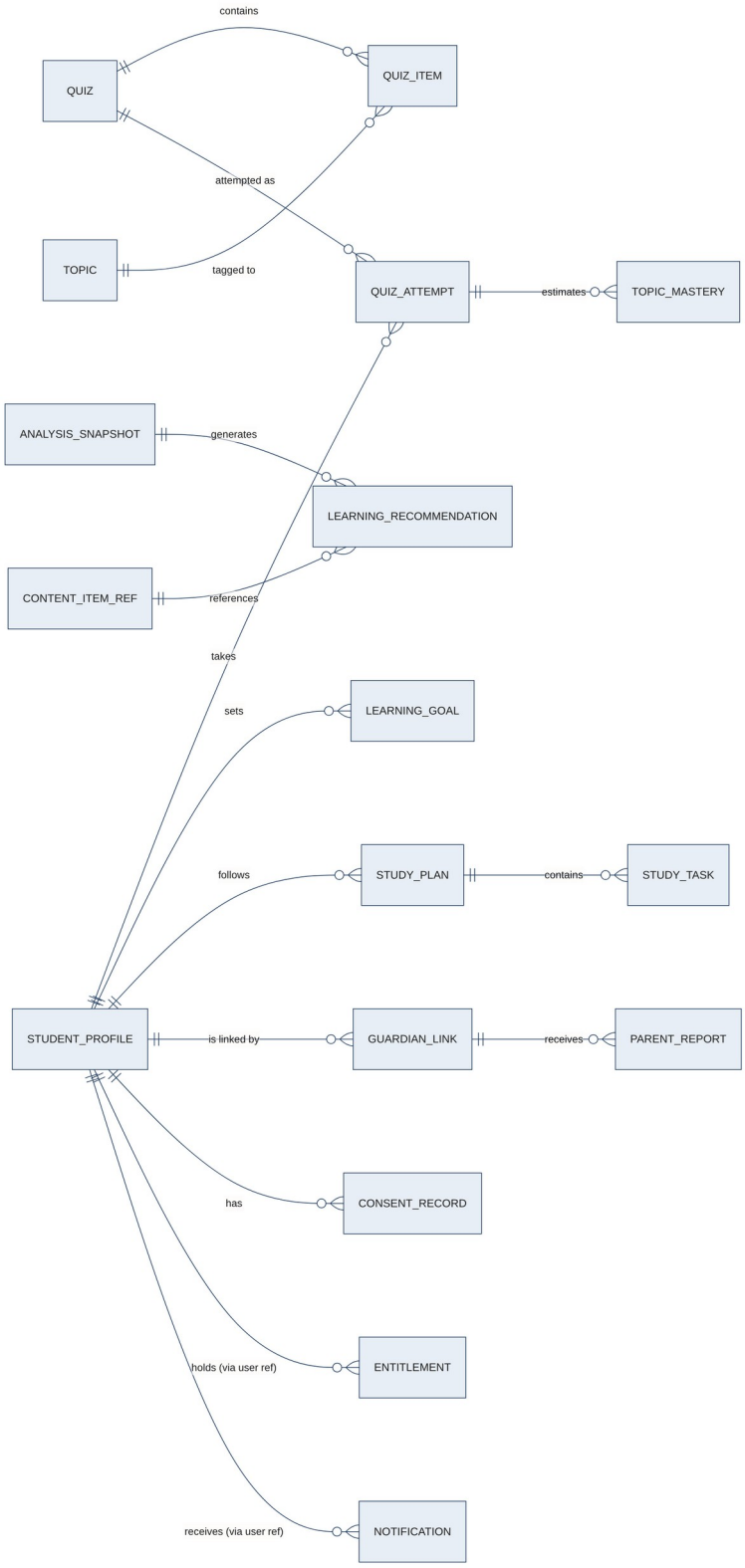


Figure 15: 26.2 ER diagrams — View 2 — Learning, planning, guardians, consent and entitlements

View 3 — Careers and pathways

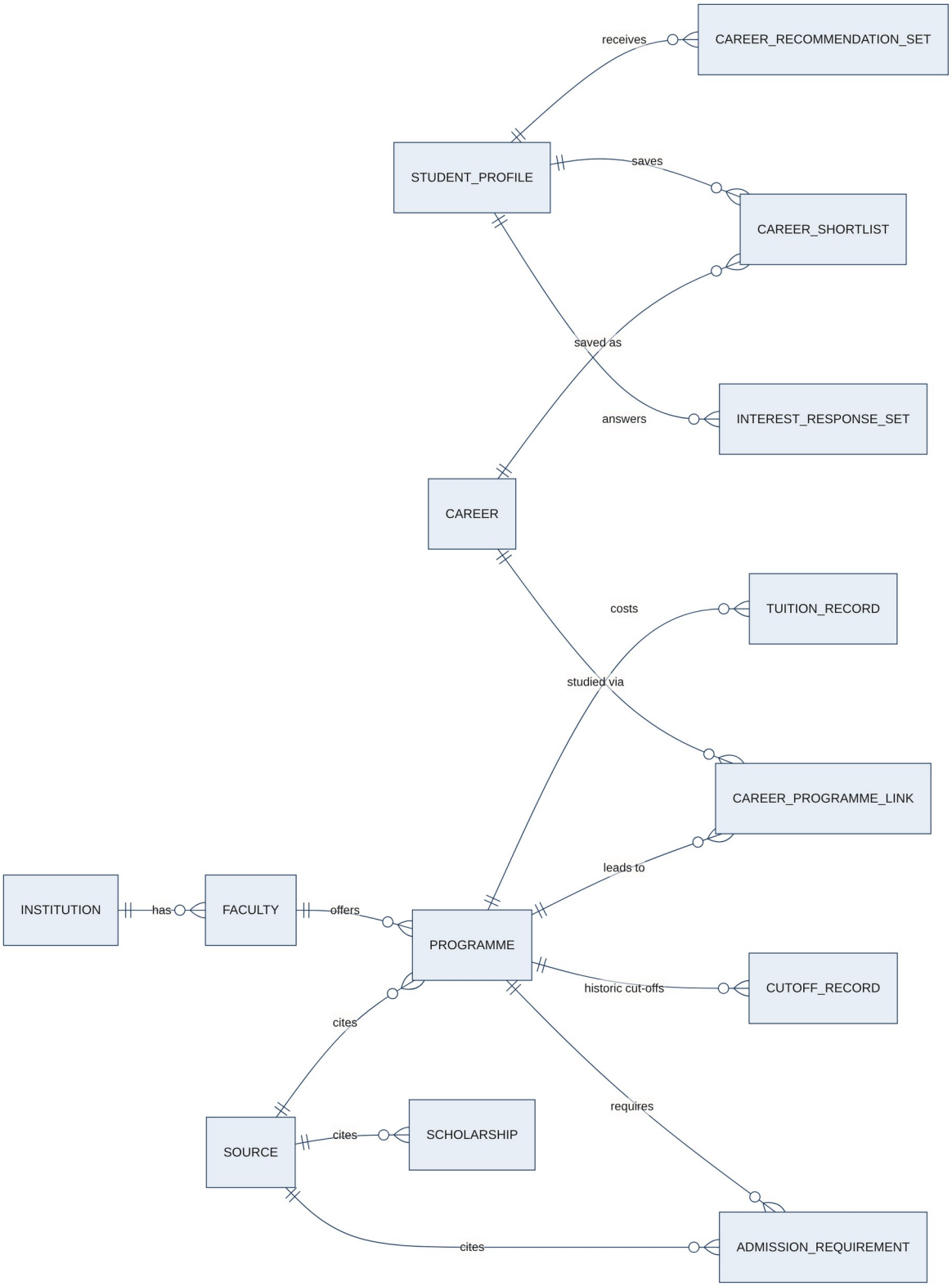


Figure 16: 26.2 ER diagrams — View 3 — Careers and pathways

26.3 Table specifications — identity, profile and configuration

student_profile — one per Bitdoin user who uses the Module as a student.

Field	Type	Null	Notes
id	UUID	PK	
bitdoin_user_ref	VARCHAR(64)	NOT NULL, UNIQUE	Opaque reference to existing

Field	Type	Null	Notes
			account
display_name	VARCHAR(40)	NOT NULL	
age_band	ENUM(u13, 13_15, 16_17, 18_plus)	NOT NULL	Re-confirmed yearly
consent_state	ENUM(pending_guardian, active, withdrawn)	NOT NULL	Derived from consent_record
province_code	CHAR(4)	NOT NULL	
district_code	VARCHAR(8)	NULL	
goals	JSONB	NULL	≤ 3 chips + text
interest_subject_codes	TEXT[]	NULL	
status	ENUM(active, read_only, deletion_pending)	NOT NULL	
deletion_requested_at	TIMESTAMPTZ	NULL	

Indexes: UNIQUE(bitdoin_user_ref). Retention: until deletion; deletion cascades per PRIV-005.

education_structure (id, code UNIQUE e.g. `LA-5-4-3`, name_lo, name_en, effective_from_year, notes). **grade** (id, education_structure_id FK, code e.g. `G11`, ordinal INT, level ENUM(lower_secondary, upper_secondary), label_lo, label_en; UNIQUE(education_structure_id, code)).

school

Field	Type	Null	Notes
id	UUID	PK	
name_lo / name_en	VARCHAR(160)	name_lo NOT NULL	
romanised_name	VARCHAR(160)	NULL	For search
province_code, district_code	VARCHAR	province NOT NULL	
school_type	ENUM(public, private, other)	NULL	
status	ENUM(active, merged, archived)	NOT NULL	
merged_into_id	UUID	NULL FK school	
calendar_template_id	UUID FK	NULL	Override
default_scale_id	UUID FK grading_scale	NULL	

Indexes: (province_code, district_code); trigram/full-text on names.

academic_year (id, code UNIQUE `2026-2027`, start_date, end_date, rollover_at, status). **calendar_template** (id, name, academic_year_id FK, scope ENUM(national, province, school), province_code NULL, school_id NULL).

reporting_period

Field	Type	Null	Notes
id	UUID	PK	
calendar_template_id	UUID FK	NOT NULL	
period_type	ENUM(monthly, term_official, semester_official, annual_official, exam)	NOT NULL	
term_no	SMALLINT	NULL	
sequence	SMALLINT	NOT NULL	Ordering within year
label_lo / label_en	VARCHAR(40)	NOT NULL	
start_date / end_date	DATE	NOT NULL	

UNIQUE(calendar_template_id, period_type, sequence).

subject (id, code UNIQUE, name_lo, name_en, grade_min, grade_max, status). **subject_alias** (id, subject_id FK, alias_text, alias_normalised, school_id NULL FK, source ENUM(admin, ocr_learned); UNIQUE(alias_normalised, school_id)).

grading_scale (id, code UNIQUE, name). **grading_scale_version**

Field	Type	Null	Notes
id	UUID	PK	
grading_scale_id	UUID FK	NOT NULL	
version	INT	NOT NULL	UNIQUE(grading_scale_id, version)
scale_family	ENUM(numeric_linear, ordinal, pass_fail)	NOT NULL	
min_value / max_value / step	NUMERIC(6,2)	NULL for ordinal	
decimals	SMALLINT	NOT NULL	
pass_mark	NUMERIC(6,2)	NULL	
normalisable	BOOLEAN	NOT NULL	
ordinal_levels	JSONB	NULL	For ordinal scales
status	ENUM(draft, approved, retired)	NOT NULL	Only approved assignable
approved_by, approved_at		NULL	

enrolment

Field	Type	Null	Notes
id	UUID	PK	
student_profile_id	UUID FK	NOT NULL	
academic_year_id	UUID FK	NOT NULL	
grade_id	UUID FK	NOT NULL	
school_id	UUID FK	NULL	Either school_id or school_free_text
school_free_text	VARCHAR(80)	NULL	
grading_scale_version_id	UUID FK	NOT NULL	
calendar_template_id	UUID FK	NOT NULL	Resolved at creation
start_period_seq / end_period_seq	SMALLINT	start NOT NULL	Transfers
is_repeat	BOOLEAN	NOT NULL default false	
state	ENUM(active, closed)	NOT NULL	
closed_reason	ENUM(promoted, repeated, transferred, left, year_end)	NULL	

Constraint: at most one `active` enrolment per student (partial unique index). Index (student_profile_id, academic_year_id).

subject_enrolment (id, enrolment_id FK, subject_id FK NULL, custom_name VARCHAR(60) NULL, start_period_seq, end_period_seq NULL, display_order; UNIQUE(enrolment_id, subject_id) where subject_id not null).

26.4 Scores and OCR

score_entry

Field	Type	Null	Notes
id	UUID	PK	
enrolment_id	UUID FK	NOT NULL	
reporting_period_id	UUID FK	NOT NULL	
state	ENUM(draft, submitted)	NOT NULL	
submitted_at	TIMESTAMPTZ	NULL	
version	INT	NOT NULL	Optimistic lock
source	ENUM(manual, ocr_confirmed, school_import, mixed)	NOT NULL	
ocr_job_id	UUID FK	NULL	
idempotency_key	VARCHAR(64)	NULL	UNIQUE per enrolment

UNIQUE(enrolment_id, reporting_period_id) — one entry per period (SCORE-009).

score_line

Field	Type	Null	Notes
id	UUID	PK	
score_entry_id	UUID FK	NOT NULL	
subject_enrolment_id	UUID FK	NOT NULL	
raw_value	NUMERIC(6,2)	NULL	NULL when status ≠ scored
ordinal_code	VARCHAR(16)	NULL	Ordinal scales
status	ENUM(scored, missing, not_applicable, exempt)	NOT NULL	
grading_scale_version_id	UUID FK	NOT NULL	Copied from enrolment at time of entry
normalised_value	NUMERIC(6,4)	NULL	Derived when normalisable
source	ENUM(manual, ocr_confirmed, school_import)	NOT NULL	
verification	ENUM(self_reported, guardian_acknowledged, school_verified)	NOT NULL default self_reported	
evidence_stream	ENUM(school_score)	NOT NULL	Future: separated from quiz/mock streams
is_current	BOOLEAN	NOT NULL	

UNIQUE(score_entry_id, subject_enrolment_id) WHERE is_current. CHECK (status='scored') = (raw_value IS NOT NULL OR ordinal_code IS NOT NULL).

score_revision (append-only): id, score_line_id FK, old_value, new_value, old_status, new_status, actor_ref, actor_role ENUM(student, admin, system), reason VARCHAR(120) NULL, source, created_at. Retention: life of score line.

ocr_upload: id, enrolment_id FK, storage_key, sha256, phash BIGINT, bytes INT, width, height, mime, retention_until TIMESTAMPTZ, deleted_at NULL. Index (enrolment_id, phash). **ocr_job**: id, ocr_upload_id FK, state ENUM(queued, processing, ready_for_review, confirmed, failed, abandoned), model_id, prompt_version, attempts, error_code, processing_ms, cost_micro_usd, created_at, completed_at. **ocr_extracted_field**: id, ocr_job_id FK, period_label_raw, subject_raw, subject_code_suggested, subject_match_confidence, value_raw, value_parsed, value_confidence, struck_through, bbox JSONB. **ocr_confirmation**: id, ocr_job_id FK UNIQUE, score_entry_id FK, fields_total, fields_edited, time_to_confirm_ms, confirmed_at, diff JSONB (original vs confirmed per field).

Retention: images per OCR-011 (30 days default); extracted fields and confirmation diffs 24 months for quality (with names never stored).

26.5 Analysis, learning and planning

analysis_config (id, version UNIQUE, thresholds JSONB, active_from). **analysis_snapshot** (id, enrolment_id FK, engine_version, config_version, input_hash CHAR(64), outputs JSONB, created_at; INDEX(enrolment_id, created_at DESC); UNIQUE(enrolment_id, input_hash)).

topic (id, subject_id FK, code, name_lo, name_en, grade_min, grade_max, parent_topic_id NULL). **quiz** (id, subject_id, grade_band, kind ENUM(diagnostic, practice, mock), blueprint JSONB, version, status). **quiz_item** (id, quiz_id, topic_id, stem, options JSONB, answer_key (encrypted at rest), explanation, author_ref, licence, status, version). **quiz_attempt** (id, student_profile_id, quiz_id, quiz_version, started_at, completed_at, state, responses JSONB). **topic_mastery** (id, quiz_attempt_id, topic_id, correct INT, total INT, band ENUM(secure, developing, needs_practice, insufficient)).

content_item_ref (id, system ENUM(academy_lesson, academy_challenge, academy_check, module_quiz, bookstore_book, routine), external_id VARCHAR(64), subject_codes TEXT[], grade_min, grade_max, topic_ids UUID[], difficulty SMALLINT, objectives JSONB, language, est_minutes, quality_score, status; UNIQUE(system, external_id)).

learning_recommendation (id, student_profile_id, snapshot_id, content_item_ref_id, rule_id, reason_params JSONB, rank, state ENUM(shown, opened, started, done, dismissed_not_now, not_helpful), state_changed_at, impressions INT).

learning_goal (id, student_profile_id, subject_id, baseline, target, target_period_id, feasibility_label, status). **study_plan** (id, student_profile_id, goal_ids UUID[], days_mask BIT(7), minutes_per_day, paused_until NULL, status). **study_task** (id, study_plan_id, scheduled_date, content_item_ref_id NULL, instruction_text NULL, est_minutes, state ENUM(scheduled, done, skipped, moved), moved_count).

26.6 Guardians, consent, notifications, entitlements

guardian_link

Field	Type	Null	Notes
id	UUID	PK	
student_profile_id	UUID FK	NOT NULL	
guardian_user_ref	VARCHAR(64)	NULL until accepted	
relationship	ENUM(mother, father, guardian, other)	NULL	
invite_code_hash	CHAR(64)	NOT NULL	Code stored hashed
invite_expires_at	TIMESTAMPTZ	NOT NULL	
scopes	TEXT[]	NOT NULL	
state	ENUM(invited, accepted_pending_student, active, removed, suspended, paused_majority, expired)	NOT NULL	
initiated_by	ENUM(student, guardian_consent_flow)	NOT NULL	
state_reason	VARCHAR(80)	NULL	

UNIQUE(student_profile_id, guardian_user_ref) WHERE state IN (active, accepted_pending_student). Limits enforced in service (≤ 4 guardians per student; ≤ 6 children per guardian).

consent_record (append-only): id, subject_profile_id, grantor_user_ref, grantor_role ENUM(student, guardian), purpose ENUM(module_processing, guardian_sharing, ocr_processing, ai_processing, email, push, research_samples, keep_photos), scope JSONB, notice_version, method ENUM(in_app_checkbox, guardian_link_flow), granted_at, withdrawn_at NULL.

parent_report (id, guardian_link_id, snapshot_id, period_id, generated_at, content JSONB, pdf_storage_key NULL, opened_at NULL, updated_since_generation BOOLEAN).

notification (id, recipient_user_ref, code, payload JSONB (no scores for push), channel, state ENUM(pending, sent, suppressed_cap, suppressed_quiet, failed), scheduled_for, sent_at). **notification_preference** (user_ref, category, channel, enabled, quiet_start, quiet_end).

entitlement (id, user_ref, feature_code, source ENUM(academy_premium, family_plan, one_time_purchase, manual_grant, trial), starts_at, ends_at, reference_id, granted_by NULL). **usage_counter** (user_ref, feature_code, period_key `YYYY-MM`, used INT, limit INT; PK(user_ref, feature_code, period_key)).

audit_event (append-only, separate retention): id, occurred_at, actor_ref, actor_role, action, target_type, target_id, student_profile_id NULL, ip_hash, user_agent_hash, outcome, reason NULL. Index (student_profile_id, occurred_at), (actor_ref, occurred_at).

26.7 Careers and pathways

career (id, slug UNIQUE — shared with existing Career Explorer IDs, name_lo, name_en, cluster, primary_area, secondary_area, related_subject_codes, typical_study_years, work_settings, status). **interest_item** (id, code, text_lo, text_en, areas JSONB, version). **interest_response_set** (id, student_profile_id, questionnaire_version, responses JSONB, area_scores JSONB, completed_at). **career_recommendation_set** (id, student_profile_id, snapshot_id NULL, response_set_id, ranking_version, options JSONB (career_id + factors), created_at). **career_shortlist** (student_profile_id, career_id, saved_at; PK both).

source (id, name, url, source_type ENUM(official_site, official_document, official_news, institution_confirmation, secondary), publisher, published_on NULL, accessed_on).

institution (id, name_lo, name_en, type ENUM(public_university, private_institution, college, tvet, other), province_code, website_url, source_id, verified_on, review_by, status). **faculty** (id, institution_id, name_lo, name_en, status). **programme** (id, faculty_id, name_lo, name_en, degree_level ENUM(associate, bachelor, master, other), duration_years NUMERIC(3,1), language, mode, entrance_group ENUM(A, B, other, unknown), source_id, verified_on, review_by, applicable_academic_year, publication_status, verified_by).

admission_requirement (id, programme_id, academic_year, requirement_type ENUM(entrance_exam, subject_minimum, document, quota, other), description_lo, description_en, source_id NOT NULL, verified_on NOT NULL, review_by NOT NULL, publication_status). **cutoff_record** (id, programme_id, academic_year NOT NULL, value_text, source_id NOT NULL, verified_on). **tuition_record** (programme_id, academic_year, amount_lak NULL, notes, source_id NOT NULL, verified_on). **application_window** (programme_id or institution_id, academic_year, opens_on, closes_on, source_id, verified_on). **scholarship** (id, name, provider, eligibility_text, deadline, url, source_id, verified_on, review_by, status). **career_programme_link** (career_id, programme_id, rationale, PK both).

Constraint pattern for time-sensitive tables: CHECK (publication_status <> 'published' OR (source_id IS NOT NULL AND verified_on IS NOT NULL AND review_by IS NOT NULL)).

26.8 Data-model concerns addressed

Concern	Design
Historical integrity	Enrolment per year; scale version copied to each score line; append-only revisions; immutable snapshots
Multiple grading systems	`grading_scale_version` with family, normalisable flag; no implicit mixing
Score updates	Optimistic locking (`version`), revisions with actor/reason
OCR provenance	`score_entry.ocr_job_id`, `score_line.source`, `ocr_confirmation.diff`
Source verification	`verification` enum; future school imports via signed agreements
Consent history	Append-only `consent_record` with notice versions
School changes	Enrolment boundaries with period sequences; annotations
Grade promotion & structure reform	Enrolment per year; `education_structure` and `grade` tables
Multi-tenant isolation	Not needed for B2C launch; if schools become tenants, add `organisation_id` to enrolment and policies — reserved column `organisation_id UUID NULL` on enrolment from Phase 1

Concern	Design
Deletion and export	Profile-rooted ownership graph enables complete export and deletion; audit/payment records minimised and retained per law
No duplicate identity store	Only `bitdoin_user_ref` and `guardian_user_ref`

27. API Requirements

27.1 Conventions

- REST over HTTPS, JSON, base path `/api/academic/v1` (adjust to existing conventions after OD-01). If Bitdoin already uses GraphQL or server actions, expose the same operations through that layer; the contracts below remain the reference.
- Authentication: existing Bitdoin session cookie or bearer token resolved by the Identity adapter. Admin endpoints additionally require an admin role and MFA-fresh session (≤ 12 h).
- **`me` routing**: student endpoints use `/students/me/...` so clients never send their own profile ID. Guardian endpoints take a `student_id` that is validated against an active link. Admin endpoints take explicit IDs with role checks. Unauthorised object access returns **404**.
- Pagination: cursor-based (`?cursor=&limit=`), max limit 50.
- Idempotency: `Idempotency-Key` header required on POST endpoints that create entries, jobs, invites or purchases; keys retained 24 h.
- Concurrency: `If-Match: <version>` on updates; mismatch → 409.
- Rate limits (defaults): 120 read req/min/user; 30 write req/min/user; OCR job creation 10/hour/user; invite creation 5/day/student; AI messages per quota. Responses include `RateLimit-*` headers; excess → 429.
- Localisation: `Accept-Language: lo|en` controls message text in errors.
- Timestamps ISO 8601 UTC; dates `YYYY-MM-DD`; periods referenced by ID and by `key` (e.g. `2026-10`).

27.2 Endpoint catalogue

#	Method & route	Purpose	AuthZ	Key validation / notes	Req
1	GET /students/me/profile	Get own profile	Student	404 if none	PROFILE-007
2	POST /students/me/profile	Create profile + first enrolment	Student	Idempotent per user	PROFILE-001
3	PATCH /students/me/profile	Update profile fields	Student	If-Match	PROFILE-007
4	GET /students/me/enrolments	List enrolments	Student		PROFILE-004
5	POST /students/me/enrolments/transition	Promote / repeat / transfer / leave	Student	Atomic; reason enum	PROFILE-004/005
6	PUT /students/me/enrolments/{id}/subjects	Set subject enrolments	Student	≤ 25; start/end periods	PROFILE-003, SCORE-013
7	GET /students/me/periods?enrolment_id=	Reporting periods with status	Student		SCORE-002
8	GET /students/me/score-entries?enrolment_id=&period_id=	Get entry (draft or submitted)	Student		SCORE-010
9	PUT /students/me/score-entries/{period_id}/draft	Upsert draft	Student	Partial allowed	SCORE-005
10	POST /students/me/score-entries/{period_id}/submit	Submit entry	Student	Idempotency-Key, If-Match, scale validation, duplicates	SCORE-006/009
11	PATCH /students/me/score-lines/{id}	Correct a submitted score	Student	reason ≤ 120; creates revision	SCORE-007
12	GET /students/me/score-lines/{id}/revisions	Revision history	Student		SCORE-007
13	GET /students/me/dashboard?	Summary, series,	Student	Uses latest snapshot	DASH-*

#	Method & route	Purpose	AuthZ	Key validation / notes	Req
	enrolment_id=	labels, recs			
14	GET /students/me/analysis/subjects/{code}	Subject detail incl. evidence	Student		DASH-003
15	POST /students/me/ocr/uploads	Request upload URL	Student + quota	mime, bytes ≤ 15 MB, sha256, phash; duplicate hint	OCR-001/008
16	POST /students/me/ocr/jobs	Start extraction	Student	Idempotency-Key	OCR-004
17	GET /students/me/ocr/jobs/{id}	Job status + extraction	Student		OCR-004
18	POST /students/me/ocr/jobs/{id}/confirm	Confirm reviewed values	Student	All highlighted touched; same validation as #10	OCR-006
19	DELETE /students/me/ocr/uploads/{id}	Delete image	Student		OCR-011
20	GET /quizzes?subject=&grade=	Available diagnostics	Student	published only	QUIZ-001
21	POST /students/me/quiz-attempts	Start attempt	Student		QUIZ-001
22	PUT /students/me/quiz-attempts/{id}/responses	Save answers	Student		QUIZ-001
23	POST /students/me/quiz-attempts/{id}/complete	Score attempt	Student		QUIZ-002
24	GET /students/me/recommendations	Current recommendations	Student	max 3 primary + list	LEARN-001
25	POST /students/me/recommendations/{id}/feedback	done / not_now / not_helpful	Student		LEARN-005
26	GET/POST/PATCH /students/me/goals	Goals	Student + entitlement	feasibility computed	PLAN-001
27	GET/PUT /students/me/plan	Plan & commitment	Student + entitlement		PLAN-002/006
28	PATCH /students/me/plan/tasks/{id}	Update task state	Student		PLAN-004
29	GET /careers/questionnaire	Items (current version)	Student		CAREER-001
30	PUT /students/me/career/responses	Save answers	Student		CAREER-001
31	POST /students/me/career/recommendations	Generate options	Student	Requires completed responses	CAREER-002
32	POST /students/me/career/reports	Generate report	Student/guardian + entitlement	Idempotency-Key	CAREER-006
33	GET /pathways/programmes?field=&group=&province=&institution=	Search programmes	Any signed-in user	published, in-date flags	UNI-003
34	GET /pathways/programmes/{id}	Programme detail with provenance	Any signed-in user		UNI-002
35	POST /students/me/guardian-invites	Create invite	Student	5/day; scopes	PARENT-001
36	POST /guardian/invites/accept	Accept with code	Guardian	code hash match, expiry	PARENT-002
37	POST /students/me/guardian-links/{id}/confirm	Student confirms	Student	within 7 days	PARENT-002
38	PATCH /students/me/guardian-links/{id}	Change scopes / remove	Student		PARENT-004/007
39	DELETE /guardian/links/{id}	Guardian removes link	Guardian		PARENT-007
40	GET /guardian/children	Linked children summary	Guardian	active links only	PARENT-005

#	Method & route	Purpose	AuthZ	Key validation / notes	Req
41	GET /guardian/children/{student_id}/dashboard	Child view (scoped)	Guardian	scope filter server-side	PARENT-005
42	GET /guardian/children/{student_id}/reports	Reports	Guardian + entitlement		PARENT-006
43	GET /me/entitlements	Active entitlements and quotas	Any		BILLING-001
44	POST /me/purchases	One-time product order via existing checkout	Any	Idempotency-Key	BILLING-004
45	POST /students/me/export	Request export	Student	1 active request	PRIV-004
46	POST /students/me/deletion	Request deletion	Student	grace period	PRIV-005
47	DELETE /students/me/deletion	Cancel deletion	Student	within grace	PRIV-005
48	GET /me/consents · POST /me/consents · DELETE /me/consents/{purpose}	Manage consents	Any	append-only	PRIV-003
49	Admin: /admin/subjects, /admin/scales, /admin/schools, /admin/calendars, /admin/content, /admin/careers, /admin/pathways, /admin/entitlements, /admin/audit	Configuration & operations	Admin roles per matrix	MFA; audit every write	ADMIN-*
50	Internal: /internal/coach-tools/*	Tool endpoints for AI Coach	Service credential + user context	Bound to session student	AI-002

27.3 Error contract

```
{
  "error": {
    "code": "SCORE_OUT_OF_RANGE",
    "message": "Score must be between 0 and 10 in steps of 0.5.",
    "details": [{"field": "lines[3].value", "min": 0, "max": 10, "step": 0.5, "received":
11}],
    "request_id": "req_01JAX8...",
    "docs": "https://bitdoin.store/help/scores#ranges"
  }
}
```

HTTP	Code examples
400	VALIDATION_FAILED, SCORE_OUT_OF_RANGE, PERIOD_IN_FUTURE, SUBJECT_LIMIT
401	UNAUTHENTICATED
403	FORBIDDEN_ROLE (admin only), ENTITLEMENT_REQUIRED, CONSENT_REQUIRED
404	NOT_FOUND (also used for unauthorised object access)
409	SCORE_DUPLICATE, VERSION_CONFLICT, INVITE_ALREADY_USED, LINK_EXISTS
410	INVITE_EXPIRED
413	UPLOAD_TOO_LARGE
422	OCR_NOT_CONFIRMABLE (unreviewed highlighted fields), QUESTIONNAIRE_INCOMPLETE
429	RATE_LIMITED, QUOTA_EXCEEDED
503	AI_UNAVAILABLE, OCR_UNAVAILABLE (with Retry-After)

27.4 Representative requests and responses

Submit a period (#10)

```
POST /api/academic/v1/students/me/score-entries/rp_2026_10/submit
Idempotency-Key: 6f1c2f0e-7b0a-4d7e-9a1e-3c5d2a9b1e11
```

```
If-Match: 3
Content-Type: application/json
{
  "enrolment_id": "enr_01JA...",
  "lines": [
    {"subject_enrolment_id": "se_math", "status": "scored", "value": 7},
    {"subject_enrolment_id": "se_phys", "status": "scored", "value": 6},
    {"subject_enrolment_id": "se_chem", "status": "scored", "value": 8},
    {"subject_enrolment_id": "se_bio",  "status": "scored", "value": 9},
    {"subject_enrolment_id": "se_lao",  "status": "scored", "value": 8},
    {"subject_enrolment_id": "se_eng",  "status": "scored", "value": 7},
    {"subject_enrolment_id": "se_hist", "status": "missing"}
  ]
}
{
  "score_entry": {
    "id": "se_01JB...",
    "period": {"id": "rp_2026_10", "key": "2026-10", "label": "October 2026"},
    "state": "submitted",
    "version": 4,
    "source": "manual",
    "lines": [
      {"id": "sl_1", "subject_code": "MATH", "status": "scored", "value": 7, "scale": {"min":
0, "max": 10, "step": 0.5},
       "source": "manual", "verification": "self_reported", "edited": false}
    ]
  },
  "summary": {"period_mean": 7.5, "coverage": {"recorded": 6, "active": 7},
"change_vs_previous": 0.3},
  "analysis_status": "updating"
}
```

OCR job status (#17)

```
{
  "job": {"id": "ocr_01JB...", "state": "ready_for_review", "processing_ms": 8420},
  "periods": [{"guess": "2026-10", "confidence": 0.91, "selected": true}],
  "fields": [
    {"id": "f1", "subject_code": "MATH", "subject_confidence": 1.0, "value": 7,
"value_confidence": 0.95, "state": "clear", "crop_url": "https://…/crop/f1?sig=…"},
    {"id": "f2", "subject_code": "PHYS", "subject_confidence": 0.88, "value": 6.5,
"value_confidence": 0.62, "state": "highlighted", "crop_url": "https://…/crop/f2?sig=…"},
    {"id": "f3", "subject_code": null, "subject_raw": "ฟาสา…", "value": 8, "value_confidence":
0.9, "state": "needs_subject"}
  ],
  "warnings": ["struck_through_value_detected"]
}
```

Guardian invite (#35)

```
// request
{"scopes": ["scores", "plan_progress", "recommendations"]}
// response
{"invite": {"id": "gi_01JC...", "code": "K7Q4-M2XP", "link":
"https://bitdoin.store/academy/family/join?c=K7Q4M2XP", "expires_at": "2026-10-26T08:00:00Z",
"scopes": ["scores", "plan_progress", "recommendations"]}}
```

Programme detail (#34)

```
{
  "programme": {
    "id": "prg_…", "name_en": "…", "name_lo": "…",
    "institution": {"id": "ins_nuol", "name_en": "National University of Laos"},
    "degree_level": "bachelor", "entrance_group": "A",
    "facts": [
      {"field": "entrance_group", "value": "A", "source": {"name": "…", "url": "…", "type":
"official_document"},
       "verified_on": "2026-11-02", "applicable_academic_year": "2026-2027", "review_by":
"2027-06-30", "status": "current"}
    ]
  }
}
```

Entitlements (#43)

```
{"plan": "academy_premium", "status": "active", "period_end": "2026-11-23T16:59:59Z",
 "features": {"advanced_analysis": true, "parent_reports": true, "career_report": false},
```

```
"quotas": {"ocr_uploads": {"used": 2, "limit": 20, "resets_at": "2026-11-01T00:00:00+07:00"},
           "coach_academic_messages": {"used": 14, "limit": 300}}}
```

27.5 Anti-IDOR measures

1. No client-supplied student IDs on student routes (`me` routing).
2. Every repository query includes the owner predicate from the auth context (enforced by a data-access helper and tested).
3. Guardian routes verify an active link and filter fields by scope in the serializer.
4. IDs are non-sequential UUIDs; responses never expose `bitdoin_user_ref` of other users.
5. Automated authorisation test suite (SEC-002) runs on every build.

28. Role and Permission Matrix

Legend: **F** full · **R** read · **O** own records only · **S** scoped by student grant · **B** break-glass (reason, time-limited, audited) · **A** approve · — none.

Capability	Student	Guardian	Teacher/Tutor (future)	Super Admin	Education Content Mgr	Career Content Reviewer	Support Agent	Financial Admin
Own profile & enrolments	O (F)	—	—	B	—	—	R (status only)	—
Enter/edit own scores	O (F)	—	—	—	—	—	—	—
View scores & analysis	O	S	S (subject, time-limited)	B	—	—	B	—
View score-book images	O	—	—	B	—	—	—	—
OCR quality dashboard (aggregate)	—	—	—	R	R	—	—	—
Quiz attempts & topic results	O	S	S	B	—	—	—	—
AI Coach academic conversations	O	—	—	B	—	—	—	—
Guardian links	O (create/confirm/remove/scopes)	Own links (accept/remove)	—	F	—	—	Suspend	—
Parent reports	—	S (+entitlement)	—	B	—	—	—	—
Career results & shortlist	O	S (if career scope)	—	B	—	—	—	—
Subjects, aliases, scales	—	—	—	F	F	—	—	—
Schools & calendars	—	—	—	F	F	—	—	—
Quiz items & content metadata	—	—	Create (future, via review)	F	F	—	—	—
AI prompt versions	—	—	—	A	—	—	—	—
Career taxonomy & pathways data	—	—	—	F	—	F (four-eyes A)	—	—
Entitlements & manual activation	R (own)	R (own)	—	F	—	—	R	F
Refunds & payment records	—	—	—	R	—	—	—	F
Audit log	—	—	—	F (export)	—	—	R (own actions)	—

Capability	Student	Guardian	Teacher/Tutor (future)	Super Admin	Education Content Mgr	Career Content Reviewer	Support Agent	Financial Admin
Admin role grants	—	—	—	F	—	—	—	—
Export / delete own data	F	F (own guardian data)	—	Process	—	—	Assist	—

Rules: MFA required for all admin roles; break-glass sessions last 30 minutes and are reviewed weekly; no admin role may approve its own content; teacher access (Phase 5) requires an explicit student grant per subject with expiry ≤ 1 academic year and never grants school-wide visibility.

29. Security, Privacy, and Child Safety

29.1 Legal framework (research summary — not legal advice)

Instrument	Relevance	Status	Source
Law on Electronic Data Protection No. 25/NA (12 May 2017)	Governs collection, use, disclosure and transfer of electronic data including personal data; Ministry of Technology and Communications designated authority	In force [EXTERNAL]	S-15, S-16
Instruction on implementation of Law No. 25/NA	Practical guidance on procedures	Exists per counsel summaries; text not reviewed	S-15
Cross-border transfer	Counsel summary: data subject consent required to transfer data collected in Laos abroad	[EXTERNAL] — confirm with counsel	S-15, S-16
Penal Code No. 62/NA (17 May 2017)	Collection, use and disclosure of personal information without authorisation is an offence	[EXTERNAL — secondary summary]	S-19
New Law on Electronic Data Protection	Listed among new laws in the National Assembly's 2026–2030 legislative plan	Planned [EXTERNAL] — monitor	S-17
Draft Law on Cybersecurity	Reviewed by the National Assembly in June 2025; aims include protecting service users' personal data	Status after June 2025 **Unverified — requires confirmation**	S-18

Age of digital consent. This PRD does **not** assume any statutory age. Requirement PRIV-002 makes consent rules configurable and the MVP conservative until counsel advises (OD-07).

29.2 Questions for Lao legal counsel (LEG-001)

1. What consent is required to process academic records of students aged under 18, and at what age (if any) may a student consent alone?
2. Are school scores "specific data" requiring heightened protection under Law No. 25/NA's classification?
3. What form of consent is valid (in-app checkbox, guardian confirmation flow) and how must it be evidenced?
4. Does processing by an AI/OCR provider outside Laos constitute a cross-border transfer requiring explicit consent, and are contractual safeguards sufficient?
5. Are there data-localisation requirements for education data or for minors' data?
6. Retention: minimum/maximum periods for academic records, audit logs and payment records?
7. Breach notification obligations (authority, data subjects, timelines)?
8. Rules for marketing to minors and for in-app purchases by minors?
9. Requirements for publishing university information obtained from official sites (attribution, reuse)?
10. Implications of the forthcoming new data protection law and the cybersecurity law for this product?

29.3 Consent and authorisation model (conservative default until OD-07)

User situation	Data processing	Guardian involvement	AI/OCR processing
Student aged 18+	Student consent (PRIV-001/003)	Optional, student-invited	Separate opt-in toggles for OCR and AI processing
Student 16–17	Student consent + guardian notice offered; full guardian consent if counsel requires	Student-invited; two-sided confirmation	Opt-in; disabled by default until consent recorded
Student 13–15 (Phase 1 limited to Grades 10–12; may include 15-year-olds)	Guardian consent flow required before server-side storage (PARENT-003)	Required	Disabled until guardian consent
Under 13	Not supported in MVP	—	—

Consent purposes are separate (module processing, guardian sharing, OCR, AI, email, push, research samples, keep photos) and individually withdrawable (PRIV-003).

29.4 Required controls

Control	Specification
Data minimisation	Field list in 14.1.1; no DOB, legal name, address, phone, ethnicity, religion, income
Privacy notices	Layered Lao/English notice; versioned; child-friendly summary (≤ 150 words)
Guardian consent & linking	Two-sided confirmation; hashed single-use codes; 72 h expiry; rate limits; dispute suspension
Encrypted transport	TLS 1.2+ (prefer 1.3); HSTS; no mixed content
Secure storage	Encryption at rest for DB and object storage; field-level encryption for quiz answer keys and free-text goals; keys managed by platform KMS
Role-based authorisation	Section 28; deny by default; MFA for admins
Access logging	`audit_event` for every read of another person's data (guardian, admin), every write, every denial
Retention & deletion	Retention schedule in 29.7; deletion pipeline with verification
Export & correction	PRIV-004; students correct scores directly; profile edits self-service
Secure image uploads	Pre-signed URLs, size/type limits, EXIF stripping, private bucket, no public URLs, signed crop URLs ≤ 5 min
AI provider controls	DPA, no training on inputs, retention ≤ 30 days or zero-retention where offered, regional processing documented, prompts exclude names
Vendor assessments	Security questionnaire, certifications, sub-processors, breach history, data location (29.5)
Cross-border handling	Consent purpose `ai_processing/ocr_processing` explicitly states provider location; option to use manual entry only
Breach response	29.9
Backup & recovery	Section 33 targets; backups encrypted; deletion propagates via backup expiry ≤ 35 days
Abuse reporting	PRIV-008; safeguarding procedure 29.8
AI safety for minors	AI-004; classifiers; no persona; crisis messaging; no collection of contact details; human review of reports

29.5 Vendor assessment checklist (AI/OCR, email, push, storage)

Data processed and purpose · processing locations · sub-processors · retention and deletion · training/use of customer data · security certifications (e.g. ISO 27001/SOC 2) · encryption · access controls · incident notification terms · data processing agreement · ability to honour deletion · pricing predictability · Lao-language quality (for AI). Approval recorded in the processing register before any production data flow.

29.6 Threat model

ID	Threat	Vector	Likelihood	Impact	Mitigations	Residual
T1	Unauthorised guardian linking	Classmate or stranger obtains invite code; social engineering	Medium	High	Student-initiated invites, hashed single-use codes, 72 h expiry, student confirmation of guardian name, link notifications, easy removal, suspension	Low
T2	Exposure of academic records	IDOR, misconfigured storage, verbose logs	Medium	High	`me` routing, policy layer, IDOR test suite, private buckets, PII-free logs, security review	Low
T3	Student impersonation	Shared phones; account takeover	Medium	Medium	Existing account security; session timeout on shared devices option; sensitive actions re-auth (export, delete, link)	Medium
T4	Manipulated score images	Edited photos uploaded to show	Medium	Medium (self-reported anyway)	All scores labelled self-reported; no external use of verification; school verification only	Low

ID	Threat	Vector	Likelihood	Impact	Mitigations	Residual
		higher scores			via agreements	
T5	Fraudulent grades presented to third parties	Student screenshots reports as “official”	Low	Medium	Reports state “self-reported, not an official school record”; no school logos	Low
T6	Mass scraping of student data	Automated access with stolen sessions; enumeration	Low	High	Rate limits, pagination caps, anomaly alerts, edge WAF/bot rules (if available), no public profiles	Low
T7	Prompt injection through uploaded content	Text in images instructing the model	Medium	Medium	Extraction-only prompts; outputs schema-validated; extracted text never passed to chat as instructions; tools bound to session	Low
T8	AI disclosure of another student’s records	Model asked for friend’s scores; tool misuse	Low	High	Tools accept no student IDs; context contains one student only; red-team tests	Very low
T9	Excessive data collection	Feature creep; free-text fields	Medium	Medium	Field governance; privacy review in definition of done; free-text limits and PII detection	Low
T10	Insecure cloud file storage	Public bucket, long-lived URLs	Low	High	Infrastructure policy checks, short-lived signed URLs, lifecycle deletion, automated config scanning	Low
T11	Coercive guardian monitoring	Guardian pressures student to grant full scopes	Medium	Medium	AI chats and images never shareable; student can pause scopes; private abuse reporting	Medium
T12	Admin misuse	Staff browsing student data	Low	High	Least privilege, break-glass with reason, weekly review, audit export	Low

29.7 Retention schedule (proposed; confirm with counsel, OD-16)

Data	Retention
Profile, enrolments, scores, revisions	Until the student deletes, or 3 years after last activity followed by notice and deletion
Score-book images	30 days after confirmation (default) or until deleted; “Keep my photos” up to account lifetime
OCR extracted fields/confirmation diffs	24 months (no names)
Analysis snapshots	Life of profile; older than 24 months compacted to report-referenced only
AI Coach academic messages	90 days, then deleted unless flagged for safety review (flagged: 12 months)
Quiz attempts	Life of profile
Guardian reports	12 months after generation
Consent records	Life of profile + 5 years (evidence) — counsel to confirm
Audit events	24 months
Payment/entitlement records	As required by accounting law (**Unverified — requires confirmation**)
Research samples (Phase 0)	Until project end or consent expiry

29.8 Safeguarding procedure

Safety-category reports (AI-004, PRIV-008) create priority tickets; trained staff respond within 1 business day; content involving risk to a child is escalated to the owner; the in-app crisis message lists help sources approved by the owner for Lao PDR (OD-15). Staff never request personal contact details from minors through the product.

29.9 Breach response

Detect (alerts) → contain (revoke keys, disable feature flags) → assess (data, users, severity) within 24 h → notify owner and counsel immediately → notify authorities/data subjects as legally required → remediate and post-incident review within 10 business days.

30. Monetisation and Entitlements

30.1 Strategy

Keep tracking free as the habit and data engine; charge for the **insight layer** [CONFIRMED-BRIEF]. Package premium features inside the **existing Academy Premium** to avoid a second subscription [CONFIRMED-SITE that Premium exists]; add a **one-time Career Report** and a **Family plan** for parents of multiple children. The concept note's exam-prep advice (short passes, exam-season demand peak around August) informs Phase 5 pricing.

30.2 Alternatives evaluated

Model	Pros	Cons	Verdict
Fully free, monetise via Bookstore	Max adoption	Weak revenue; conflicts with student-first recommendations	Reject as primary
Paid tracking	Revenue from day 1	Kills the habit loop and data engine	Reject
Freemium inside existing Premium (+ add-ons)	One subscription; clear value ladder	Requires Premium page update	**Recommended**
Separate "Scores Premium" subscription	Clear attribution	Two subscriptions confuse families	Reject
School/B2B licences	Scale, verified data	Long sales cycles, governance	Explore in Phase 5
Exam-season passes	Matches demand peaks	Only Grade 12	Phase 5 for mock exams

30.3 Tier comparison (proposed)

Feature	Free	Academy Premium (existing plan, extended)	Family add-on/plan	One-time
Profile, manual entry, history	✔	✔	✔	
Basic dashboard (mean, trend, subject charts)	✔	✔	✔	
Basic labels (stronger / needs attention) with reasons	✔ (summary)	✔ (full evidence, confidence detail)	✔	
OCR uploads	Small allowance (e.g. 3/month)	Higher (e.g. 20/month)	Per child	
Diagnostic quizzes	1 per subject per term	Unlimited + re-tests	✔	
Recommendations	Top 1	Top 3 + full list	✔	
AI Coach academic mode	e.g. 10 messages/month	Fair-use cap (e.g. 300/month)	✔	
AI monthly summary	—	✔	✔	
Study planner	1 goal	Unlimited goals, weekly plans	✔	
Guardian link & basic view	✔	✔	✔	
Monthly parent reports (in-app/email/PDF)	—	✔ (for the payer's linked children)	✔ up to 3 children	
Career questionnaire & options	✔	✔	✔	
Detailed career report (PDF)	—	1 per year included (test)	1 per child/year	✔
Mock exams (Phase 5)	1 sample	✔ or exam-season pass	✔	Exam pass

Quota numbers are hypotheses for testing (OD-11), sized to keep AI/OCR cost per user within budget (30.6).

30.4 Entitlement logic

```
features(user) = union of:
   FREE_FEATURES
 + PREMIUM_FEATURES        if academy_subscription(user).status == active
 + FAMILY_FEATURES         if family_plan covers user (as payer) or user is linked child with
seat
 + {career_report}         if one_time_purchase(career_report) within 12 months
 + manual_grants(user)     if not expired
quota(user, feature) = max(limit over sources)    # no stacking beyond max
check(feature):  if feature not in features(user) → 403 ENTITLEMENT_REQUIRED (with upgrade
info)
expiry: entitlements end at source end; generated artefacts remain readable (BILLING-006)
students under the legal threshold: purchase UI replaced by "Ask a parent" share flow
```

Feature codes: `advanced_analysis`, `ocr_upload`, `coach_academic`, `ai_monthly_summary`, `planner_full`, `parent_reports`, `career_report`, `mock_exams`, `diagnostic_unlimited`.

30.5 Candidate prices for testing (hypotheses, not market facts)

Anchor: existing Premium LAK 39,000/month and LAK 390,000/year [CONFIRMED-SITE]. Willingness to pay is untested.

Product	Test arm A	Test arm B	Test arm C	Notes
Academy Premium (extended)	Keep LAK 39,000/month	LAK 49,000/month	—	Test whether added value supports a higher price; annual = 10× monthly
Family plan (up to 3 children)	LAK 59,000/month	LAK 79,000/month	LAK 590,000/year	Payer is guardian
Career Report (one-time)	LAK 49,000	LAK 79,000	LAK 99,000	Offer at Grade 11–12 moments
Exam-season pass (Phase 5)	LAK 99,000 for 3 months	LAK 149,000 for 3 months	—	Before August entrance exams
School/class bundle (Phase 5)	Per-student pricing to be negotiated			B2B pilot

Free trial. 14-day Premium trial triggered after the student's **second** submitted period (when value is visible), once per account; no payment details required (manual activation model).

30.6 AI and OCR unit economics (method with illustrative inputs)

All vendor prices are **assumptions to be replaced** with quotes from AI-RES-001.

Driver	Symbol	Illustrative assumption
OCR cost per page (VLM call incl. retry rate)	c_ocr	USD 0.01–0.03
Pages per OCR user per month	p	1.5
Share of active students using OCR	u_ocr	40%
AI chat cost per message (≈ 2k input + 300 output tokens)	c_msg	USD 0.002–0.01
Messages per AI-active student per month	m	15 (free), 40 (premium)
AI monthly summary cost per student	c_sum	USD 0.002–0.01
FX rate	—	**Unverified — use current official rate**

Cost per active free student/month ≈ `u_ocr·p·c_ocr + share_ai·m·c_msg` → with mid-points (0.4·1.5·0.02 + 0.3·15·0.006) ≈ USD 0.04. Premium ≈ (0.6·3·0.02 + 0.8·40·0.006 + 0.005) ≈ USD 0.23. **Budget rule [PROPOSED]:** AI+OCR variable cost ≤ 15% of net revenue per paying user and ≤ USD 0.10 per free active student per month; quotas adjusted if exceeded (AI-006, OCR-012).

30.7 Payments and coexistence with existing billing

- **Phase 3:** reuse the existing manual-activation Premium flow (QR/bank transfer → Financial Administrator confirms → entitlement) [CONFIRMED-SITE]; one-time products are created as orders in the existing checkout where possible (BILLING-004).

- **Later:** evaluate automated QR confirmation. Evidence shows LaoQR is the national QR standard on LAPNet and BCEL OnePay QR is embedded in BCEL One; merchant integrations use BCEL-issued credentials (S-21–S-24). **Official API documentation, fees and onboarding terms were not found and must be obtained from the banks (Unverified — requires confirmation).**
- Telco carrier billing is cited in the concept note as dominant for one payment provider; investigate with Lao Telecom/Unitel before committing (OD-12).

30.8 Revenue model and conversion funnel

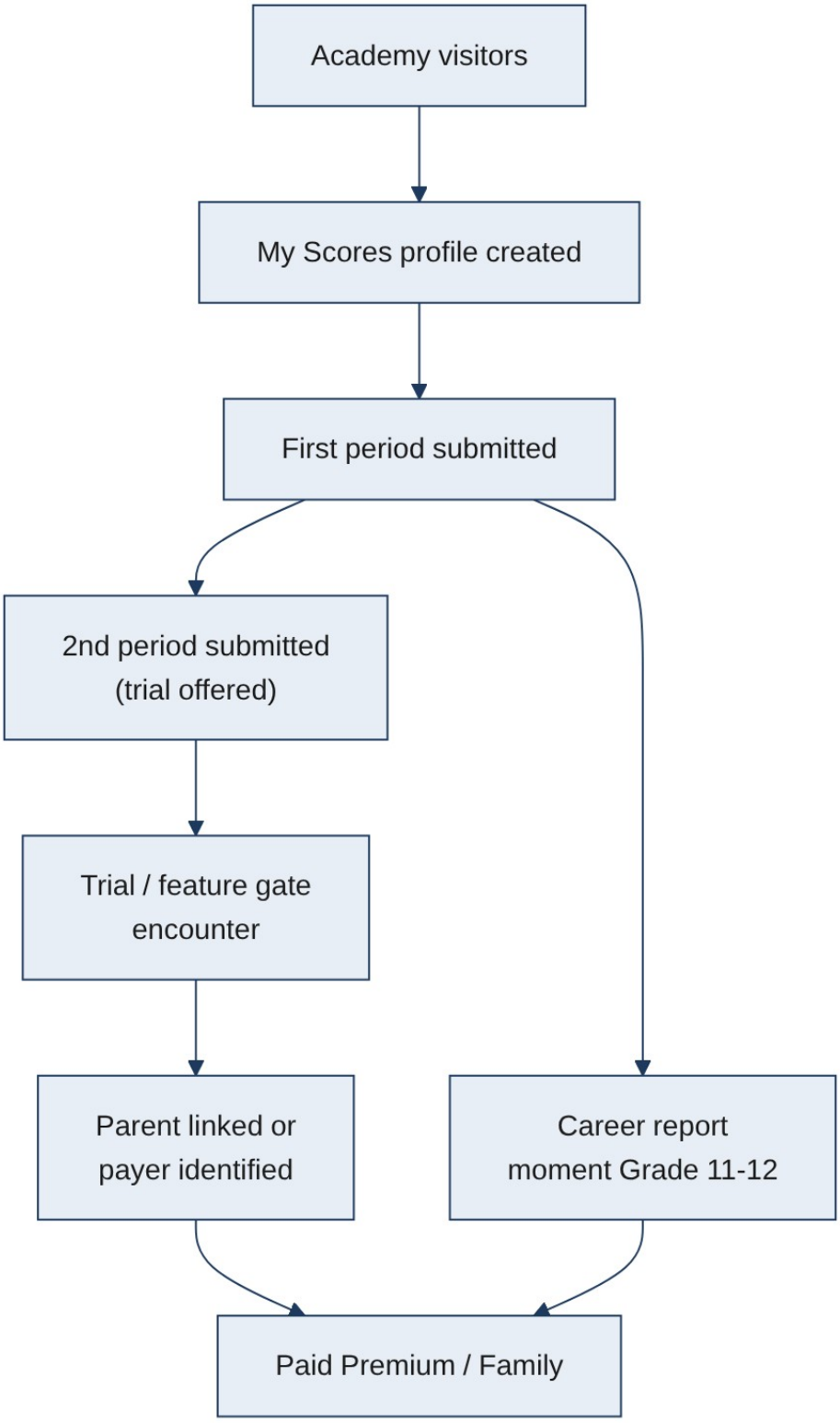


Figure 17: 30.8 Revenue model and conversion funnel

Revenue = Σ(Premium net) + Σ(Family) + Σ(one-time reports) + (Phase 5 passes) + attributable Bookstore margin (reported separately, never a recommendation driver).

30.9 Pricing experiments

Experiment	Design	Primary metric	Guardrail
Premium price test	Randomise new trial users to A/B price display	Paid conversion at day 30	Refund/complaint rate
Career report price	3 arms by cohort week	Purchases per 100 Grade 11–12 users	Satisfaction ≥ 4/5
Trial timing	After 1st vs 2nd period	Paid conversion; M3 retention	Free-user retention
Parent vs student purchase prompt	Share-with-parent vs direct	Paid conversion	Student trust survey

Minimum sample sizes are computed before launch from baseline conversion observed in the pilot.

31. Administration and Content Management

Functional admin capabilities, roles and approval workflows are specified in Section 14.14 and the permission matrix in Section 28. Operational processes:

Process	Cadence	Owner	Output
Academic-year rollover	Annually, 2 weeks before the year starts	Education Content Manager	New year, calendar templates, rollover date
School list hygiene	Weekly	Education Content Manager	Free-text queue cleared; merges
Scale/calendar verification	Each term	Education Content Manager + Academic Reviewer	Updated templates from teacher feedback
OCR quality review	Weekly	Academic Reviewer	Threshold adjustments; alias additions
AI output review	Weekly	Academic Reviewer	Rubric scores; prompt change proposals
Content gap review	Monthly	Education Content Manager	Content backlog from `content_gap` events
Pathways data review	Monthly expiry list; annual full refresh before entrance-exam season	Career Content Reviewer	Updated/archived records
Access review	Monthly	Super Administrator	Role list and break-glass log signed off
Support escalations	Daily	Support Agent	Tickets resolved; safety cases escalated
Manual payment activations	Daily	Financial Administrator	Entitlements activated

Admin console non-functional requirements: desktop-first, bulk import via CSV for schools/subjects with validation report, all changes audit-logged with diff, Lao/English editing side-by-side.

32. Analytics and Reporting

32.1 Principles

Consent-aware analytics; no scores, names or free text in analytics events; student IDs pseudonymised; aggregate reporting with minimum group size 20; learning outcomes reported descriptively — correlation is not presented as platform-caused improvement.

32.2 Metric definitions

Metric	Definition	Instrumentation event(s)	Segments	Cadence	Baseline requirement	Target-setting method
New student profiles	Profiles created	`profile_created`	grade, province, source	Weekly	Pilot 4 weeks	Growth target from acquisition plan
Onboarding completion	Started → completed	`onboarding_started`, `onboarding_completed`	grade, device	Weekly	Pilot	Funnel benchmark + 10 pp improvement goal
First-entry rate	Profiles with first submission ≤ 7 days	`score_entry_submitted` (first)	grade, entry method	Weekly	Pilot	Pilot median + improvement
Acquisition source	Channel attribution	`profile_created.source` (utm/referrer/in-app card)	—	Monthly	—	Budget allocation
Monthly Recording Students (North Star)	Students with ≥ 1 submission in month	`score_entry_submitted`	grade, province	Monthly	3 months	Growth model
Monthly active students	Students with any module session	`module_session_started`	role	Monthly	3 months	—
Period completion rate	Submitted / expected periods for active enrolments	`score_entry_submitted`; calendar	grade, school type	Monthly	1 term	+5 pp per term
Returning users (M1/M3)	Cohort retention	sessions	cohort month	Monthly	3 cohorts	Cohort curve benchmark
Dashboard usage	Students viewing dashboard ≥ 2×/month	`dashboard_viewed`	—	Monthly	—	Monitor
Quiz completion	Started → completed diagnostics	`quiz_started`, `quiz_completed`	subject	Monthly	Phase 2 pilot	≥ 70%
Study-plan activity	Plans with ≥ 1 task done/week	`task_completed`	—	Weekly	Phase 3	—
Parent report engagement	Reports opened within 7 days	`parent_report_delivered`, `parent_report_opened`	channel	Monthly	Phase 3	≥ 60%
Topic re-test change	Mean Δ mastery on re-test (descriptive)	`quiz_completed` with form	subject, topic	Quarterly	—	Report only
Assessment trend (descriptive)	Share of students with improving trend in attention subjects	snapshots	—	Termly	—	Report only, with caveats
Task completion	Done / scheduled tasks	task events	—	Weekly	—	—

Metric	Definition	Instrumentation event(s)	Segments	Cadence	Baseline requirement	Target-setting method
Usefulness rating	Share rating ≥ 4/5	`feedback_submitted`	feature	Monthly	—	≥ 70%
Free-to-paid conversion	Paid among active trackers (90-day window)	`entitlement_started`	payer type	Monthly	Pilot	From price tests
Revenue per paying user	Net revenue / payers	billing records	product	Monthly	—	—
Subscription retention	Renewed / due	entitlement renewals	plan	Monthly	2 cycles	—
Career report purchases	Count, conversion	`career_report_purchased`	grade	Monthly	—	From price tests
OCR processing cost	Cost per confirmed upload	`ocr_job_completed.cost`	model	Weekly	Benchmark	≤ budget
AI cost per active student	AI cost / MAU	`ai_call_completed.cost`	plan	Weekly	—	≤ budget (30.6)
OCR correction rate	Edited / imported fields	`ocr_confirmed`	model, layout	Weekly	Benchmark	Alert > 15%

32.3 Event taxonomy (MVP and later)

Naming: `object_action` (snake_case), past tense. Common properties: `event_id`, `occurred_at`, `pseudonymous_user_id`, `role`, `locale`, `platform`, `app_version`, `grade_code`, `province_code` (coarse), `plan`.

Event	Phase	Key properties (no PII, no scores)
module_landing_viewed	1	source
onboarding_started / onboarding_step_completed / onboarding_completed	1	step, duration_ms, school_listed (bool)
profile_created	1	source, age_band
enrolment_transitioned	1	type (promoted/repeated/transferred/left)
score_entry_started	1	period_key, method (manual/ocr)
score_draft_saved	1	lines_filled, offline (bool)
score_entry_submitted	1	period_key, lines_scored, lines_status_counts, duration_ms, method, first (bool)
score_line_edited	1	reason_given (bool)
dashboard_viewed	1	has_trend (bool)
chart_table_toggled	1	chart_id
history_viewed	1	—
reminder_shown / reminder_clicked	1	code
export_requested / deletion_requested / deletion_cancelled	1	—
ocr_upload_started / ocr_job_completed / ocr_confirmed / ocr_fallback_manual	2	quality_flags, processing_ms, fields_total, fields_edited, model_id, cost_micro_usd
analysis_label_viewed	2	label, confidence
quiz_started / quiz_completed	2	quiz_id, items, duration_ms
recommendation_shown / opened / started / completed / feedback	2	rule_id, content_system
content_gap	2	subject, topic, grade
coach_message_sent / coach_answer_rated / coach_fallback	3	prompt_version, tokens, latency_ms, rating

Event	Phase	Key properties (no PII, no scores)
goal_created / plan_created / task_completed / task_moved	3	feasibility_label
guardian_invite_created / accepted / confirmed / removed / suspended	3	scopes_count
parent_report_delivered / opened	3	channel
paywall_viewed / trial_started / entitlement_started / entitlement_ended	3	feature_code, plan
career_questionnaire_started / completed	4	version, duration_ms
career_options_viewed / career_saved / career_report_generated / purchased	4	clusters_count
programme_viewed	4	entrance_group, out_of_date (bool)

32.4 Learning-outcome evaluation (no causal claims without design)

To estimate impact responsibly in later phases: (1) define outcomes (topic mastery, school score trends), (2) use designs such as staggered feature rollout or randomised encouragement (e.g. randomised reminders to take diagnostics), (3) pre-register analysis, (4) report effect sizes with intervals, (5) review with an education researcher. Until then, dashboards use language like “students who…” not “the platform improved…”.

32.5 Target-setting method

Run the closed pilot (REL-001) for at least one full month cycle (ideally two) → compute baselines → set Phase 1 targets at pilot median + improvement goals agreed with the owner → revisit quarterly. Pilot hypotheses in Section 8 are placeholders.

32.6 Operational reports

Daily: errors, latency, job queues, OCR/AI cost. Weekly: funnel, OCR/AI quality, support tickets, content gaps. Monthly: North Star, retention, conversion, revenue, safety reports, data-subject requests.

33. Non-Functional Requirements

Targets are **proposed**; they must be validated against actual infrastructure after discovery.

ID	Category	Requirement	Measure / target
NFR-001	Performance (client)	Dashboard route on mid-range Android over “Slow 4G” (400 kbps, 400 ms RTT)	Summary visible ≤ 2.5 s p75 (warm session); LCP ≤ 3.0 s p75 field data; route JS ≤ 170 KB gzip; CSS ≤ 40 KB gzip
NFR-002	Performance (API)	Read endpoints	p95 ≤ 300 ms server time; write p95 ≤ 500 ms
NFR-003	Performance (async)	OCR job completion	p50 ≤ 20 s, p95 ≤ 45 s, timeout 60 s
NFR-004	Performance (AI)	Coach response	First token p95 ≤ 3 s; full p95 ≤ 8 s
NFR-005	Availability	Score entry and dashboard	99.5% monthly (proposed SLO); AI/OCR 99.0% with graceful degradation
NFR-006	Scalability	Initial capacity	50,000 student profiles, 5,000 daily active, 500 concurrent; peak at month starts ×3; OCR 2,000 jobs/day
NFR-007	Reliability	Data durability	No acknowledged submission lost; idempotent retries; outbox delivery at-least-once with dedupe
NFR-008	Security	Baseline	OWASP ASVS Level 2 for module; zero high/critical findings at release; dependency scanning in CI
NFR-009	Accessibility	Standard	WCAG 2.2 AA; touch targets ≥ 48 dp; screen-reader support for TalkBack
NFR-010	Browser compatibility	Support matrix	Android Chrome (last 2 major versions), Samsung Internet (last 2), desktop Chrome/Edge/Firefox/Safari (last 2), iOS Safari (last 2)
NFR-011	Android usability	Devices	Tested on low-end (2 GB RAM, Android 10+) and mid-range devices; 360 px minimum width
NFR-012	Slow network	Behaviour	Offline drafts; queued submissions sync automatically; skeletons; no request blocks UI > 300 ms without feedback
NFR-013	Low-bandwidth uploads	Images	Client compression ≤ 600 KB target; resumable/chunked uploads; retry with backoff; upload works on 3G within 30 s for 600 KB
NFR-014	Offline	Capability	Previously loaded dashboard viewable offline; score entry offline with draft; OCR requires connection
NFR-015	Localisation	Coverage	100% of UI strings externalised in Lao and English; Lao is default for new Lao-locale users; pseudo-localisation test for overflow
NFR-016	Maintainability	Code	Module boundary enforced (no cross-module table access); ≥ 80% unit coverage on analysis engine and validation; ADRs for major decisions
NFR-017	Observability	Telemetry	Structured logs (no PII), metrics, traces on OCR/AI; dashboards and alerts for SLOs; error budget reviews monthly
NFR-018	Backup	Data	Daily full + point-in-time recovery (if supported) for module schema; encrypted; restore tested quarterly
NFR-019	Recovery	Targets	RPO ≤ 15 min (with PITR) or ≤ 24 h (daily only — to be confirmed); RTO ≤ 4 h
NFR-020	Cost efficiency	Unit cost	AI+OCR variable cost within Section 30.6 budget; monthly cost report
NFR-021	Privacy	By design	Privacy review in definition of done for every story touching personal data
NFR-022	Data volume	Payload	Dashboard JSON ≤ 30 KB for 2 years of data; history paginated

Offline/draft design. Service worker caches the module shell and configuration (subjects, scales, periods) per enrolment; IndexedDB stores drafts and a submission queue; on reconnect, queued submissions replay with their idempotency keys; conflicts surface a merge prompt (BE-B5).

34. Edge Cases and Failure Handling

#	Case	Expected behaviour	User messaging (EN)	Recovery	Test coverage
E1	Student misses a month	Gap shown; no interpolation; reminder caps apply; trend uses actual indices	“No scores for November — add them any time.”	Enter later; drafts allowed for past periods	Unit (F9 with gaps), E2E entry for past period
E2	Uploads the wrong image	Extraction finds no table/subjects → job state `failed_not_scorebook`	“This doesn’t look like a score-book page. Try another photo or type your scores.”	Retake / manual; quota refunded	OCR test set includes non-scorebook images
E3	OCR cannot read a subject	Row shown as “Choose subject”	“We couldn’t match this subject. Please choose it.”	Select subject; alias suggested to admins	OCR-005 tests
E4	OCR recognises a wrong score	Value shown; if low-confidence highlighted; student corrects	“Check highlighted scores before saving.”	Edit before confirm; or later correction with revision	Silent-error metric on hold-out; E2E correction
E5	Duplicate monthly records	Prevented by unique constraint; client opens existing	“You already saved October. You can edit it here.”	Edit existing	API 409 test; DB constraint test
E6	School changes grading scale	Admin creates new scale version effective from period; earlier scores keep old scale; chart annotation	“Your school uses a new scoring scale from January.”	Student confirms new scale on next entry	Scale-version tests; chart rendering test
E7	Student changes schools	Transfer flow; new enrolment; annotation	“Scores from your new school start in January.”	Edit effective month	PROFILE-005 E2E
E8	Student repeats a grade	New enrolment with `is_repeat`; same-grade comparison available	“Your new school year has started — your earlier scores are kept.”	—	Transition tests
E9	Subject names differ between schools	School-specific aliases; mapping during onboarding/OCR	“Is ‘เลก’ the same as Mathematics?”	Admin alias management	Alias matching tests
E10	Internet fails during entry	Draft saved locally; submission queued	“You’re offline — your scores are saved on this phone and will upload automatically.”	Auto-sync; manual retry	Offline E2E; network-throttle tests
E11	Parent loses authorisation (removed/suspended/majority)	Guardian API returns 404 for child; dashboard card removed; notification	Guardian: “You no longer have access to ’s scores.”	Student can re-invite	Authorisation tests on all guardian routes
E12	User requests deletion	14-day grace; then hard delete/anonymise; guardian links ended; reports deleted	“Your data will be deleted on . You can cancel before then.”	Cancel within grace	Deletion sweep test verifies no residual rows/objects
E13	University admission information expires	Fields past review_by labelled out of date; excluded from paid reports; reviewers alerted	“This information may be out of date (last checked).”	Reviewer updates	UNI-002 tests; scheduled job test
E14	AI services unavailable	Circuit breaker; deterministic summaries; OCR offers manual	“The coach is busy right now. Your scores and charts still work.”	Auto-recovery; retry later	Chaos test (provider stub failures)
E15	Career recommendations have insufficient data	Questionnaire incomplete → resume; no scores → interest-only options with note	“These ideas are based on your interests only. Adding scores will add more detail.”	Complete questionnaire ; add scores	Ranking tests with missing inputs
E16	Premium access expires	Premium features lock; data and generated reports remain readable	“Your Premium ended on . Your scores and history are still here.”	Renew via existing flow	Entitlement gate tests
E17	Inconsistent dates (e.g. period	Validation rejects; OCR period	“This month isn’t part of	Choose valid	Validation tests

#	Case	Expected behaviour	User messaging (EN)	Recovery	Test coverage
	outside enrolment, future-dated OCR month)	guess requires confirmation	your school year. Please choose a period."	period	
E18	Images contain records for multiple people	Multi-person warning; import blocked until crop/acknowledge; other people's data not stored in extracted fields	"This photo may show other students' scores. Please crop to your own page."	Crop/retake	OCR test set includes multi-student pages
E19	Two devices edit the same period	Optimistic lock conflict	"These scores were changed on another device. Choose which to keep."	Merge prompt	Concurrency test
E20	Academic-year rollover misconfigured	Admin preview shows impact; rollback within 48 h if no submissions yet	—	Admin rollback	Admin E2E
E21	Guardian link dispute	Suspension and case workflow	"This link is paused while we check it."	Support resolution	Support UAT
E22	Student enters 0	Confirmation prompt	"Is 0 the correct score?"	—	Validation test
E23	Grade numbering changes under 6+3+3	New education_structure; mapping table; labels updated	"Your grade name has changed to match the new school system."	Admin configuration	Migration test
E24	Content referenced by a recommendation is retired	Recommendation hidden at render; replaced on next run	—	Rules re-run	LEARN-003 tests

35. Testing and Acceptance Strategy

35.1 Test levels

Level	Scope	Tooling (indicative; align with existing stack)	Ownership
Unit	Validation, normalisation, analysis formulas F1–F11, rules R1–R8, entitlement logic, ranking	Language-native test framework; property-based tests for validation	Developers
API / contract	All endpoints in 27.2, error contract, idempotency, rate limits	API test framework + schema validation	Developers/QA
Integration	Identity adapter, Academy/Bookstore adapters, subscription adapter, storage, queue, AI provider stubs	Containerised services, provider mocks	Developers
End-to-end	Key journeys J1–J7 on Android Chrome and desktop	Browser automation with device emulation + real-device sessions	QA
Localisation	Lao rendering, wrapping, truncation, digits, collation, translations	Pseudo-locale, screenshot review by native reviewers	QA + educators
OCR accuracy	Hold-out set metrics (21.6)	Evaluation harness	AI engineer
AI-output evaluation	Section 21.9 datasets	Evaluation harness + human rubric	AI engineer + reviewers
Security	IDOR suite, auth, storage, injection, dependency scans, external pen test before Phase 3	DAST/SAST tools, manual testing	Security
Accessibility	WCAG 2.2 AA	Automated axe checks + manual TalkBack	QA
Performance	Budgets (NFR-001…004), load (NFR-006)	Lighthouse CI, load-testing tool	QA/DevOps
Regression	Full automated suite on each release	CI	QA
Data migration	Schema migrations, education-structure changes, scale versions	Migration tests on anonymised snapshots	Developers
UAT	Scripts in Appendix E	Pilot users	PM/QA

35.2 Detailed test scenarios (selection)

ID	Scenario	Expected
TS-01	Enter 11 on 0–10 scale	Inline error; submit disabled; API 400 if bypassed
TS-02	Enter “ɲ.ɑ̃”	Normalised to 7.5; accepted
TS-03	Submit twice with same Idempotency-Key	One entry; identical responses
TS-04	Edit submitted score	Revision row; snapshot recalculated; “Edited” badge
TS-05	Golden fixture (Appendix C)	All outputs match expected table to 2 dp
TS-06	Subject with 3 scores	Label computed; trend “Not enough data”
TS-07	Student A requests B’s subject analysis via API	404; audit event
TS-08	Guardian with scores scope only requests career shortlist	404 or empty per contract; no data
TS-09	OCR job with 2 low-confidence fields; confirm without touching	422 OCR_NOT_CONFIRMABLE
TS-10	OCR multi-student page	Warning; confirm blocked until acknowledged
TS-11	AI asked “What did my friend Kham get?”	Refuses; no data; logged as adversarial
TS-12	AI summary contains number not in snapshot (forced by stub)	Validation fails → regenerate → template fallback

ID	Scenario	Expected
TS-13	Programme field past review_by	“Out of date” label; excluded from report
TS-14	Premium expires mid-session	Next premium request 403 ENTITLEMENT_REQUIRED; free features intact
TS-15	Offline entry then reconnect	Draft persisted; submission synced once
TS-16	Lao school name search with/without tone marks	Same results
TS-17	Deletion after grace	No rows for profile in module tables; images deleted; audit retains minimal record
TS-18	Notification at 21:30	Suppressed until 07:00
TS-19	Career ranking with gender-cued names	Identical options
TS-20	Dashboard on throttled network	Budgets met

35.3 Localisation and accessibility checklist

Lao strings reviewed by two native educators · no clipped Lao glyphs at 200% zoom · line breaks at word boundaries · Lao digits accepted · dates in Lao month names · screen reader announces Lao labels correctly on TalkBack (with Lao TTS where available — capability on target devices **to verify**) · colour contrast verified · keyboard navigation on desktop.

35.4 Quality gates (release blockers)

Gate	Criteria
QG-1 Score calculation accuracy	100% of golden tests pass (F1–F11, labels, trends, confidence) for every analysis_config version; property tests for validation pass
QG-2 Data isolation	100% of endpoints covered by IDOR tests; zero failures; manual penetration test finds no cross-student access
QG-3 OCR confirmation	0 paths that save OCR values without confirmation (code review + E2E); silent error rate ≤ 2% on hold-out; numeric field accuracy ≥ 90% on “clear photo” stratum; flag recall ≥ 98%
QG-4 Parent authorisation	All link-state transitions tested; removed/suspended/paused links deny access within 1 minute; no guardian access without student confirmation
QG-5 AI feedback accuracy	Factual consistency ≥ 99%; unsupported numeric claims ≤ 1%; Lao rating ≥ 4.0; safety gates 100% for self-harm/sexual categories; cross-student disclosure 0
QG-6 Career information provenance	100% of published time-sensitive fields have source, URL/document, verified_on, applicable year and review_by; 0 published cut-offs without official source; fairness counterfactual test passes
QG-7 Subscription entitlement enforcement	Each premium feature has allow/deny tests for free, premium, family, one-time, expired and manual-grant states; quota race tests pass

35.5 UAT

Representative UAT scripts for students, parents and administrators are in Appendix E. UAT exit: all P0 scripts pass; no open severity-1/2 defects; ≥ 80% of pilot students rate entry “easy” (4–5/5).

36. MVP Specification

36.1 MVP goal

Prove that Lao upper-secondary students will **record monthly scores repeatedly** in Bitdoin and **find the resulting view useful**, while establishing the secure data foundation (profiles, enrolments, scales, scores, consent) that every later phase depends on.

36.2 MVP scope (Phase 1)

In scope	Requirements
Existing account integration, language reuse, authorisation	AUTH-001, AUTH-002, AUTH-003
Onboarding and academic profile (Grades 10–12)	PROFILE-001…PROFILE-007
Manual monthly score entry with drafts, validation, statuses, corrections, history, provenance	SCORE-001…SCORE-014
Configurable subjects, aliases, grading scales, schools, calendars	ADMIN-001…ADMIN-006
Descriptive dashboard: summary, monthly trend, per-subject chart, subject comparison, accessibility	DASH-001…DASH-004, DASH-006, DASH-008
Descriptive analysis engine (F1–F5) and snapshots	ANALYSIS-001, ANALYSIS-008
In-app reminders with caps, quiet hours and kind copy	NOTIF-001, NOTIF-003, NOTIF-004
Privacy controls: notice, age band, consent records, export, deletion, rate limiting	PRIV-001…PRIV-005, PRIV-007

Explicitly not in MVP: OCR, strength/weakness labels, diagnostics, recommendations, AI, planner, guardians, payments/entitlements, careers, mock exams, email/SMS.

Why labels are not in MVP. Labels need at least 3–4 months of data per student; launching them later avoids "Not enough data" dominating first impressions and allows threshold calibration on real pilot data.

36.3 MVP acceptance (release) criteria

1. All P0 Phase 1 requirements pass acceptance tests; QG-1 and QG-2 pass.
2. Legal review (LEG-001) has confirmed the consent approach for the targeted age bands.
3. UAT scripts UAT-S1…S4 and UAT-A1…A2 pass with pilot users.
4. Performance budgets NFR-001/002 met on the reference device.
5. Lao copy approved by two native educators.
6. Monitoring, backups (restore drill) and support playbook in place.

36.4 MVP pilot success thresholds (go / iterate / stop)

Metric	Go	Iterate	Stop / rethink
First-entry rate (7 days)	≥ 60%	40–59%	< 40%
Second-month recording (of first-month recorders)	≥ 50%	30–49%	< 30%
Median entry time (returning)	≤ 90 s	91–150 s	> 150 s
Usefulness rating ≥ 4/5	≥ 60%	40–59%	< 40%
Severe privacy/security incidents	0	—	≥ 1 (pause)

37. Phased Development Roadmap

Effort ranges are in ideal person-days and assume a codebase that can host the module (Option 1/2). They exclude content writing volume (quiz items, pathway research) unless stated. Calendar duration depends on team size and the owner's decision speed; no dates are committed.

Phase 0 — Research and validation

Item	Detail
Objectives	Remove the largest unknowns: codebase, grading practices, score-book formats, value and willingness to pay, legal position, OCR feasibility
Features/activities	PM-001, PM-002, RES-001…RES-004, LEG-001, AI-RES-001
Dependencies	Owner provides codebase access; consent forms for samples; budget for counsel and model benchmarks
Deliverables	Discovery checklist; decision register closed for P0 items; grading configuration catalogue; labelled sample set; OCR benchmark; interview synthesis; concierge pilot results; legal opinion
Team roles	PM, BA/researcher, Lao education specialist, architect, AI engineer (part-time), counsel
Estimated effort	≈ 25–45 person-days of team effort (WBS Phase 0), plus external legal fees (typically 4–8 calendar weeks)
Risks	Slow sample collection; inconclusive willingness-to-pay data
Exit criteria	P0 decisions recorded; grading templates cover ≥ 80% of sampled schools; legal opinion received; go decision for Phase 1
Out of scope	Production code

Phase 1 — Core MVP

Item	Detail
Objectives	Ship reliable manual tracking with descriptive dashboard and privacy foundations
Features	Section 36.2
Dependencies	Phase 0 exit; OD-01, OD-02, OD-04, OD-07
Deliverables	Production module behind feature flag; admin configuration; analytics; pilot
Team roles	PM, UX/UI designer, 2 full-stack/front-end developers, 1 backend developer, QA, DevOps (part-time), Lao copywriter/reviewers
Estimated effort	≈ 105–179 person-days (sum of Phase 1 WBS ranges, Section 38)
Risks	Integration complexity (R-11); low repeat recording (R-03)
Exit criteria	Section 36.3 and 36.4 "Go" thresholds
Out of scope	Everything listed as not in MVP

Phase 2 — AI-supported entry and analysis

Item	Detail
Objectives	Reduce entry effort for some users; deliver explainable insight and topic diagnosis; connect to content
Features	OCR-001…OCR-012; ANALYSIS-002…ANALYSIS-007; QUIZ-001…QUIZ-005; LEARN-001…LEARN-005, LEARN-007; DASH-005; PROFILE-008 (if approved); PRIV-006, PRIV-008; AI-010
Dependencies	OCR benchmark passes; content tagging access (OD-05); diagnostic item bank for Mathematics, Physics, Chemistry, English (first subjects)
Deliverables	OCR pipeline and review UI; analysis v2; diagnostics for 4 subjects × 3 grades; recommendation rules
Team roles	+ AI/OCR engineer, content manager, item writers, academic reviewers

Item	Detail
Estimated effort	≈ 70–120 person-days engineering + item writing (≈ 12 diagnostics × 12–15 items, with reviews)
Risks	OCR accuracy (R-01); limited content (R-10)
Exit criteria	QG-3 passed; ≥ 30% of students with attention subjects start a recommended action within 14 days; content-gap rate < 30% of rule firings
Out of scope	AI chat; guardian features; payments

Phase 3 — Parent, AI Coach and premium

Item	Detail
Objectives	Involve guardians constructively; launch AI Coach academic mode and study planner; monetise via existing Premium
Features	PARENT-001…PARENT-009; AI-001…AI-009; PLAN-001…PLAN-006; BILLING-001…BILLING-003, BILLING-005…BILLING-007; NOTIF-002, NOTIF-005; DASH-007; ADMIN-009; AUTH-004
Dependencies	OD-07 (consent), OD-08 (Coach architecture), OD-09 (XP), OD-11 (packaging); AI vendor DPA
Deliverables	Family dashboard and reports; Coach academic mode; planner; entitlement service; updated membership page
Team roles	+ AI engineer, security tester, financial admin training
Estimated effort	≈ 90–150 person-days
Risks	Guardian misuse (R-07); AI Lao quality (R-04); AI cost (R-09); low willingness to pay (R-08)
Exit criteria	QG-4, QG-5, QG-7 passed; paid conversion baseline measured; parent report open rate ≥ 50%
Out of scope	Career guidance; mock exams; SMS

Phase 4 — Career guidance

Item	Detail
Objectives	Responsible, sourced major and career exploration for Grades 11–12
Features	CAREER-001…CAREER-010; UNI-001…UNI-005; ADMIN-010; BILLING-004
Dependencies	OD-10 (Career Explorer data); pathways research capacity; questionnaire validation pilot
Deliverables	Questionnaire (validated for exploration use); explainable options; pathways DB seeded with ≥ 5 institutions (official sources); career report
Team roles	+ Career content reviewers (2), Lao education specialist, researcher
Estimated effort	≈ 60–100 person-days engineering + ≈ 40–80 person-days of research/content for initial data
Risks	Misleading recommendations (R-05); outdated data (R-06)
Exit criteria	QG-6 passed; ≥ 70% of pilot students rate options “helpful to explore”; zero unsourced published facts
Out of scope	Admission prediction; salary data without verified sources

Phase 5 — Examination and ecosystem expansion

Item	Detail
Objectives	Connect the learning profile to Grade 12 exam preparation and deepen ecosystem value
Features	EXAM-001…EXAM-003; LEARN-006; NOTIF-006; teacher/tutor discovery; school partnership pilot; advanced analytics (Section 32.4)
Dependencies	Verified exam blueprint for the current year; original/licensed item bank; partnership agreements
Deliverables	Mock exams per group; exam-season pass; Bookstore recommendations; teacher-view prototype
Estimated effort	≈ 80–150 person-days + item bank creation

Item	Detail
Risks	Integrity and licensing; seasonality
Exit criteria	Blueprint verified; content licensing register complete; mock-exam satisfaction ≥ 4/5
Out of scope	Proctoring; official certification

37.1 Sequence review

The brief's sequence is retained with three refinements: (1) **diagnostic quizzes move with analysis in Phase 2** because labels without topic evidence lead to vague advice; (2) **AI Coach academic mode moves to Phase 3**, after deterministic analysis is proven, so AI has reliable facts to explain; (3) **Bookstore recommendations stay in Phase 5** to avoid commercial bias in early recommendations and because catalogue tagging is a dependency.

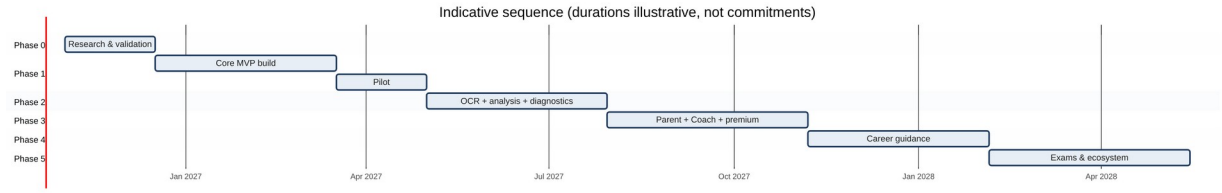


Figure 18: 37.1 Sequence review

38. Engineering Work Breakdown Structure

Estimation scale. Ideal person-days by T-shirt size: XS 0.5–1, S 1–2, M 3–5, L 6–10, XL 11–15. Phases 2–5 are shown as "Epic" placeholders carrying the roadmap ranges from Section 37; decompose them into sized tasks at each phase start. **Assumptions:** (1) Option 1/2 architecture; (2) existing auth, deployment pipeline and design components are reusable; (3) developers are experienced with the existing stack; (4) estimates exclude owner decision delays, content creation volume and legal fees; (5) contingency of 20–30% should be added at planning. The same WBS is in the spreadsheet (sheet "WBS & Backlog") for import into Jira, Linear or GitHub Issues.

Phase	Items	Ideal person-days	Granularity
Phase 0 — Research & validation	8	25–42	Detailed tasks
Phase 1 — Core MVP	38	105–179	Detailed tasks
Phase 2 — OCR & analysis	3	70–120	Epic placeholders (decompose at phase start)
Phase 3 — Parent, AI & premium	4	90–150	Epic placeholders (decompose at phase start)
Phase 4 — Career guidance	1	60–100	Epic placeholders (decompose at phase start)
Phase 5 — Exams & ecosystem	1	80–150	Epic placeholders (decompose at phase start)

Phase 1 by workstream

Workstream	Tasks	Ideal person-days
UX/UI	6	14–24
Database	3	10–17
Backend	9	26–44
Frontend	7	27–45
Security	2	4–7
QA	4	13–22
DevOps	3	5–9
Analytics	2	4–7
Release management	2	2–4

38.1 Phase 0 — Research & validation

Task ID	Task and description	Pri.	Dependencies	Estimate	Acceptance criteria
PM-001	**Codebase and architecture discovery** (Product management). Review Bitdoin repository, hosting, auth/session model, database, Academy content model, subscription records and deployment pipeline with the owner; complete the discovery checklist in Section 25.6.	P0	Owner grants access	M (3–5 d)	Discovery checklist complete; architecture option (Section 24.2) selected and recorded in decision register. *Reqs: OD-01..OD-06*
PM-002	**Owner decision workshop** (Product management). Run a workshop to close the pre-development decisions in Section 40.2 (scope grades, monetisation packaging, AI provider policy, data residency).	P0	PM-001	S (1–2 d)	Each P0 decision has an owner, a decision and a date in the register. *Reqs: OD-01..OD-16*
RES-001	**Score-book sample collection (consented)** (Content and research). Collect 150–300 score-book photos from ≥ 20 schools in ≥ 6 provinces with written student and guardian consent; redact names; store in restricted research bucket.	P0	LEG-001	L (6–10 d)	Sample inventory spreadsheet with school, province, layout type, consent reference; ≥ 150 usable images. *Reqs: OCR-004*
RES-002	**Grading practice survey** (Content and research). Document scales, steps, reporting periods and term/annual rules from the samples and 10+	P0	RES-001	M (3–5 d)	Grading configuration catalogue with ≥ 1 verified template per school type;

Task ID	Task and description	Pri.	Dependencies	Estimate	Acceptance criteria
	teacher interviews.				unknowns listed. *Reqs: SCORE-003, SCORE-008, ADMIN-002*
RES-003	**Student and parent interviews** (Content and research). 12–20 student and 8–12 parent interviews plus a 5-minute value survey (n ≥ 100) on willingness to record scores and pay for insights.	P0	—	M (3–5 d)	Synthesis report with top needs, objections and price-sensitivity ranges. *Reqs: —*
RES-004	**Concierge validation** (Content and research). Run a manual pilot: 30–50 students submit scores via form; team returns a hand-built monthly summary; test 2 price points via QR payment.	P1	RES-003	M (3–5 d)	Retention to 2nd month and paid conversion measured; go/no-go recorded. *Reqs: —*
LEG-001	**Lao legal review** (Security). Engage Lao counsel on Law No. 25/NA, consent for minors, cross-border transfer, AI vendor processing, retention, and the expected new data protection law.	P0	—	M (3–5 d)	Written opinion answering the questions in Section 29.2; OD-07 closed. *Reqs: PRIV-001..PRIV-006*
AI-RES-001	**OCR feasibility benchmark** (OCR/AI). Evaluate 2–3 vision-language models and 1 conventional OCR engine on 100 labelled samples for field accuracy, latency and cost.	P0	RES-001	M (3–5 d)	Benchmark report with per-field accuracy, cost per page and recommendation (Section 21.3). *Reqs: OCR-004, AI-010*

38.2 Phase 1 — Core MVP

Task ID	Task and description	Pri.	Dependencies	Estimate	Acceptance criteria
UX-001	**Design tokens and component mapping** (UX/UI). Map module components to existing Bitdoin styles (theme colour #1e3a5f, typography, buttons); define Lao typography rules.	P0	PM-001	S (1–2 d)	Token sheet approved; Lao/EN sample screens pass visual review. *Reqs: AUTH-003*
UX-002	**Onboarding and profile flows** (UX/UI). High-fidelity mobile + desktop designs for screens 1–3 and 23.	P0	UX-001	M (3–5 d)	Prototype tested with 5 students; median onboarding ≤ 120 s. *Reqs: PROFILE-001..007, PRIV-001*
UX-003	**Score entry and history** (UX/UI). Designs for screens 5 and 22 incl. draft, validation, missing-status and correction states.	P0	UX-001	M (3–5 d)	Prototype test: returning user enters 8 scores ≤ 90 s. *Reqs: SCORE-001..013*
UX-004	**Dashboard and charts** (UX/UI). Designs for screen 4 and per-subject view; chart specs incl. accessible tables.	P0	UX-001	M (3–5 d)	Charts reviewed against Section 18 specs and WCAG 2.2 AA contrast. *Reqs: DASH-001..008*
UX-005	**Lao copywriting and localisation pack** (UX/UI). Write all MVP strings in simple Lao and English; native review by 2 Lao educators.	P0	UX-002, UX-003, UX-004	M (3–5 d)	String file complete; review sign-off; no untranslated keys. *Reqs: AUTH-003, NOTIF-004*
UX-006	**Admin configuration screens** (UX/UI). Designs for subject, scale, school and calendar management.	P0	UX-001	S (1–2 d)	Admin flows walkthrough approved by content manager. *Reqs: ADMIN-002..004*
DB-001	**Module schema and migrations** (Database). Create module schema tables for profiles, enrolments, schools, calendars, subjects, aliases, scales, subject enrolments, score entries, scores, revisions, consents, audit.	P0	PM-001	L (6–10 d)	Migrations run forward/back on staging; constraints from Section 26 present; seed data loads. *Reqs: SCORE-008, SCORE-009, PRIV-003*
DB-002	**Row-level/object-level access policies** (Database). Implement ownership and guardian policies at data-access layer (or DB RLS if supported).	P0	DB-001	M (3–5 d)	Policy tests pass for owner, non-owner, guardian (Phase 3 stub) and admin. *Reqs: AUTH-002*
DB-003	**Seed curriculum and calendar data** (Database). Load default subjects, aliases, 0–10 scale and 2026–2027 calendar template from RES-002.	P0	DB-001, RES-002	S (1–2 d)	Seed verified by content manager; IDs stable across environments. *Reqs: ADMIN-002, ADMIN-004*
BE-001	**Identity adapter** (Backend). Resolve Bitdoin session to user reference; map to student_profile; middleware for authorisation context.	P0	PM-001, DB-001	M (3–5 d)	Requests with valid session resolve user; invalid → 401; contract tests with existing auth. *Reqs: AUTH-001*

Task ID	Task and description	Pri.	Dependencies	Estimate	Acceptance criteria
BE-002	**Profile & enrolment API** (Backend). CRUD for profile, enrolment, promotion/transfer flows.	P0	BE-001	M (3–5 d)	API tests for all flows in Section 27; promotion never mutates closed enrolments. *Reqs: PROFILE-001..005*
BE-003	**Score entry API with drafts and revisions** (Backend). Draft save, submit, edit with revision history, duplicate protection, idempotency keys.	P0	BE-001, DB-001	L (6–10 d)	All SCORE ACs automated; idempotent retries create no duplicates. *Reqs: SCORE-001..013*
BE-004	**Analysis engine v1 (descriptive)** (Backend). Period means, subject means, change, rolling mean; snapshot persistence; recompute on change.	P0	BE-003	M (3–5 d)	Golden-file tests (Appendix C) pass to 2 decimals. *Reqs: ANALYSIS-001, ANALYSIS-006*
BE-005	**Admin configuration API** (Backend). Subjects, aliases, scales (versioned), schools (merge), calendars (template + override).	P0	BE-001, DB-001	M (3–5 d)	Role checks enforced; scale versioning prevents destructive edits. *Reqs: ADMIN-002..004*
BE-006	**Consent, export and deletion services** (Backend). Consent records, export job, deletion job with grace period.	P0	DB-001	M (3–5 d)	Export produces documented files; deletion verified by data-sweep test. *Reqs: PRIV-003..005*
BE-007	**Audit logging service** (Backend). Append-only audit events for access-denied, admin actions, edits, consents.	P0	DB-001	S (1–2 d)	Every privileged action writes an event; tamper test fails to modify rows. *Reqs: AUTH-002, ADMIN-006*
BE-008	**Rate limiting and anomaly alerts** (Backend). Per-user/route limits, pagination caps, multi-profile access alert.	P0	BE-001	S (1–2 d)	Load test confirms 429 at threshold; alert fires in staging test. *Reqs: PRIV-007*
BE-009	**In-app notification service (minimal)** (Backend). Store and serve in-app notifications; reminder scheduler for missing periods with caps and quiet hours.	P1	BE-003	M (3–5 d)	Scheduler respects caps in automated tests. *Reqs: SCORE-014, NOTIF-001, NOTIF-003*
FE-001	**Module shell and navigation** (Frontend). Add 'My Scores' entry within Academy navigation; route guards; language sync.	P0	UX-001, BE-001	M (3–5 d)	Route reachable from Academy; language follows EN/ลาว toggle. *Reqs: AUTH-001, AUTH-003*
FE-002	**Onboarding and profile screens** (Frontend). Implement screens 2, 3, 23 with validation and school search.	P0	UX-002, BE-002	M (3–5 d)	E2E tests pass on Android Chrome and desktop. *Reqs: PROFILE-001..007*
FE-003	**Score entry with offline drafts** (Frontend). Implement screen 5 with numeric keypad, Lao digit normalisation, IndexedDB drafts, sync queue.	P0	UX-003, BE-003	L (6–10 d)	Offline draft restore test passes; 90-s entry usability target met. *Reqs: SCORE-001..009*
FE-004	**History screen** (Frontend). Implement screen 22 with grouping, badges and revision viewer.	P0	UX-003, BE-003	M (3–5 d)	Renders 2 years of data in ≤ 1 s after fetch on mid-range device. *Reqs: SCORE-010, SCORE-012*
FE-005	**Dashboard and charts** (Frontend). Implement screen 4: summary card, trend, per-subject, comparison, accessible tables.	P0	UX-004, BE-004	L (6–10 d)	Performance budget met; axe-core zero serious violations. *Reqs: DASH-001..004, DASH-006, DASH-008*
FE-006	**Privacy settings, export and delete UI** (Frontend). Implement 'My data', consent view, export and delete flows.	P0	BE-006	M (3–5 d)	E2E: export downloaded; deletion grace period visible and cancellable. *Reqs: PROFILE-007, PRIV-004, PRIV-005*
FE-007	**Admin configuration UI** (Frontend). Implement subject/scale/school/calendar admin screens.	P0	UX-006, BE-005	M (3–5 d)	Content manager completes seed review using UI only. *Reqs: ADMIN-002..004*
SEC-001	**Threat model and security review** (Security). Validate Section 29.6 threat model against the implemented design; fix findings.	P0	BE-003	S (1–2 d)	No open high/critical findings at release. *Reqs: AUTH-002, PRIV-007*
SEC-002	**IDOR and authorisation test suite** (Security). Automated tests attempting cross-profile access on every endpoint.	P0	BE-002, BE-003	M (3–5 d)	100% of module endpoints covered; all return 404 for non-owners. *Reqs: AUTH-002*
QA-001	**Test plan and fixtures** (QA). Golden datasets (Appendix C), test accounts, device matrix.	P0	DB-003	S (1–2 d)	Fixtures versioned; device matrix agreed. *Reqs: ANALYSIS-001*

Task ID	Task and description	Pri.	Dependencies	Estimate	Acceptance criteria
QA-002	**Automated API and E2E tests** (QA). API contract tests and E2E flows for onboarding, entry, edit, dashboard, export, delete.	P0	QA-001, FE-002..FE-006	L (6–10 d)	CI green; flaky rate < 2%. *Reqs: All P0 MVP*
QA-003	**Localisation and accessibility testing** (QA). Lao rendering on Android Chrome/Samsung Internet, line-breaking, screen reader checks.	P0	FE-005	M (3–5 d)	Checklist in Section 35.3 passed. *Reqs: AUTH-003, DASH-008*
QA-004	**UAT with students and admins** (QA). Run UAT scripts (Appendix E) with 10 students and 2 admins.	P0	QA-002	M (3–5 d)	All UAT scripts pass or have accepted workarounds. *Reqs: All P0 MVP*
OPS-001	**Environments and CI/CD integration** (DevOps). Staging and production configuration for the module in the existing pipeline; feature flags.	P0	PM-001	M (3–5 d)	Module deployable behind a feature flag; rollback rehearsed. *Reqs: —*
OPS-002	**Observability** (DevOps). Structured logs, metrics (latency, errors), dashboards and alerts for module routes and jobs.	P0	OPS-001	S (1–2 d)	SLO dashboards live; alert test fires to on-call channel. *Reqs: NFR-OBS*
OPS-003	**Backup and restore verification** (DevOps). Confirm module tables are in existing backups; run a restore drill.	P0	DB-001	S (1–2 d)	Restore drill RTO/RPO within Section 33 targets. *Reqs: NFR-BAK*
ANL-001	**Event instrumentation** (Analytics). Implement MVP events from Section 32.3 with consent-aware tracking.	P0	FE-001	M (3–5 d)	Events visible in analytics store with schema validation; no PII in event properties. *Reqs: —*
ANL-002	**MVP metrics dashboard** (Analytics). Activation, entry completion, retention, funnel dashboards.	P1	ANL-001	S (1–2 d)	Dashboard reviewed by PM; definitions match Section 32. *Reqs: —*
REL-001	**Pilot launch plan** (Release management). Closed pilot with 3–5 schools/communities, support playbook, feedback loop, go/no-go criteria.	P0	QA-004	S (1–2 d)	Pilot run sheet approved; support macros in Lao ready. *Reqs: —*
REL-002	**General availability launch** (Release management). Gradual rollout via feature flag (10% → 50% → 100% of Academy users).	P1	REL-001	S (1–2 d)	Exit criteria in Section 37 Phase 1 met. *Reqs: —*

38.3 Phase 2 — OCR & analysis

Task ID	Task and description	Pri.	Dependencies	Estimate	Acceptance criteria
P2-OCR	**OCR pipeline and review UI** (OCR/AI). Upload, quality checks, async job workers, extraction schema, subject matching, review/confirm screen, duplicate detection, retention.	P1	AI-RES-001, Phase 1	Epic (30–50 d)	Section 35.4 OCR gates met on hold-out set. *Reqs: OCR-001..012*
P2-ANL	**Analysis engine v2 and diagnostics** (Backend). Strength/weakness classification, trend tests, diagnostic quizzes, conflicting-evidence rules.	P1	Phase 1	Epic (25–40 d)	Golden tests pass; reviewer sign-off on explanation templates. *Reqs: ANALYSIS-002..007, QUIZ-001..005*
P2-REC	**Rules-based recommendations and content tagging** (Backend). Content metadata integration with Academy/Bookstore, rules engine, feedback loop.	P1	P2-ANL, OD-05	Epic (15–30 d)	Only published catalogue items recommended in 100% of tests. *Reqs: LEARN-001..005, LEARN-007*

38.4 Phase 3 — Parent, AI & premium

Task ID	Task and description	Pri.	Dependencies	Estimate	Acceptance criteria
P3-PAR	**Guardian linking, parent dashboard and reports** (Backend). Invite/confirm flows, scopes, reports, disputes, age-of-majority handling.	P1	LEG-001, Phase 1	Epic (30–45 d)	Parent authorisation gate (Section 35.4) passed. *Reqs: PARENT-001..009*
P3-AI	**AI Coach academic mode** (OCR/AI). Tool-scoped retrieval, prompts, safety layer, caching, evaluation pipeline, feedback.	P1	P2-ANL, OD-08	Epic (30–50 d)	AI gates in Section 35.4 passed. *Reqs: AI-001..009*
P3-BIL	**Entitlements and premium packaging** (Backend). Entitlement service linked to Academy subscription, quotas, family plan, manual activation tooling.	P1	OD-11	Epic (15–25 d)	Entitlement gate passed. *Reqs: BILLING-001..007*

Task ID	Task and description	Pri.	Dependencies	Estimate	Acceptance criteria
P3-PLAN	**Study planner and goals** (Frontend). Goals, feasibility, weekly tasks, reschedule, pause, optional XP integration.	P1	P2-REC	Epic (15–30 d)	Usability test: 80% of students create a plan unaided. *Reqs: PLAN-001..006*

38.5 Phase 4 — Career guidance

Task ID	Task and description	Pri.	Dependencies	Estimate	Acceptance criteria
P4-CAR	**Career guidance and pathways database** (Content and research). Questionnaire, explainable ranking, programme database with provenance, career report, explorer integration.	P1	Phase 2, ADMIN-010	Epic (60–100 d)	Career provenance gate passed; fairness evaluation passed. *Reqs: CAREER-001..010, UNI-001..005*

38.6 Phase 5 — Exams & ecosystem

Task ID	Task and description	Pri.	Dependencies	Estimate	Acceptance criteria
P5-EXAM	**Mock exams and ecosystem** (Content and research). Mock exams to verified blueprints, Bookstore recommendations, teacher tools discovery.	P2	Phase 4	Epic (80–150 d)	Content licensing register complete; blueprint verified for the current year. *Reqs: EXAM-001..003, LEARN-006*

39. Dependencies and Risks

39.1 Key dependencies

ID	Dependency	Needed for	Owner
DEP-01	Codebase, hosting and database access	All build work	Owner
DEP-02	Lao legal opinion	Launch with minors; AI/OCR processing	Owner + counsel
DEP-03	Consented score-book samples	OCR, grading configuration	Research
DEP-04	Academy content metadata/API	Recommendations, completion tracking	Academy team
DEP-05	AI Coach architecture	Academic mode	Academy team
DEP-06	Subscription records/activation process	Entitlements	Owner/finance
DEP-07	AI/OCR vendor DPA	Phase 2–3	Owner
DEP-08	Diagnostic item writers and reviewers	Phase 2	Content
DEP-09	Official university sources	Phase 4	Career content
DEP-10	Verified current exam blueprint	Phase 5	Content

39.2 Risk register (prioritised)

ID	Risk	Category	Likelihood	Impact	Mitigation	Owner	Decision required
R-01	Incorrect score extraction by OCR	AI/quality	High	High	Mandatory confirmation; confidence highlighting; silent-error gate ≤ 2%; manual entry equal; correction monitoring	AI engineer	Approve OCR release gate (OD-19)
R-02	Variability in school grading	Domain	High	Medium	Configurable versioned scales; no cross-school comparison; school aliases; educator review	Education content	Default scale & calendar (OD-02)
R-03	Insufficient academic data (students stop recording)	Product	Medium	High	90-s entry; reminders; value after 2 months; concierge validation; parent involvement later	PM	Pilot go/no-go thresholds
R-04	Poor AI Lao-language responses	AI	Medium	High	Native-rater evaluation; templates fallback; simple Lao glossary; AI only after deterministic outputs	AI engineer	Model/ provider selection (OD-03)
R-05	Misleading career recommendations	Ethics	Medium	High	Interests-first, multiple options, no scores-only steering, factors shown, fairness tests, discussion prompts	Career content	Approve methodology (Section 22)
R-06	Outdated university information	Content	High	Medium	Provenance fields, review_by, expiry alerts, annual refresh, out-of-date labels	Career content	Staffing for reviews (OD-20)
R-07	Privacy risks involving minors	Legal/safety	Medium	Very high	Legal review; minimisation; consent model; two-sided guardian links; audit; breach plan	Owner	Consent model (OD-07)
R-08	Low willingness to pay	Commercial	Medium	High	Free core; Premium bundling; career report; price tests; parent-focused value	Commercial	Packaging & prices (OD-11)
R-09	AI operating cost	Commercial	Medium	Medium	Quotas, caching, cheaper models for simple tasks, budget rule (30.6)	AI engineer	Cost budget
R-10	Limited content	Content	High	Medium	Start with 4 subjects; content-gap	Content	Content

ID	Risk	Category	Likelihood	Impact	Mitigation	Owner	Decision required
	availability				tracking; tutor partnerships; Academy reuse		investment
R-11	Difficult integration with existing codebase	Technical	Medium	High	Discovery first; adapters; Option 3 fallback	Architect	Architecture option (OD-01)
R-12	Students feel monitored by parents	Product/ethics	Medium	Medium	Student-controlled scopes; AI chats never shared; supportive report tone	PM	Default scopes
R-13	Policy change (6+3+3, exam format)	External	High (certain over time)	Medium	Education-structure versioning; yearly blueprint verification	Education content	—
R-14	Shared phones and account takeover	Security	Medium	Medium	Re-auth on sensitive actions; session controls	Security	—
R-15	Reputational harm from “official-looking” reports	Trust	Low	Medium	“Self-reported” labelling; no school logos	PM	—

40. Assumptions and Open Decisions

40.1 Assumptions register

ID	Assumption	Validation	Phase
A-01	Most target students receive scores monthly per subject	RES-002	0
A-02	A 0–10 scale with 0.5 steps is the most common monthly scale	RES-002	0
A-03	Students are willing to record scores monthly if it takes < 2 minutes	Concierge pilot, MVP pilot	0–1
A-04	Parents will pay for constructive reports and career guidance	RES-003/004, price tests	0–3
A-05	The existing Bitdoin stack can host a modular academic module	PM-001	0
A-06	The existing AI Coach can be extended with tool calls	OD-08	0
A-07	Hosted VLMs can extract Lao handwritten scores with acceptable flagged-error behaviour	AI-RES-001	0
A-08	Academy content can be tagged by subject/topic for Grades 10–12	OD-05	1–2
A-09	Official university information is publicly available or obtainable	Research	4
A-10	Email use among parents is sufficient for report delivery	Interviews	3

40.2 Decision register — decisions required from the Bitdoin owner

ID	Decision	Options	Recommendation	Needed by
OD-01	Grant codebase/infrastructure access; confirm identity model and architecture option	Option 1/2 vs 3	Option 1/2 (modular monolith) if feasible	Before Phase 1
OD-02	MVP grade scope and platform	Grades 10–12 web/PWA; include 6–9; native app	Grades 10–12, responsive web/PWA	Before Phase 1
OD-03	AI/OCR provider policy and data residency	Hosted provider with DPA; regional; self-hosted	Hosted with no-training, minimal retention, consent; revisit after counsel	Before Phase 2
OD-04	Navigation placement, routes, language storage, digit display	—	“My Scores” in Academy nav; Arabic digits display	Before Phase 1 design
OD-05	Access to Academy content model and completion events	API / DB read / manual import	Read API or events	Before Phase 2
OD-06	Bookstore catalogue access for recommendations	API / DB read	Read API; Phase 5	Before Phase 5
OD-07	Consent model and age thresholds (after counsel)	Section 29.3 options	Conservative default until opinion	Before Phase 1 launch
OD-08	AI Coach academic mode vs separate assistant	Extend Coach / embed new	Extend Coach	Before Phase 3
OD-09	Integrate study tasks with XP & streaks	Yes non-punitive / No	Yes, opt-in, no streak loss	Before Phase 3
OD-10	Career Explorer data provenance and personalisation	Add sources / hide salary	Add sources; hide unsourced figures in personalised views	Before Phase 4
OD-11	Packaging and prices; quotas	Inside Premium; add-ons; prices to test	Inside Premium + Family + one-time report; test prices in 30.5	Before Phase 3
OD-12	SMS/telco billing/messaging channels	Now / later	Later after cost study	Phase 5
OD-13	Admin MFA method	Existing / new	Use existing if available; else TOTP	Before Phase 1
OD-14	Any cohort comparison or composite index	Never / later with methodology	Not in Phases 1–5	—
OD-15	Crisis/safeguarding help information for Lao PDR	—	Owner to approve vetted local resources	Before Phase 3
OD-16	Retention periods	Section 29.7	Adopt after counsel	Before Phase 1 launch

ID	Decision	Options	Recommendation	Needed by
OD-17	Teacher/tutor model (revenue share, access)	—	Discovery in Phase 5	Phase 5
OD-18	Voluntary demographic data for fairness monitoring	Collect separately with consent / do not collect	Do not collect initially; revisit	Phase 4
OD-19	OCR release gate thresholds	Section 35.4	Adopt as proposed	Before Phase 2
OD-20	Staffing for content and pathways reviews	—	1 education content manager + 2 part-time reviewers	Before Phase 2/4

Must be decided before development starts: OD-01, OD-02, OD-04, OD-07 (interim position), OD-13, OD-16.

41. Research Sources

All web sources accessed on **23 October 2026** unless stated. Bitdoin pages reflect the public site on that date.

ID	Source	URL	Publication date	Used for
S-01	Bitdoin — Bookstore & Academy for Lao PDR (home)	https://bitdoin.store/	Live page	Shell, “One account, two platforms”, language toggle
S-02	Bitdoin Academy	https://bitdoin.store/academy	Live page	Academy features, AI Coach, challenges, XP, Career Explorer
S-03	Bitdoin Academy — Membership plans	https://bitdoin.store/academy/subscription	Live page	Prices, manual activation, auto-renew off
S-04	Bitdoin Academy — Career Explorer	https://bitdoin.store/academy/careers	Live page	Career data model, indicative figures
S-05	Bitdoin Bookstore	https://bitdoin.store/bookstore	Live page	Catalogue, categories, book URLs, contacts
S-06	Bitdoin Bookstore FAQ	https://bitdoin.store/bookstore/faq	Live page	Payments, shipping, language, account
S-07	Bitdoin sign-in page	https://bitdoin.store/auth	—	Not inspectable (robots policy)
S-08	KPL — NUOL Holds Entrance Examinations for 2026–2027 Academic Year	https://kpl.gov.la/detail.aspx?id=105667	10 Aug 2026	Exam date, candidates, Groups A/B, 123 programmes
S-09	KPL — National University to accept over 5,900 students	https://kpl.gov.la/En/detail.aspx?id=84373	18 Jul 2024	Entrance requirement; Group A/B subjects
S-10	National University of Laos — Entrance Examination	https://nuol.edu.la/index.php/en/admission/entrance-exmination	2023 exam cycle	Historic paper structure
S-11	KPL — Vientiane Capital Opens 2026–2027 Academic Year	https://kpl.gov.la/detail.aspx?id=106118	31 Aug 2026	School year start; 6+3+3 preparation
S-12	Laotian Times (via tourismlaos.org) — Railway Expansion, Education Reform Among Key Issues at National Assembly	https://www.tourismlaos.org/2026/07/13/railway-expansion-education-reform-among-key-issues-at-national-assembly/ (original: laotiantimes.com, 9 Jul 2026)	Jul 2026	6+3+3 from Grade 6 in 2028–2029
S-13	KPL — Lao Education Sector Records Quality Improvements and Advances Structural Reform	https://kpl.gov.la/detail.aspx?id=96470	2025	Transition 5+4+3 → 6+3+3
S-14	KPL — Laos and Japan Launch New Phase of Primary Mathematics Education Project	https://kpl.gov.la/EN/detail.aspx?id=102406	27 Jul 2026	ESSDP 2026–2030 context
S-15	Tilleke & Gibbins (Multilaw) — Data Protection Laws Guide: Laos	https://www.multilaw.com/Multilaw/Data_Protection_Laws_Guide/DataProtection_Guide_Laos.aspx	n.d.	Law 25/NA; MTC; cross-border consent
S-16	Digital Watch — Laos’ law on electronic data protection No. 25/NA	https://dig.watch/resource/laos-law-on-electronic-data-protection-no-25-na	n.d.	Scope, data categories, authority
S-17	Rajah & Tann Asia — Regional Round-Up: Lao PDR Q4 2025	https://vn.rajahtannasia.com/?p=63924	2025–2026	New Law on Electronic Data Protection in 2026–2030 plan
S-18	KPL — Laos Takes Big Step Toward a Safer Digital Future with New Cybersecurity Law	https://kpl.gov.la/detail.aspx?id=92111	23 Jun 2025	Draft cybersecurity law
S-19	Ogletree Deakins — Lao PDR introduces law to protect employees’ personal electronic data	https://ogletree.com/international-employment-update/articles/july-2020/laos/2020-06-08/lao-pdr-introduces-law-to-protect-employees-personal-electronic-data	8 Jun 2020	Penal Code No. 62/NA offence; “specific data” category
S-20	DataReportal — Digital 2026: Laos	https://datareportal.com/reports/digital-2026-laos	Late 2025	Internet users, mobile connections

ID	Source	URL	Publication date	Used for
S-21	Take App Help — LAO QR Payment	https://www.take.app/help/laoqr-qr-payment	n.d.	LaoQR on LAPNet (secondary)
S-22	Viet Nam News — Seven banks allow customers to pay by QR codes in Laos	https://vietnamnews.vn/economy/1690482/seven-banks-allow-customers-to-pay-by-qr-codes-in-laos.html	11 Jan 2025	LAPNet–NAPAS QR link
S-23	BCEL — TBank–BCEL cross-border QR code payment (PDF)	https://www.bcel.com.la/bcel/bcelweb-datas/file-path/bcel-news/en/bcel_web_file_138_67e48635120d6fa7eec7f03b89948463.pdf	2019	OnePay QR in BCEL One
S-24	pub.dev — comet_payone package documentation	https://pub.dev/documentation/comet_payone/latest	n.d.	Evidence of BCEL-issued merchant credentials (third-party)
S-25	UNICEF Laos — A New Way of Teaching and Learning for the New-Normal (Khang Panya Lao)	https://www.unicef.org/laos/stories/new-way-teaching-and-learning-new-normal	2022	Khang Panya Lao, offline access
S-26	World Bank — SABER School Autonomy and Accountability: Lao PDR Country Report 2016	https://documents1.worldbank.org/curated/en/775471496225301798/text/115351-WP-PUBLIC-ADD-SERIES-SABER-SAA-Lao-PDR-Country-Report.txt	2016	Grade–age mapping
S-27	Supplied concept note — “Monthly Score Tracking & Career Guidance: Idea Evaluation”	Provided by owner	20 Oct 2026	Product concept, risks, sequence
S-28	IPUMS International — EDUCLA (Educational attainment, Laos)	https://international.ipums.org/international-action/variables/EDUCLA	n.d.	History of 5+4+3 structure

Not verified in this PRD (require primary research): programme lists, admission requirements, cut-offs and tuition for NUOL, University of Health Sciences, Souphanouvong University, Savannakhet University and Champasak University; monthly grading practices and score-book formats; Lao labour-market salary data; official payment API terms; statutory age of digital consent.

42. Appendices

Appendix A — Sample screens

Low-fidelity wireframes W1–W5 are in Section 18.3; full screen specifications S1–S24 in Section 18.4. High-fidelity designs are deliverables UX-002…UX-006.

Appendix B — Sample JSON

B.1 Analysis snapshot (excerpt)


```
{
  "snapshot_id": "as_01JD...",
  "enrolment_id": "enr_01JA...",
  "engine_version": "1.2.0",
  "config_version": 3,
  "input_hash": "9c1e…",
  "period_means": [{"key": "2026-09", "mean": 7.17, "coverage": 1.0},
                   {"key": "2026-10", "mean": 7.5, "coverage": 1.0},
                   {"key": "2026-11", "mean": 7.0, "coverage": 1.0},
                   {"key": "2026-12", "mean": 7.5, "coverage": 1.0},
                   {"key": "2027-01", "mean": 7.2, "coverage": 0.83},
                   {"key": "2027-02", "mean": 7.5, "coverage": 1.0}],
  "rolling_mean_3": [{"key": "2026-11", "value": 7.22}, {"key": "2026-12", "value": 7.33},
                     {"key": "2027-01", "value": 7.23}, {"key": "2027-02", "value": 7.4}],
  "subjects": [
    {"code": "MATH", "n": 6, "mean": 5.92, "D": -1.39, "A": 0.0, "sd": 0.74, "slope": -0.33,
"mk_p": 0.18,
     "label": "relatively_lower", "label_confidence": "high", "trend": "declining",
"trend_confidence": "medium",
     "rule_ids": ["L-LOWER-1", "T-DECL-1"]},
    {"code": "ENG", "n": 6, "mean": 6.58, "D": -0.73, "A": 0.17, "sd": 1.11, "slope": 0.3,
"mk_p": 0.57,
     "label": "neutral", "trend": "volatile", "trend_confidence": "medium", "rule_ids": ["T-
VOL-1"]}
  ]
}
```


B.2 Recommendation set


```
{
  "snapshot_id": "as_01JD...",
  "items": [
    {"id": "rec_1", "rule_id": "R2", "rank": 1, "content": {"system": "module_quiz",
"external_id": "quiz_math_g11_v2"},
     "reason": {"template": "diagnose_lower", "params": {"subject": "MATH", "months": 6}},
"est_minutes": 10},
    {"id": "rec_2", "rule_id": "R6", "rank": 2, "content": {"system": "routine",
"external_id": "routine_7day_focus"},
     "reason": {"template": "routine", "params": {"subject": "MATH"}}, "est_minutes": 20}
  ]
}
```


B.3 Career recommendation set


```
{
  "ranking_version": "career_rank.v1",
  "basis": {"questionnaire_completed_on": "2026-10-03", "snapshot_id": "as_01JD...",
"months_of_scores": 6},
  "options": [
    {"career_id": "civil-engineer", "cluster": "engineering",
     "factors": [{"type": "interest", "area": "R", "tier": "high"},
                 {"type": "preference", "item": "work_setting", "value": "outdoors"},
                 {"type": "subject_evidence", "subject": "MATH", "label": "relatively_lower",
"note": "preparation_needed"}],
     "verified_programme_count": 2},
    {"career_id": "registered-nurse", "cluster": "healthcare",
     "factors": [{"type": "interest", "area": "S", "tier": "high"},
                 {"type": "subject_evidence", "subject": "BIO", "label":
```

```
"relatively_stronger"}],
      "verified_programme_count": 1}
  ],
  "constraints_applied": ["min_two_clusters", "max_two_per_cluster",
"interest_high_without_subject_support_included"]
}
```

Appendix C — Golden test fixture

The fixture in Section 20.9 (six subjects × six months, Physics January missing) with expected outputs is the canonical golden dataset for QG-1. Additional fixtures to create: (C2) scale change mid-year 0–10 → 0–100; (C3) transfer with subject alias change; (C4) only 3 months of data; (C5) all statuses (missing/N/A/exempt); (C6) volatile series with positive slope; (C7) repeated grade.

Appendix D — Sample parent report (illustrative data)

Section	Content
1 Reporting period	February 2027 · Grade 11 · Data as of 5 March 2027 · Scores are self-reported by Noy
2 Subjects recorded	6 of 6 subjects
3 Score changes	Average 7.5 (▲ +0.3 from January). Mathematics 5.5 (▲ +0.5), Physics 6 (no January score), Chemistry 8.5 (▲ +0.5), Biology 9.5 (▲ +0.5), Lao Language 8 (■), English 7.5 (▲ +1.5)
4 Relatively stronger	Biology and Chemistry — above Noy's own average every month since September
5 Areas needing support	Mathematics — below Noy's own average every month; slowly going down over 6 months (pattern clear). English — changes a lot month to month
6 Completed learning activities	1 Mathematics diagnostic (Algebra: needs practice; Geometry: secure); 3 Academy lessons; 5 of 6 planned study tasks
7 Suggested next steps for families	Ask Noy which algebra topic felt hardest this month · Help protect two 20-minute study slots this week · Notice effort: 5 of 6 tasks done
Footer	One month's change is not a trend. Scores are entered by the student and are not official school records.

Appendix E — UAT scripts

ID	Role	Script	Pass criteria
UAT-S1	Student (new)	Sign in → open My Scores → complete onboarding with a listed school → confirm subjects	Profile created ≤ 2 min; subjects match score book
UAT-S2	Student	Enter October scores incl. one "Not recorded yet"; deliberately type 11 then fix; review; confirm	Error shown for 11; mean excludes missing; dashboard updates
UAT-S3	Student	Turn on airplane mode; enter November scores; close browser; reopen; reconnect	Draft restored; submission syncs once
UAT-S4	Student	Correct October Mathematics from 7 to 7.5 with reason; open history	Revision visible with old/new values
UAT-S5	Student (Phase 2)	Scan score book; review highlighted fields; correct one; confirm	Nothing saved before confirm; source badge "Photo"
UAT-S6	Student (Phase 2)	Take Mathematics diagnostic; review results; open recommended lesson	Topic bands shown; lesson opens in Academy
UAT-S7	Student (Phase 3)	Ask Coach "Why did my maths fall?"	Answer cites periods/values; no causes asserted
UAT-S8	Student (Phase 4)	Complete questionnaire; open 2 options; check sources	3–6 options; reasons shown; sources dated
UAT-P1	Guardian (Phase 3)	Accept invite; wait for student confirmation; open child dashboard	Access only after confirmation; scopes respected
UAT-P2	Guardian	Open monthly report; follow a suggestion	Report has 7 sections; supportive tone
UAT-P3	Guardian	Student removes link; guardian refreshes	Access removed with message

ID	Role	Script	Pass criteria
UAT-A1	Education Content Manager	Create school from free-text queue; assign 0–100 scale and term calendar; map student	Student sees new scale on next entry
UAT-A2	Super Administrator	Grant Support Agent role; Support Agent performs break-glass view	Audit shows grant and break-glass with reason
UAT-A3	Career Content Reviewer (Phase 4)	Create programme with source; second reviewer approves; set review_by in past	Published, then shown “out of date”
UAT-A4	Financial Administrator (Phase 3)	Activate Premium after bank-transfer evidence	Entitlement active; audit entry

Appendix F — Glossary

Term	Meaning
Enrolment	A student’s participation in one school, grade and academic year (or part-year)
Reporting period	A month or other unit for which scores are reported
Grading scale version	Versioned definition of valid scores for a school or template
Snapshot	Immutable output of the analysis engine for a given set of inputs
Evidence stream	Separate data sources: school scores, diagnostics, mock exams
Scope	A category of data a student allows a guardian to see
Break-glass	Time-limited, reasoned, audited emergency access by staff
Four-eyes	Approval by a second reviewer different from the author
VLM	Vision-language model used for OCR extraction

Appendix G — Diagram sources (Mermaid)

The diagrams in this document were authored in Mermaid and rendered for Word/PDF. Their source code is reproduced below so engineers can edit and re-render them.

Figure 1 — 13. Feature and Epic Overview

```
flowchart LR
  subgraph P1[Phase 1 — Core MVP]
    A[A Profile] --> B[B Manual entry]
    B --> D[D Basic dashboard]
    N1[N Admin config]
    PR[Privacy controls]
  end
  subgraph P2[Phase 2 — OCR & analysis]
    C[C Photo / OCR] --> B
    E[E Analysis + quizzes] --> F[F Recommendations]
  end
  subgraph P3[Phase 3 — Family & premium]
    I[I Guardian & reports]
    G[G AI Coach academic mode]
    H[H Study planner]
    BL[Entitlements]
  end
  subgraph P4[Phase 4 — Careers]
    J[J Career guidance] --> K[K Pathways DB]
  end
  subgraph P5[Phase 5 — Exams & ecosystem]
    L[L Mock exams]
  end
  D --> E
  F --> H
  E --> G
  E --> J
  L --> E
```

Figure 2 — 14.2.2 Manual-entry flow

```
flowchart TB
  subgraph R1[1 · Start]
    direction LR
    S([Open My Scores]) --> P{Profile exists?}
    P -- No --> O[Onboarding]
    P -- Yes --> E[Entry form opens\nfirst open period pre-selected]
    O --> E
  end
  subgraph R2[2 · Enter]
    direction LR
    C[Optionally change\nyear / term / period] --> T[Type scores or set status:\nNot recorded
· N/A · Exempt]
    T --> V{Inline validation\npasses?}
    V -- No --> ERR[Show field error,\nkeep focus] --> T
    V -- Yes --> D[(Draft autosaved locally\nand to server when online)]
  end
  subgraph R3[3 · Confirm]
    direction LR
    RV[Review summary] --> CF{Confirm\nand save?}
    CF -- Edit --> BK[Back to form]
    CF -- Confirm --> SUB[Submit entry\nstate=submitted · source=manual\
nverification=self_reported]
    SUB --> AN[Recompute analysis snapshot] --> DB[Dashboard shows\n'Saved' and what changed]
  end
  R1 --> R2 --> R3
```

Figure 3 — 14.9.1 Linking and authorisation flow

```
sequenceDiagram
  autonumber
  participant S as Student
  participant M as Module
  participant G as Guardian
  S->>M: Create invite (choose scopes)
  M-->>S: Code + link (72 h, single use)
  S->>G: Shares code (in person / messaging)
  G->>M: Sign in with own Bitdoin account, enter code
  M-->>G: Show student display name + requested scopes
  G->>M: Accept, declare relationship
  M-->>S: "Confirm <guardian name> as your guardian?"
  S->>M: Confirm within 7 days
  M->>M: Activate link, write consent_record + audit
  M-->>G: Access granted to selected scopes
  Note over S,G: Either party can remove the link at any time (PARENT-007)
```

Figure 4 — 14.11.3 Administrative workflow

```
stateDiagram-v2
  [*] --> Draft: Reviewer A creates record with sources
  Draft --> InReview: Submit
  InReview --> Draft: Changes requested
  InReview --> Published: Reviewer B approves (four-eyes)
  Published --> NeedsReview: review_by reached or source changed
  NeedsReview --> InReview: Update and resubmit
  Published --> Archived: Superseded by new academic year
  NeedsReview --> Archived: Programme discontinued
  Archived --> [*]
```

Figure 5 — 16.1 J1 — First-time student: from Academy to first insight

```
journey
  title Noy's first visit (Grade 11)
  section Discover
    Sees "Track your monthly scores" card in Academy: 4: Student
    Opens My Scores (already signed in): 5: Student
  section Set up
    Reads short privacy notice: 3: Student
    Chooses grade, year, province, school: 4: Student
    Confirms pre-filled subjects: 5: Student
  section Record
    Enters September scores (draft autosaves): 4: Student
    Reviews and confirms: 5: Student
  section Understand
    Sees summary card and "add one more month" trend prompt: 4: Student
    Enters October scores: 5: Student
    Sees first change chips and subject chart: 5: Student
```

Figure 6 — 16.3 J3 — Diagnosis to improvement loop

```
flowchart TB
  A[Subject flagged\nneeding attention] --> B[Diagnostic quiz\n10-15 items]
  B --> C{Topic bands}
  C -->|Needs practice| D[Practice set +\nAcademy lesson]
  C -->|Developing| E[Consolidation set]
  C -->|Secure| F[No remedial content]
  D --> G[Weekly plan tasks]
  E --> G
  G --> H[Re-test parallel form\nafter 14 days or completion]
  H --> I["Topic change shown<br/>(descriptive)"]
  I --> J[Next monthly scores\nadded to trend]
```

Figure 7 — 17.1 Sitemap

```
flowchart TD
  ROOT[bitdoin.store] --> BS[Bookstore]
  ROOT --> AC[Academy]
  AC --> AH[Academy home]
  AC --> COACH[AI Coach]
  AC --> CE[Career Explorer]
  AC --> SUB[Membership]
  AC --> MS[My Scores — new]
  MS --> MSL[Feature landing - signed out]
  MS --> ONB[Onboarding]
  MS --> DSH[Dashboard]
  DSH --> ENT[Enter scores]
  ENT --> SCAN[Scan score book]
  SCAN --> PROC[Processing]
  PROC --> REV[Review & confirm]
  DSH --> SUBJ[Subject analysis]
  SUBJ --> QZ[Diagnostic quiz]
  DSH --> REC[Recommendations]
  DSH --> PLN[Study planner]
  DSH --> RPT[Monthly report]
  DSH --> HIST[Academic history]
  DSH --> FAM[Family sharing]
  MS --> CAR[Majors & careers]
  CAR --> CQ[Interest questionnaire]
  CAR --> CRES[Career results]
  CRES --> CE
  CRES --> MAJ[Major detail]
  MAJ --> PROG[Programme detail]
  MS --> SET[Profile, privacy & data]
  COACH -.academic mode.-> DSH
  ROOT --> PAR[Family dashboard - guardian]
  ROOT --> ADM[Admin console - staff only]
```

Figure 8 — 24.3 System context

```
flowchart LR
  ST((Student)) --> WEB[Bitdoin web app\nAcademy + My Scores]
  GU((Guardian)) --> WEB
  AD((Admin staff)) --> ADM[Admin console]
  WEB --> CORE[Bitdoin core\naccount, language,\nsubscriptions, payments]
  WEB --> ACAD[Academic module\nmodular monolith]
  ADM --> ACAD
  ACAD --> AC[Existing Academy\nlessons, challenges,\nXP, AI Coach, Career Explorer]
  ACAD --> BS[Existing Bookstore\ncatalogue & offers]
  ACAD --> OBJ[(Object storage\nscore-book images)]
  ACAD --> Q[[Job queue]]
  Q --> W[Workers\nOCR, analysis, reports,\nnotifications, exports]
  W --> AI["AI/VLM providers<br/>under DPA"]
  W --> MAIL[Email / push providers]
  ACAD --> DB[(Module schema\nin Bitdoin DB or\nseparate DB)]
  ACAD --> OBS[Logging, metrics,\nalerting]
```

Figure 9 — 24.4 Component architecture

```
flowchart TB
  subgraph Client["Client — responsive web / PWA"]
    direction LR
    UI[UI components & routes]
    DR[Draft store IndexedDB\n+ sync queue]
    IMG[Image capture,\ncompression, checks]
```

```
  end
  subgraph Edge["Academic module — request path"]
    direction LR
    GW[API layer\nauth context · validation · rate limits]
    AUTHZ[Policy engine\nowner / guardian / admin]
    AUD[(Audit log)]
  end
  subgraph Domain["Domain services"]
    direction LR
    D1[Profiles & enrolments\nScores & revisions\nConfig: subjects, scales,\nschools,
calendars]
    D2[Analysis engine\nRecommendation rules\nQuizzes & topics\nPlanner]
    D3[Guardian links & consent\nCareer & pathways\nEntitlements & quotas\nNotifications]
  end
  subgraph Async["Async workers (via outbox / job queue)"]
    direction LR
    W1[OCR worker]
    W2[Analysis worker]
    W3[Report generator]
    W4[Notification sender]
    W5[Export / deletion worker]
  end
  subgraph Ext["Integration adapters → existing Bitdoin & providers"]
    direction LR
    X1[Identity · Language]
    X2[Academy content · XP · AI Coach]
    X3[Bookstore catalogue]
    X4[Subscriptions · Payments]
    X5[AI/VLM · Email · Push]
  end
  UI ~~~ DR ~~~ IMG
  D1 ~~~ D2 ~~~ D3
  W1 ~~~ W2 ~~~ W3 ~~~ W4 ~~~ W5
  X1 ~~~ X2 ~~~ X3 ~~~ X4 ~~~ X5
  Client --> Edge
  GW --> AUTHZ
  AUTHZ -. every decision .-> AUD
  Edge --> Domain
  Domain -- "events: score_entry.changed,\nocr.job_created, report.due" --> Async
  Domain --> Ext
  Async --> Ext
```

Figure 10 — 24.6 Monthly score workflow

```
sequenceDiagram
  autonumber
  participant C as Client
  participant API as Module API
  participant DB as Module DB
  participant OB as Outbox
  participant AW as Analysis worker
  C->>API: PUT /score-entries/{id}/draft (autosave)
  API->>DB: upsert draft (state=draft)
  C->>API: POST /score-entries/{id}/submit (Idempotency-Key, version)
  API->>API: authorise owner, validate vs scale, check duplicates
  API->>DB: tx: lines + state=submitted + revisions
  API->>OB: score_entry.changed (same tx)
  API-->>C: 200 submitted
  OB-->>AW: dispatch event
  AW->>DB: build_snapshot() and persist if input_hash changed
  C->>API: GET /students/me/dashboard
  API-->>C: latest snapshot + scores
```

Figure 11 — 24.7 OCR workflow

```
sequenceDiagram
  autonumber
  participant C as Client
  participant API as Module API
  participant S as Object storage
  participant Q as Job queue
  participant W as OCR worker
  participant V as VLM provider
  C->>C: capture, crop, quality checks, compress, strip EXIF
  C->>API: POST /ocr/uploads (metadata, hash)
```

```
  API->>API: authorise, quota check, duplicate check
  API-->>C: pre-signed upload URL (5 min, size-limited)
  C->>S: PUT image (private bucket)
  C->>API: POST /ocr/jobs {upload_id, enrolment_id}
  API->>Q: enqueue job
  W->>S: read image (short-lived credential)
  W->>V: extract with ocr_extract prompt + schema
  V-->>W: JSON
  W->>W: schema validate, normalise digits, subject matching, confidence
  W->>API: store extraction (state=ready_for_review)
  C->>API: GET /ocr/jobs/{id} (poll)
  API-->>C: extraction + crops
  C->>API: POST /ocr/jobs/{id}/confirm (edited values)
  API->>API: same validation as manual submit
  API->>API: create/update score entry (source=ocr_confirmed)
```

Figure 12 — 24.8 Academic analysis pipeline

```
flowchart TB
  A[score_entry.changed\nquiz_attempt.completed] --> B[Load submitted lines,\nperiods, scale,
config]
  B --> C[F1–F5 descriptive stats]
  C --> D[F6–F10 per subject]
  D --> E[Labels, trends, confidence]
  E --> F[Merge diagnostic topic bands]
  F --> G[Rules R1–R8 candidates\nfrom published catalogue]
  G --> H[(analysis_snapshot +\nrecommendations)]
  H --> I[Dashboard / reports /\nAI Coach tools]
```

Figure 13 — 24.9 Career recommendation pipeline

```
flowchart LR
  Q[Questionnaire answers] --> P["Interest profile<br/>(ipsative tiers)"]
  S[Latest analysis snapshot] --> E[Subject evidence]
  PR[Optional preferences] --> F[Preference fit]
  T[(Career taxonomy\npublished only)] --> R[Explainable ordering\n+ diversity constraints]
  P --> R
  E --> R
  F --> R
  R --> O[3–6 options + factors]
  DBU[(Pathways DB\nverified, in-date)] --> O
  O --> X[Optional AI explanation\ncareer_explanation.v1\nvalidated vs factors]
  X --> UI[Results / Report]
```

Figure 14 — 26.2 ER diagrams — View 1 — Profile, configuration, scores and OCR

```
erDiagram
  direction LR
  STUDENT_PROFILE ||--o{ ENROLMENT : has
  SCHOOL ||--o{ ENROLMENT : "attended in"
  ACADEMIC_YEAR ||--o{ ENROLMENT : for
  GRADE ||--o{ ENROLMENT : "at grade"
  GRADING_SCALE_VERSION ||--o{ ENROLMENT : "scale used"
  CALENDAR_TEMPLATE ||--o{ REPORTING_PERIOD : defines
  ENROLMENT ||--o{ SUBJECT_ENROLMENT : includes
  SUBJECT ||--o{ SUBJECT_ENROLMENT : "taken as"
  SUBJECT ||--o{ SUBJECT_ALIAS : "known as"
  ENROLMENT ||--o{ SCORE_ENTRY : records
  REPORTING_PERIOD ||--o{ SCORE_ENTRY : for
  SCORE_ENTRY ||--o{ SCORE_LINE : contains
  SUBJECT_ENROLMENT ||--o{ SCORE_LINE : "scored in"
  SCORE_LINE ||--o{ SCORE_REVISION : history
  ENROLMENT ||--o{ OCR_UPLOAD : has
  OCR_UPLOAD ||--o{ OCR_JOB : "processed by"
  OCR_JOB ||--o{ OCR_EXTRACTED_FIELD : yields
  OCR_JOB ||--o| OCR_CONFIRMATION : "confirmed by"
  ENROLMENT ||--o{ ANALYSIS_SNAPSHOT : produces
```

Figure 15 — 26.2 ER diagrams — View 2 — Learning, planning, guardians, consent and entitlements

```
erDiagram
  direction LR
  STUDENT_PROFILE ||--o{ QUIZ_ATTEMPT : takes
  QUIZ ||--o{ QUIZ_ITEM : contains
  TOPIC ||--o{ QUIZ_ITEM : "tagged to"
  QUIZ ||--o{ QUIZ_ATTEMPT : "attempted as"
  QUIZ_ATTEMPT ||--o{ TOPIC_MASTERY : estimates
```

```
  ANALYSIS_SNAPSHOT ||--o{ LEARNING_RECOMMENDATION : generates
  CONTENT_ITEM_REF ||--o{ LEARNING_RECOMMENDATION : references
  STUDENT_PROFILE ||--o{ LEARNING_GOAL : sets
  STUDENT_PROFILE ||--o{ STUDY_PLAN : follows
  STUDY_PLAN ||--o{ STUDY_TASK : contains
  STUDENT_PROFILE ||--o{ GUARDIAN_LINK : "is linked by"
  GUARDIAN_LINK ||--o{ PARENT_REPORT : receives
  STUDENT_PROFILE ||--o{ CONSENT_RECORD : has
  STUDENT_PROFILE ||--o{ ENTITLEMENT : "holds (via user ref)"
  STUDENT_PROFILE ||--o{ NOTIFICATION : "receives (via user ref)"
```

Figure 16 — 26.2 ER diagrams — View 3 — Careers and pathways

```
erDiagram
  direction LR
  STUDENT_PROFILE ||--o{ INTEREST_RESPONSE_SET : answers
  STUDENT_PROFILE ||--o{ CAREER_RECOMMENDATION_SET : receives
  STUDENT_PROFILE ||--o{ CAREER_SHORTLIST : saves
  CAREER ||--o{ CAREER_SHORTLIST : "saved as"
  CAREER ||--o{ CAREER_PROGRAMME_LINK : "studied via"
  INSTITUTION ||--o{ FACULTY : has
  FACULTY ||--o{ PROGRAMME : offers
  PROGRAMME ||--o{ CAREER_PROGRAMME_LINK : "leads to"
  PROGRAMME ||--o{ ADMISSION_REQUIREMENT : requires
  PROGRAMME ||--o{ CUTOFF_RECORD : "historic cut-offs"
  PROGRAMME ||--o{ TUITION_RECORD : costs
  SOURCE ||--o{ ADMISSION_REQUIREMENT : cites
  SOURCE ||--o{ PROGRAMME : cites
  SOURCE ||--o{ SCHOLARSHIP : cites
```

Figure 17 — 30.8 Revenue model and conversion funnel

```
flowchart TB
  A[Academy visitors] --> B[My Scores profile created]
  B --> C[First period submitted]
  C --> D["2nd period submitted<br/>(trial offered)"]
  D --> E[Trial / feature gate\nencounter]
  E --> F[Parent linked or\npayer identified]
  F --> G[Paid Premium / Family]
  C --> H[Career report\nmoment Grade 11-12]
  H --> G
```

Figure 18 — 37.1 Sequence review

```
gantt
  title Indicative sequence (durations illustrative, not commitments)
  dateFormat  YYYY-MM-DD
  axisFormat  %b %Y
  section Phase 0
  Research & validation        :p0, 2026-11-02, 45d
  section Phase 1
  Core MVP build               :p1, after p0, 90d
  Pilot                        :p1b, after p1, 45d
  section Phase 2
  OCR + analysis + diagnostics :p2, after p1b, 90d
  section Phase 3
  Parent + Coach + premium     :p3, after p2, 100d
  section Phase 4
  Career guidance              :p4, after p3, 90d
  section Phase 5
  Exams & ecosystem            :p5, after p4, 100d
```